

codex-windows / 9aed6235-275c-4ce8-9a82-1ed7f8b297e1

Metadata

```
- Source: `codex-windows`  
- Kind: `codex`  
- Source file: `C:\Users\avidu\codex\archived_sessions\rollout-2026-07-02T11-41-33-019f2174-3d66-70b2-98c9-c41db7f55d96.jsonl`  
- SHA-256: `78f6a2eeab4b3da1a0b565df07e61bec4e5e5810061adfd1baff75e8f9c4f1fb`  
- Source modified: `2026-07-02T06:11:34+00:00`  
- Imported at: `2026-07-05T16:48:16+00:00`  
- cli_version: `0.0.0`  
- cwd: `C:\Users\avidu\Projects\badminton-highlight-indexer`  
- session_id: `9aed6235-275c-4ce8-9a82-1ed7f8b297e1`  
- source: `vscode`
```

Transcript

1. user (2026-07-02T06:11:34.063Z)

Hello!

2. assistant (2026-07-02T06:11:34.063Z)

```
[external_agent_tool_call: Skill]  
input: {"skill": "new-session-hello"}  
[/external_agent_tool_call]
```

3. assistant (2026-07-02T06:11:34.063Z)

```
[external_agent_tool_result]  
Launching skill: new-session-hello  
[/external_agent_tool_result]
```

4. assistant (2026-07-02T06:11:34.063Z)

```
[external_agent_tool_call: Read]  
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md  
[/external_agent_tool_call]
```

5. assistant (2026-07-02T06:11:34.063Z)

```
[external_agent_tool_result]  
<system-reminder>[Truncated: PARTIAL view ? showing lines 1-247 of 334 total (28664 tokens, cap 25000). Call Read with offset=248 limit=247 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>
```

```
1      # CODE_MAP.md  
2  
3      **Purpose:** cold-start navigation map for the badminton-highlight-indexer ? a self-  
hosted video ? highlight-reel pipeline (CV/AI rally detection ? SQLite ? ffmpeg stitch) with a  
segmenter-agnostic eval/calibration stack, typed config, and per-video observability reporting.  
4  
5      **Regenerate:** rerun the `build-code-map` workflow (parallel subsystem readers ? this  
doc). Re-run after any registry/contract/WSL-interface change, or after a directory restructure  
(this revision follows #40).  
6  
7      ? **2026-06-10 audit-hardening sweep changed several behaviors documented below ? trust  
git history over stale lines until the next full regen:** `add_video` is now an **UPSERT** (no  
FK-cascade wipe on re-ingest; segmentation is destroy-AFTER-confirm via  
`segment_watermarks`/\`prune_segments`); detector artifacts are keyed by **video_id**, not
```

```

basename; **`indexing.wasb.*` is now a real typed config** (was silently dropped); `timeout_sec`
defaults **1800s** (was 0/hang-forever); SQLite opens with **WAL + busy_timeout**;
`match_intervals` requires real overlap (iou>0); `run_regression` baseline lives in tracked
`eval_baselines/` and **exits 3 on a missing baseline**; the API validates
`video_path`/`output_filename`. New tests: `test_api_endpoints`, `test_compiler`,
`test_wasb_runner`, `test_reingest_safety`, `test_detector_keying`, `test_config_wasb`,
`test_path_safety`, `test_reliability_hardening` (+ collector `test_licensing_gate`,
`test_import_guard`, `test_cvat_robustness`).
8
9     **LEGEND**
10    - `@path` = file (always repo-relative)
11    - `::sym` = symbol (class/func/const) in the nearest preceding `@path`
12    - `->` = dataflow / call / "produces"
13    - `S` = data contract (canonical shape)
14    - `?` = gotcha / footgun
15    - `?` = registry / plugin extension point
16    - `WSL` = runs inside WSL2 conda env, NOT the Windows Python process
17
18    ---
19
20    ## 1. PIPELINE
21
22    ### 1A. Runtime: video file -> highlight reel
23    ```
24    typed config load (built-in defaults -> config.json overrides; absent/parse-fail -> all-
defaults, never fatal)
25                                @backend/config/loader.py::load_config ->
@backend/config/models.py::AppConfig
26    -> video_id = MD5(whole file)                                @backend/utils/hashing.py::compute_video_id
(single shared helper, 64KB chunks; imported by main + all run_*.py ? no more per-entrypoint
copies)
27    -> validate
@backend/pipeline/validators/base.py::registry.run_all_with_context(video_path, global_config)
28    FormatValidator -> ctx['ffprobe_meta'] -> ResolutionFPSValidator -> PTSContinuity
-> SceneCut -> Blur -> Relevance (? ACTUAL registration/exec order ? see
@backend/pipeline/validators/__init__.py)
29    [validation FAIL -> persist status="failed", return early; report STILL emitted
in finally]
30    -> db.add_video(video_id, filepath, status, [r.dict()])
@backend/infrastructure/database.py
31    -> seg_name = req.segmenter or global_config.indexing.default_segmenter (fallback
'motion')
32    -> seg = segmenter_registry.get(seg_name)(db, global_config) ?
@backend/pipeline/segmenters/base.py::segmenter_registry (400 + available list if missing)
33    -> segments = seg.process_video(video_id, video_path, player_pool?, max_frames?)
34    [STOP POINT: segments-only ? predictions now in DB; eval/regression operate here
without stitching]
35    -> finally: emit_indexer_report(...) @backend/reporting/__init__.py (? ALWAYS runs
? even on validation failure / exception; NEVER raises; grafts response["reports"])
36    -> select segment ids -> [(start,end)] (+ stitching padding)
37    -> VideoCompiler.compile_highlights(filepath, time_pairs, ou?
[/external_agent_tool_result]

```

6. assistant (2026-07-02T06:11:34.063Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md
[/external_agent_tool_call]

```

7. assistant (2026-07-02T06:11:34.063Z)

```

[external_agent_tool_result]
1    # Next steps (live) ? owned-model + multisport
2
3    **Updated:** 2026-06-10 (end of the audit?M0?Gemini-battery session). **Authoritative

```

```

roadmap:**
4     `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-
blow:** memory
5     `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
6
7     ## Where we are (one paragraph)
8     ADR-008 ratified ("repo split driven by productionization, not isolation"): training
lives in-repo (`training/`)
9     behind a CI-enforced import wall; the featurizer is single-sourced
(`backend/features/rally_seq.py`, train==serve);
10    the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest
output/human_lovo_manifest.json --floor
11    eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0
(LOVO **0.626**, floor passed,
12    sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned
corpus is in the collector**
13    (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet
reclassification). The **Gemini
14    battery is done** (see table) ? the moat is quantified.
15
16    ## The quality table that now drives strategy (IoU 0.5 vs human golden labels)
17    | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
18    |---|---|---|
19    | Heuristic (velocity windowing) | 0.657 | 0.157 |
20    | **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |
21    | Two-stage WASB+Gemini refine | 0.83 | ? |
22    | **Gemini wrapper** (`gemini-flash-latest`) | **0.93** | **0.66** |
23
24    **Reading:** clean footage is commoditized (serve it with Gemini for cents). **Multi-
court drops the commodity
25    wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game
pickups + dead-time merges,
26    the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-
growth-sized gap, not an
27    architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG
tapes (uploads candidate clips
28    only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry
load-bearing.
29    Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json,
gemini_wrapper_2026-06-10.json).
30
31    ## Prioritized next steps
32    1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move;
every lever below feeds on it.
33    Priority order for new videos: **(a) multi-court badminton** (the target distribution
AND where the ship target
34    lives), **(b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1
0.909; pickleball labels are
35    corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c)
capture variety per
36    `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training pool**
(consent guardrail). Per video:
37    label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ?
re-run the Gen-0 harness.
38    The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is
reachable this month.
39    2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480
(necessary), **multi-court
40    reference = wrapper 0.66** (the number that makes the owned model worth serving),
clean-footage ceiling = 0.93
41    (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70
on multi-court folds with
42    paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.
43    3. **M0.4 next increment ($0/local):** swap the Gen-0 LogReg for

```

```
**ActionFormer/ASFormer** (MIT, minutes on the
44      2070S ? still lightweight, NOT the heavyweight Gen-2 stack) inside the existing
harness; per-video threshold
45      calibration (the LOVO-vs-oracle gap is visible per fold in the harness output).
46      4. **Multisport thread:** ingest Rooro pickleball/TT CSVs (`import-local --normalize`);
merged-prototype LOVO across
47      badminton+TT once TT has ?3 matches; identify a clean pickleball b?
[/external_agent_tool_result]
```

8. assistant (2026-07-02T06:11:34.064Z)

```
[external_agent_tool_call: mcp_workspace_bash]
command: cd /sessions/great-sharp-gauss/mnt/badminton-highlight-indexer && git log --oneline -15
&& echo "---BRANCH---" && git branch --show-current && echo "---STATUS---" && git status --short
[/external_agent_tool_call]
```

9. assistant (2026-07-02T06:11:34.064Z)

```
[external_agent_tool_result]
05bf49e docs: field-research recruiter flyer (C13 interviews + Phase-2 recording rules) (#86)
747a7ad docs: NEXT_STEPS session wrap ? quality table, ship-target 0.66, corpus-growth
priorities (#85)
d19ef36 eval: record the Gemini-wrapper baselines (the commoditization/moat reference) (#84)
2a101cd fix(provider): 6 generate attempts w/ exponential backoff+jitter (503 waves outlast 3x
linear) (#83)
f01387e test(reporting): pin the CI-visible half of the oversize channel (no obsreport needed)
(#82)
3377486 test(reporting): pin the CI-visible half of the oversize channel (no obsreport needed)
(#81)
a2e65d7 fix(gemini): re-encode AI chunks (4K oversize skip = silent 0-segments) + loud report
flag (#80)
6353bfb fix(gemini): honor indexing.ai_max_workers (was hardcoded 5); ship serial default (#79)
4ca7526 chore(config): ai_model -> gemini-flash-latest (auto-tracking alias) (#78)
d51efe3 feat(training): Gen-0 TAL harness ? one-command train -> honest LOVO -> ship-gate ->
artifact (M0.4 scaffold) (#77)
bfb5432 fix(tests): featurizer purity test skips on hosts without numpy/scipy (#76)
d7f9896 feat(features): promote the rally_seq featurizer to backend/features + erect the ADR-008
import wall (#75)
d9d6600 feat(eval): canonical GT loader ? close the golden-CSV two-loader fork (ADR-008 prereq)
(#74)
6c2cee0 docs: ADR-008 ? owned-model topology: artifact boundary now, repo split at Gen-2 (#73)
87c10c9 docs: refresh NEXT_STEPS after the audit-hardening sweep + corpus reclass (#72)
---BRANCH---
master
---STATUS---
M .github/workflows/ci.yml
M CLAUDE.md
M README.md
M TEST_RUN_SUMMARY.md
M backend/annotations/__init__.py
M backend/annotations/automated_provider.py
M backend/annotations/base.py
M backend/annotations/file_provider.py
M backend/api/static/app.js
M backend/api/static/index.html
M backend/api/static/style.css
M backend/config/__init__.py
M backend/config/loader.py
M backend/config/models.py
M backend/eval/annotations.py
M backend/eval/audio/__init__.py
M backend/eval/audio/e1_probe.py
M backend/eval/batch.py
M backend/eval/calibrate_local.py
M backend/eval/calibrate_wasb.py
```

```
M backend/eval/calibration.py
M backend/eval/distill_local.py
M backend/eval/eval_partial_labels.py
M backend/eval/export_wasb_segments.py
M backend/eval/gemini_refine.py
M backend/eval/gt_loader.py
M backend/eval/harness.py
M backend/eval/metrics.py
M backend/eval/rally_seq_proto.py
M backend/eval/regression.py
M backend/features/__init__.py
M backend/features/rally_seq.py
M backend/infrastructure/database.py
M backend/main.py
M backend/pipeline/audio/__init__.py
M backend/pipeline/audio/hit_detector.py
M backend/pipeline/detectors/__init__.py
M backend/pipeline/detectors/base.py
M backend/pipeline/detectors/rally_gate.py
M backend/pipeline/detectors/tracknet_runner.py
M backend/pipeline/detectors/wasb_infer.py
M backend/pipeline/detectors/wasb_runner.py
M backend/pipeline/segmenters/__init__.py
M backend/pipeline/segmenters/base.py
M backend/pipeline/segmenters/gemini.py
M backend/pipeline/segmenters/motion.py
M backend/pipeline/segmenters/tracknet_hybrid.py
M backend/pipeline/segmenters/wasb_hybrid.py
M backend/pipeline/segmenters/yolo_hybrid.py
M backend/pipeline/stitcher/compiler.py
M backend/pipeline/validators/__init__.py
M backend/pipeline/validators/base.py
M backend/pipeline/validators/blur.py
M backend/pipeline/validators/format.py
M backend/pipeline/validators/pts_continuity.py
M backend/pipeline/validators/relevance.py
M backend/pipeline/validators/resolution_fps.py
M backend/pipeline/validators/scene_cut.py
M backend/providers/gemini.py
M backend/reporting/__init__.py
M backend/reporting/harvest.py
M backend/reporting/run_signals.py
M backend/reporting/spec.yaml
M backend/results.py
M backend/tools/export.py
M backend/tools/rally_slicer.py
M backend/utils/hashing.py
M backend/utils/paths.py
M backend/utils/video.py
M config.json
M docs/BACKLOG.md
M docs/CODE_MAP.md
M docs/C?
```

[/external_agent_tool_result]

10. assistant (2026-07-02T06:11:34.064Z)

```
[external_agent_tool_call: mcp_workspace_bash]
command: cd /sessions/great-sharp-gauss/mnt/badminton-highlight-indexer && git diff --stat |
tail -3 && echo "---one file diff---" && git diff CLAUDE.md | head -20 && echo "---artifacts---"
&& ls -R artifacts/ 2>/dev/null | head -20
[/external_agent_tool_call]
```

11. assistant (2026-07-02T06:11:34.064Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\HIGH_RISK_REMEDIATION_PLAN.md
[/external_agent_tool_call]
```

12. assistant (2026-07-02T06:11:34.064Z)

```
[external_agent_tool_result]
training/gen0/harness.py          | 432 +++----
training/requirements-train.txt   | 28 +-
164 files changed, 20708 insertions(+), 20708 deletions(-)
---one file diff---
diff --git a/CLAUDE.md b/CLAUDE.md
index f081b93..a9c47e8 100644
--- a/CLAUDE.md
+++ b/CLAUDE.md
@@ -1,55 +1,55 @@
-# CLAUDE.md ? project memory / agent handoff
-
-Local, AI-powered **badminton (? racket-sports) highlight indexer + rally detector**:
-FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid **Gemini** as the cloud AI. Sibling
-repos: `sports-data-collector`, a VLC-based `rally-annotator`.
-
-## Read these first
-- **`docs/README.md`** ? doc index. **`docs/CODE_MAP.md`** ? code navigation + data
- contracts (the cold-start map; read before touching code).
-- **Authoritative owned-model roadmap:** `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ the
- live `docs/NEXT_STEPS.md`). These SUPERSEDE the 2026-06-07 strategy docs below where they
- disagree (licensing, Gemma-4 stance, base model, conversational-QA).
-- **Strategy (post-scaffolding):** `docs/COMMERCIALIZATION.md`, `docs/PMF_MARKET_RESEARCH.md`,
- `docs/PLATFORM_ARCHITECTURE.md`, `docs/DEFERRED_HARDENING_PLAN.md`. Also `docs/ROADMAP.md`
- (offline-segmenter narrative) and `docs/BACKLOG.md`.
---artifacts---
artifacts/:
rally_tal_gen0

artifacts/rally_tal_gen0:
0.1.0

artifacts/rally_tal_gen0/0.1.0:
manifest.json
weights.json

[/external_agent_tool_result]
```

13. assistant (2026-07-02T06:11:34.064Z)

```
[external_agent_tool_result]
1      # High-risk remediation plan (for owner approval)
2
3      **Source:** 2026-06-10 ultracode 8-auditor deep audit (`wf_2a64bc88-2a8`), all findings
adversarially verified.
4      **Status:** AWAITING OWNER APPROVAL ? no PR opens until you pick a scope + push
preference.
5      **Conventions:** branch off `master`, ONE PR per item, no stacking (per CLAUDE.md).
Owner-gated work
6      (M0.x owned-model, async-job slice, repo rename) is intentionally **excluded** ? this is
repo-health only.
7
8      PR-1 (compiler SyntaxError + import smoke test + CI) is **already done and merged
(#63)**.
9
10     ---
11
12     ## Tier 0 ? guardrail violation (do first, it's the project's whole premise)
13
```

```

14     ### PR-A · Collector licensing gate ? stop certifying broadcast video as commercial-
clean · CRITICAL
15     **Repo:** sports-data-collector
16     **Problem:** `src/cli/ingest.py:12` defaults `--license MIT`. The ShuttleSet
**annotation** license gets stamped
17     onto the **YouTube broadcast video** it annotates; `is_commercial` then gates export.
The committed
18     `data/eval_export/manifest.json` ships 24 videos / 1915 segments as `is_commercial=true`,
license_type=MIT`,
19     including `shuttleaset_21` = a YONEX Thailand Open broadcast. This directly violates the
"training data must be
20     commercial-clean" guardrail the owned model rests on.
21     **Change:** (1) remove the `MIT` default ? make `--license` required or default to an
explicit `UNKNOWN`/`UNVERIFIED`
22     that is **excluded** from `is_commercial` export; (2) separate `annotation_license` from
`video_license` (the rights
23     that actually gate training are the video's); (3) re-audit the 24 exported entries,
demote broadcast-sourced rows out
24     of the commercial export; (4) add a test asserting an `UNKNOWN`-license video never
appears in a commercial export.
25     **Verify:** `pytest` (collector); export a fixture with a broadcast row ? assert
excluded.
26
27     ---
28
29     ## Tier 1 ? correctness & data integrity (silent corruption of results/corpus)
30
31     ### PR-B · Detector artifacts keyed by video_id, not file basename · HIGH
32     **Repo:** indexer · `backend/pipeline/detectors/wasb_runner.py`, `wasb_infer.py`,
`tracknet_runner.py`
33     **Problem:** WSL staging, `_frames`, `_wasb.csv`, and the cache manifest are
keyed by file basename.
34     Two different videos named `match.mp4` (common phone names) ? `extract_frames` sees
enough PNGs, skips re-extraction,
35     and `manifest_compatible` returns true ? WASB emits video A's trajectory as video B's,
persisted under B's MD5. Silent
36     training-data poisoning; everything else in the system is MD5-keyed.
37     **Change:** key staging/frames/cache/output paths by `video_id` (MD5) or include a
content hash in the cache manifest.
38     **Verify:** unit test ? two distinct files, same basename, B's frame count ? A's ?
assert B does not reuse A's frames/CSV.
39
40     ### PR-C · Re-ingest must not destroy prior good results · HIGH
41     **Repo:** indexer · `backend/main.py:112,123,369`,
`backend/infrastructure/database.py:107`
42     **Problem:** `add_video` is `INSERT OR REPLACE` with FK CASCADE; it runs **before** the
segmenter and on the
43     failed-validation path, so re-submitting an already-good video (transient ffmpeg
hiccup, ffmpeg off PATH, tightened
44     thresholds) cascade-deletes its play/candidate segments ? including non-recomputable
`human_label`/`notes` ? and a
45     subsequent empty run leaves zero. Destroy-before-confirm.
46     **Change:** compute the new segments first, then replace within a transaction (or use
the collector's UPSERT pattern
47     that preserves children); never wipe on the failed/early-return path.
48     **Verify:** test ? good video ? re-POST that now fails validation ? prior segments still
present.
49
50     ### PR-D · Nested config sections silently drop unknown keys · HIGH
51     **Repo:** indexer · `backend/config/models.py`
52     **Problem:** only `AppConfig` sets `extra='allow'`; nested
`IndexingConfig`/`ValidationConfig` default to `ignore`, so
53     the entire `indexing.wasb` ?
[/external_agent_tool_result]

```

14. assistant (2026-07-02T06:11:34.064Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-6be2-4d26-b022-b61c256ea645\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\MEMORY.md
[/external_agent_tool_call]
```

15. assistant (2026-07-02T06:11:34.064Z)

```
[external_agent_tool_result: error]
`C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-6be2-4d26-b022-b61c256ea645\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\MEMORY.md` is outside this session's connected folders, so Read can't reach it. If this is a user project or working folder, request it with the `request_cowork_directory` tool ? the user will be asked to approve it. Don't request system or application-internal directories.
[/external_agent_tool_result]
```

16. assistant (2026-07-02T06:11:34.064Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\MEMORY.md
[/external_agent_tool_call]
```

17. assistant (2026-07-02T06:11:34.064Z)

```
[external_agent_tool_result: error]
File does not exist. Note: your current working directory is
C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\local_0303e96c-dada-4fe1-8bf4-03585eb32cde\outputs.
[/external_agent_tool_result]
```

18. assistant (2026-07-02T06:11:34.065Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\crlf-mount-artifact.md
[/external_agent_tool_call]
```

19. assistant (2026-07-02T06:11:34.065Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\crlf-mount-artifact.md (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

20. assistant (2026-07-02T06:11:34.065Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\MEMORY.md
[/external_agent_tool_call]
```

21. assistant (2026-07-02T06:11:34.065Z)

```
[external_agent_tool_result]
```


File created successfully at: C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\MEMORY.md (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

22. assistant (2026-07-02T06:11:34.065Z)

```
[external_agent_tool_call: AskUserQuestion]
input: {"questions":[{"question":"What should we work on this session?","header":"Session focus","multiSelect":false,"options":[{"label":"M0.4 increment","description":"Swap Gen-0 LogReg for ActionFormer/ASFormer inside the existing harness + per-video threshold calibration ($0/local, NEXT_STEPS #3). Picking this = your greenlight."},{label":"Async job model","description":"The committed next PLATFORM slice (DEFERRED_HARDENING_PLAN #16, single-tenant). Picking this = your greenlight."},{label":"Ship-gate decision","description":"Work through and document the ship-gate target (suggested: LOVO F1@tIoU0.5 ? 0.70 on multi-court folds, paired-sign p<0.05) as an ADR."},{label":"Housekeeping","description":"Clean up the executed remediation-plan scratch file, prep repo rename, corpus/eval support tooling ? no gated work."}]}]}
[/external_agent_tool_call]
```

23. assistant (2026-07-02T06:11:34.065Z)

```
[external_agent_tool_result]
Your questions have been answered: "What should we work on this session?"="[No preference]". You can now continue with these answers in mind.
[/external_agent_tool_result]
```

24. assistant (2026-07-02T06:11:34.065Z)

No problem ? context is loaded and I'm ready whenever you are. Nothing gated will start without your say-so. If you just want to chat through something first (e.g., the ship-gate shape, or how to prioritize corpus growth vs. M0.4), happy to do that too ? just tell me where to dig in.

25. user (2026-07-02T06:11:34.065Z)

I am working with your compatriot Claude Code on developing this project. As you can imagine it is being developed as a CLI but the eventual goal is to have a decently simple UI on top of it to expose to users. Can you deeply study them and the docs regarding the current state and the roadmap and the vision and keep in mind that this will be used by sports enthusiasts and coaches to come up with some good proposals for UI and make sure to chunk the work in a quick and dirty alpha phase and a more polished beta. I have also acquired a domain named khelstura.guru for hosting this so I am also looking at your help to completely manage the hosting and give me good alternatives as to set it up for you to take over and create this UI for me and link/connect it with the CLI.

26. assistant (2026-07-02T06:11:34.065Z)

This is a meaty one ? UI strategy + hosting takeover for khelstura.guru. Before I dive into the deep study, four quick decisions that materially shape the proposals:

27. assistant (2026-07-02T06:11:34.065Z)

```
[external_agent_tool_call: AskUserQuestion]
input: {"questions":[{"question":"Where should the processing backend live for the alpha? (The WASB/owned-model path needs your GPU+WSL box; the Gemini segmenter path only needs ffmpeg + an API key, so it can run on cheap cloud CPU.)","header":"Alpha backend","multiSelect":false,"options":[{"label":"Local box + tunnel (Recommended)","description":"UI hosted on the domain, backend stays on your GPU box exposed via a secure tunnel (e.g. Cloudflare Tunnel). $0 compute, full pipeline available, alpha-appropriate."},{label":"Cheap cloud VPS","description":"UI + API on a ~$5/mo VPS running the Gemini-only path. Always-on without your box, but no WASB/owned-model and upload bandwidth costs."},{label":"Hybrid","description":"Cloud VPS for UI/API/Gemini path + your GPU box attached as an optional worker for WASB jobs."},{label":"Full cloud"}]}]}
[/external_agent_tool_call]
```

```
GPU","description":"Everything cloud-hosted including GPU inference. Most expensive (~$100+/mo or per-hour on-demand), only worth it at beta+."}}],{"question":"What monthly hosting budget are you comfortable with through alpha and into beta?","header":"Budget","multiSelect":false,"options":[{"label":"~$0?5/mo","description":"Free tiers + maybe a tiny VPS. Forces the local-box-tunnel architecture."},{label":"~$10?25/mo","description":"Small VPS + storage/CDN headroom. Comfortable alpha, lean beta."},{label":"~$25?100/mo","description":"Room for managed services, object storage for videos, occasional on-demand GPU."},{label":"Cost-optimal at each phase","description":"You decide per-phase what's justified; I propose the cheapest thing that works and flag upgrade triggers."}}],{"question":"Who will actually use the alpha?","header":"Alpha users","multiSelect":false,"options":[{"label":"Just me","description":"Single-user; auth can be a simple allowlist; no multi-tenancy concerns at all."},{label":"Me + 2?3 coaches","description":"The C13 interview / camera-pilot academies. Email-allowlist auth, single shared workspace or simple per-user sepa?[/external_agent_tool_call]
```

28. assistant (2026-07-02T06:11:34.065Z)

```
[external_agent_tool_result]
Your questions have been answered: "Where should the processing backend live for the alpha? (The WASB/owned-model path needs your GPU+WSL box; the Gemini segmenter path only needs ffmpeg + an API key, so it can run on cheap cloud CPU.)"="Local box + tunnel (Recommended)", "What monthly hosting budget are you comfortable with through alpha and into beta?"="~$10?25/mo", "Who will actually use the alpha?"="~10?25 testers", "Where is khelstura.guru registered? (Determines the DNS takeover steps I write up.)"="GoDaddy". You can now continue with these answers in mind.
[/external_agent_tool_result]
```

29. assistant (2026-07-02T06:11:34.065Z)

```
[external_agent_tool_call: ToolSearch]
input: {"query":"select:TaskCreate,TaskUpdate,WebSearch,mcp__cowork__present_files,mcp__cowork__send_user_message","max_results":5}
[/external_agent_tool_call]
```

30. assistant (2026-07-02T06:11:34.066Z)

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

31. assistant (2026-07-02T06:11:34.066Z)

```
[external_agent_tool_call: TaskCreate]
description: Extract personas, workflows, pricing findings, v1 scope, guardrails, and UI-relevant insights from the strategy docs via a read-only agent.
[/external_agent_tool_call]
```

32. assistant (2026-07-02T06:11:34.066Z)

```
[external_agent_tool_result]
Task #1 created successfully: Deep-study product/vision docs (PMF, commercialization, roadmap)
[/external_agent_tool_result]
```

33. assistant (2026-07-02T06:11:34.066Z)

```
[external_agent_tool_call: TaskCreate]
description: Extract current API contracts, existing static UI state, async job model plan (#16), auth/tenancy/storage plans, and security gaps via a read-only agent.
[/external_agent_tool_call]
```

34. assistant (2026-07-02T06:11:34.066Z)

```
[external_agent_tool_result]
Task #2 created successfully: Deep-study technical surface (API, static UI, async-job plan, platform architecture)
```

[/external_agent_tool_result]

35. assistant (2026-07-02T06:11:34.066Z)

[external_agent_tool_call: TaskCreate]

description: Verify current free tiers, pricing, upload limits, tunnel/auth options. GCP credits available; DO preferred alternative; budget \$10?25/mo; local GPU box + tunnel for alpha backend.

[/external_agent_tool_call]

36. assistant (2026-07-02T06:11:34.066Z)

[external_agent_tool_result]

Task #3 created successfully: Research hosting options & current pricing (GCP, DigitalOcean, Cloudflare)

[/external_agent_tool_result]

37. assistant (2026-07-02T06:11:34.066Z)

[external_agent_tool_result]

Task #7 created successfully: Verify proposals against guardrails + present deliverables

[/external_agent_tool_result]

38. assistant (2026-07-02T06:11:34.066Z)

[external_agent_tool_call: TaskCreate]

description: UI proposal for sports enthusiasts + coaches: screens, flows, correction-loop-as-corpus-engine, dependencies (async job slice), effort estimates. Saved to the indexer repo docs/.

[/external_agent_tool_call]

39. assistant (2026-07-02T06:11:34.066Z)

[external_agent_tool_result]

Task #4 created successfully: Write docs/UI_PROPOSAL.md ? alpha (quick & dirty) + beta (polished) chunks

[/external_agent_tool_result]

40. assistant (2026-07-02T06:11:34.066Z)

[external_agent_tool_call: TaskCreate]

description: Recommended architecture + 2-3 alternatives with monthly costs, GoDaddy DNS steps, access/token setup so Claude can manage deploys, upload-path design for large videos.

[/external_agent_tool_call]

41. assistant (2026-07-02T06:11:34.066Z)

[external_agent_tool_result]

Task #5 created successfully: Write docs/HOSTING_PLAN.md ? khelstura.guru takeover + alternatives

[/external_agent_tool_result]

42. assistant (2026-07-02T06:11:34.066Z)

[external_agent_tool_call: TaskCreate]

description: Self-contained HTML mockup of the alpha screens (upload ? job ? rally review ? highlights) to make the proposal concrete.

[/external_agent_tool_call]

43. assistant (2026-07-02T06:11:34.066Z)

[external_agent_tool_result]

Task #6 created successfully: Build clickable alpha UI mockup (single HTML file)

[/external_agent_tool_result]

44. assistant (2026-07-02T06:11:34.066Z)

```
[external_agent_tool_call: TaskCreate]
description: Cross-check against CLAUDE.md guardrails (licensing, minors/consent, v1 scope,
owner-gating), budget fit, and present files.
[/external_agent_tool_call]
```

45. assistant (2026-07-02T06:11:34.066Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"1","status":"in_progress"}
[/external_agent_tool_call]
```

46. assistant (2026-07-02T06:11:34.066Z)

```
[external_agent_tool_result]
Updated task #1 status
[/external_agent_tool_result]
```

47. assistant (2026-07-02T06:11:34.066Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"2","status":"in_progress"}
[/external_agent_tool_call]
```

48. assistant (2026-07-02T06:11:34.067Z)

```
[external_agent_tool_result]
Updated task #2 status
[/external_agent_tool_result]
```

49. assistant (2026-07-02T06:11:34.067Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"3","status":"in_progress"}
[/external_agent_tool_call]
```

50. assistant (2026-07-02T06:11:34.067Z)

```
[external_agent_tool_result]
Updated task #3 status
[/external_agent_tool_result]
```

51. assistant (2026-07-02T06:11:34.067Z)

```
[external_agent_tool_call: Agent]
description: Study product/vision docs
[/external_agent_tool_call]
```

52. assistant (2026-07-02T06:11:34.067Z)

```
[external_agent_tool_call: Agent]
description: Study technical API/UI surface
[/external_agent_tool_call]
```

53. assistant (2026-07-02T06:11:34.067Z)

```
[external_agent_tool_call: WebSearch]
input: {"query":"Google Cloud Run pricing free tier 2026 scale to zero"}
[/external_agent_tool_call]
```

54. assistant (2026-07-02T06:11:34.067Z)

```
[external_agent_tool_call: WebSearch]
input: {"query":"Cloudflare Tunnel free custom domain Cloudflare Access free plan 50 users 2026"}
[/external_agent_tool_call]
```

55. assistant (2026-07-02T06:11:34.067Z)

```
[external_agent_tool_call: WebSearch]
input: {"query":"DigitalOcean droplet pricing 2026 App Platform Spaces object storage cost"}
[/external_agent_tool_call]
```

56. assistant (2026-07-02T06:11:34.067Z)

```
[external_agent_tool_call: WebSearch]
input: {"query":"Cloudflare free plan request body upload size limit 100MB 2026"}
[/external_agent_tool_call]
```

57. assistant (2026-07-02T06:11:34.067Z)

```
[external_agent_tool_result]
Web search results for query: "Cloudflare free plan request body upload size limit 100MB 2026"
```

```
Links: [{"title":"Limits · Cloudflare Workers docs","url":"https://developers.cloudflare.com/workers/platform/limits/"},"{"title":"Limits · Cloudflare Pages docs","url":"https://developers.cloudflare.com/pages/platform/limits/"},"{"title":"Increase Maximum Upload Size - General - Cloudflare Community","url":"https://community.cloudflare.com/t/increase-maximum-upload-size/175622"},"{"title":"Cloudflare upload limit - Application Performance - Cloudflare Community","url":"https://community.cloudflare.com/t/cloudflare-upload-limit/770768"},"{"title":"Limit on uploads via Cloudflare (self-hosted) - Developer help - Ghost Forum","url":"https://forum.ghost.org/t/limit-on-uploads-via-cloudflare-self-hosted/59756"},"{"title":"Max file size upload error - Application Performance - Cloudflare Community","url":"https://community.cloudflare.com/t/max-file-size-upload-error/728692"},"{"title":"Cloudflare Free Plan Limits 2026: Free vs Pro vs Business Pricing · BetterLink Blog","url":"https://eastonddev.com/blog/en/posts/dev/20251201-cloudflare-pricing-compare/"},"{"title":"Maximum Upload Size is limit? - DNS & Network - Cloudflare Community","url":"https://community.cloudflare.com/t/maximum-upload-size-is-limit/418490"},"{"title":"Uploading Large Files - General - Cloudflare Community","url":"https://community.cloudflare.com/t/uploading-large-files/627287"},"{"title":"Limits · Cloudflare R2 docs","url":"https://developers.cloudflare.com/r2/platform/limits/"}]
```

Based on the search results, the request body size limit for Cloudflare's Free plan is 100 MB. This applies to uploads and API payloads. The 100MB body size limit applies to both Free and Pro plans, while the limit is 200MB on Business and 500MB on Enterprise.

The information I found is current as of June 2026, confirming these limits are still in effect for 2026.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

```
[/external_agent_tool_result]
```

58. assistant (2026-07-02T06:11:34.067Z)

```
[external_agent_tool_result]
Web search results for query: "Google Cloud Run pricing free tier 2026 scale to zero"
```

```
Links: [{"title":"Cloud Run pricing | Google Cloud","url":"https://cloud.google.com/run/pricing"},"{"title":"GCP Cloud Run Pricing: Pay-Per-Request Serverless Containers (2026)","url":"https://cloudpricecheck.com/gcp/cloud-run-pricing"},"{"title":"Google Cloud Run Pricing in 2025: A Comprehensive
```

```
Guide","url":"https://cloudchipr.com/blog/cloud-run-pricing"},"title":"Free Google Cloud
features and trial offer | Google Cloud Free Program | Google Cloud
Documentation","url":"https://docs.cloud.google.com/free/docs/free-cloud-
features"},"title":"Cloud Run | Google
Cloud","url":"https://cloud.google.com/run"},"title":"Google Cloud Run: Pricing and Cost
Optimization - ProsperOps","url":"https://www.prosperops.com/blog/google-cloud-run-pricing-and-
cost-optimization/"},"title":"Google Cloud for Beginners: Free Tier, Pricing & How to Start in
2026","url":"https://gturanker.org/articles/google-cloud-for-beginners/"},"title":"Best
practices for cost-optimized Cloud Run services | Google Cloud
Documentation","url":"https://docs.cloud.google.com/run/docs/tips/services-cost-
optimization"},"title":"Google Cloud Run Pricing Savings
Guide","url":"https://www.pump.co/blog/google-cloud-run-pricing/"},"title":"Best Google Cloud
Run Alternatives in Europe (2026) | DanubeData |
DanubeData","url":"https://danubedata.ro/blog/best-google-cloud-run-alternatives-europe-2026"]}]
```

I've successfully performed the web search for "Google Cloud Run pricing free tier 2026 scale to zero". Here are the key findings:

Free Tier

Cloud Run's free tier includes 2M requests and 360K vCPU-seconds per month. More specifically, the free tier covers the first 180,000 vCPU-seconds per month for CPU usage, the first 360,000 GiB-seconds per month for memory usage, and the first 2 million requests per month. This free tier never expires and renews monthly.

Scale to Zero

Cloud Run gives you the ability to scale to zero when there are no requests coming to your website. When idle, instances will go down to zero, meaning you won't be charged if the application takes a break.

Pricing Model

Cloud Run offers pay-per-use pricing with an always free tier, rounded up to the nearest 100 millisecond?if you don't use it, you don't pay for it. Additionally, Cloud Run users have two different pricing structures to choose from: on-demand or committed use discounts (CUDs).

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

[/external_agent_tool_result]

59. assistant (2026-07-02T06:11:34.067Z)

[external_agent_tool_result]

Web search results for query: "Cloudflare Tunnel free custom domain Cloudflare Access free plan 50 users 2026"

Links: [{"title":"Cloudflare Tunnel Review (2026) - MakerStack","url":"https://makerstack.co/reviews/cloudflare-tunnel-review/"},"title":"Quick Tunnels · Cloudflare One docs","url":"https://developers.cloudflare.com/cloudflare-one/networks/connectors/cloudflare-tunnel/do-more-with-tunnels/trycloudflare/"},"title":"Cloudflare Tunnel Pricing 2026: Plans, Hidden Costs & Cheaper Alternatives","url":"https://toolradar.com/tools/cloudflare-tunnel/pricing"},"title":"Pricing","url":"https://www.cloudflare.com/plans/"},"title":"Free Plan Overview | Cloudflare","url":"https://www.cloudflare.com/plans/free/"},"title":"Free, Stable, and Custom: How to Use Cloudflare Tunnel with Your Own Domain (Better Than ngrok) | BlockQueue Systems Limited","url":"https://blockqueue.io/blog/2025-08-13-cloudflared-an-nginx-alternative"},"title":"Cloudflare Tunnel Alternative: When You Need TCP, UDP, or No Domain | Localtonet Blog","url":"https://localtonet.com/blog/cloudflare-tunnel-alternative"},"title":"Cloudflare Tunnel in 2026: Expose localhost Without Opening Ports or Buying an IP","url":"https://recca0120.github.io/en/2026/04/14/cloudflare-tunnel-2026/"},"title":"Cloudflare Zero Trust Free Plan 2026: What 50 Users

```
Get","url":"https://costbench.com/software/business-vpn/cloudflare-zero-trust/free-plan/"]]
```

Based on the search results, here's what I found about Cloudflare Tunnel and Cloudflare Access:

Cloudflare Tunnel - Free Custom Domain Support

You can create unlimited tunnels, route traffic to any number of local services, and use custom domains you manage on Cloudflare DNS. Cloudflare Tunnel works on every Cloudflare plan, including the free tier. However, the only requirement is that your domain DNS is managed by Cloudflare.

Cloudflare Access - Free Plan with 50 User Limit

The Zero Trust features are also free for up to 50 users. Cloudflare Zero Trust's free plan includes: Up to 50 users, Zero Trust Network Access (ZTNA), Secure Web Gateway (SWG), DNS filtering, and Basic device posture checks.

Pricing Beyond 50 Users

Enterprise organizations needing more than 50 users, dedicated support, or SLAs will need custom pricing. Zero Trust Access is free up to 50 users, then Cloudflare One pay-as-you-go is approximately \$7/user/month.

In summary, both Cloudflare Tunnel (with custom domains via Cloudflare DNS) and Cloudflare Access are completely free for teams under 50 users in 2026.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

[/external_agent_tool_result]

60. assistant (2026-07-02T06:11:34.067Z)

[external_agent_tool_result]

Web search results for query: "DigitalOcean droplet pricing 2026 App Platform Spaces object storage cost"

Links: [{"title":"Droplet Pricing | DigitalOcean","url":"https://www.digitalocean.com/pricing/droplets"}, {"title":"Budget-Friendly Cloud Server Pricing | DigitalOcean","url":"https://www.digitalocean.com/pricing"}, {"title":"DigitalOcean Droplets | Scalable Cloud Compute Starting at \$4/mo","url":"https://www.digitalocean.com/products/droplets"}, {"title":"Spaces Pricing | DigitalOcean Documentation","url":"https://docs.digitalocean.com/products/spaces/details/pricing/"}, {"title":"DigitalOcean Pricing Calculator: Guide + Real Costs","url":"https://cloudpricecalculators.com/digitalocean-calculator/"}, {"title":"Pricing Calculator | DigitalOcean","url":"https://www.digitalocean.com/pricing/calculator"}, {"title":"Spaces Object Storage Pricing | DigitalOcean","url":"https://www.digitalocean.com/pricing/spaces-object-storage"}, {"title":"DigitalOcean pricing, tiers and limits in 2026 | HamsterStack","url":"https://hamsterstack.com/pricing/digitalocean/"}, {"title":"DigitalOcean Pricing 2026: Free & Droplets Plans from \$4/mo","url":"https://comparedge.com/tools/digitalocean/pricing"}, {"title":"App Platform Pricing | DigitalOcean","url":"https://www.digitalocean.com/pricing/app-platform"}]

Based on the search results, here's the current DigitalOcean pricing information for 2026:

Droplet Pricing

Effective January 1, 2026, DigitalOcean moved to per-second billing (with a minimum charge of 60 seconds or \$0.01, whichever is higher) for Droplets. DigitalOcean Droplets are available starting from as low as \$4 per month.

App Platform Pricing

App Platform begins at \$5/month. The App Platform Free Tier allows you to build and deploy up to 3 apps with static sites for free, with each free app including 1 GiB of outbound data transfer per month. App Platform starts at \$5/mo per container.

Spaces Object Storage Pricing

The base price of a Spaces Standard Storage subscription is \$5.00 per month. The base subscription includes 250 GiB of storage across all of your buckets, and additional storage beyond this allotment costs \$0.02 per GiB per month.

Spaces subscriptions include 1,024 GiB of outbound data transfer (from Spaces buckets to the internet), which is shared between all buckets, and additional outbound transfer is \$0.01 per GiB.

Additionally, Spaces Cold Storage is a low-cost storage class optimized for data that is accessed infrequently and retained for long periods, with storage costs at \$0.007 per GiB per month and retrieval at \$0.01 per GiB retrieved.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.
[/external_agent_tool_result]

61. assistant (2026-07-02T06:11:34.067Z)

[external_agent_tool_result]
Perfect. Now I have all the information I need. Let me synthesize the findings into a structured markdown report.

Technical Integration Synthesis: Badminton Highlight Indexer for Web UI + Hosting

1. Current API Surface

****Endpoints (all blocking/synchronous in current form):****

Method	Path	Request body	Response	Duration	Notes
POST	/api/process	{video_path: str, player_pool?: List[str], max_frames?: int, segmenter?: str}	{video_id, status, message, results[], play_segments[], reports}	~5?60 min (GPU-dependent)	Validate ? segment ? report. Full pipeline in-process. Returns 200 on completion. `reports` dict includes `technical_md`, `customer_md`, `json` paths.
POST	/api/query	{video_id: str, server?: str, receiver?: str, duration_min?: float, event_type?: str}	{play_segments[]}	<100ms	Filters on server/receiver names, duration, or event type. SQL-driven.
POST	/api/compile	{video_id: str, segment_ids: List[int], output_filename: str}	{status, filepath}	~10?30 sec (ffmpeg)	Extract time pairs, call FFmpeg stitcher, return artifact path.
GET	/api/config	? AppConfig (Pydantic model)		<10ms	Returns the typed configuration (indexing, validation, output, security settings).
GET	/	? HTML		<10ms	Serves backend/api/static/index.html.

****File input model:**** Videos are referenced by ****local filesystem path**** (string), passed in the request. No upload endpoint exists. The path is validated via `validate_input_path()` in `backend/utils/paths.py`:

- Rejects null bytes always.
- When `config.security.allowed_video_root` is set (nil by default, single-user mode), enforces containment via `resolve_under_root()` (real-path + commonpath check; prevents `..` escape).
- ****No enforced size limits**** documented; ffmpeg and provider (Gemini) have their own caps.

****Output naming:**** `output\_filename` in `/api/compile` is validated via `safe\_output\_filename()` ? rejects absolute paths, dir separators, `..`, `.`, and null bytes. Guarantees a bare filename safe for `os.path.join(output\_dir, name)`.

****Video identification:**** Every video is keyed by ****MD5 of the entire file**** (streamed, 64 KB chunks) via `compute\_video\_id(filepath)` in `backend/utils/hashing.py`. Used as the primary key (`videos.id`). Re-ingest with the same file produces the same `video\_id` and triggers replace-on-conflict behavior.

2. Existing Static UI

****Location:**** `backend/api/static/` (index.html, app.js, style.css).

****Screens & workflows:****

- ****Header:**** status badge ("Local Server Active").
- ****Sidebar (two glass panels):****
 - Video input: text field for local path, comma-separated player pool (defaults: "Avi, Suresh, Ramesh, Deepak"), "Validate & Index Video" button.
 - Validation checklist: dynamic list of validation-result items (pass/fail icons, messages, optional metric details). Populated after `/api/process` returns.
- ****Workspace (main area):****
 - Rally database explorer: query bar (server, receiver, min-duration dropdowns + event-type select), "Filter Index" button. Results show as selectable rally cards (title, start/end times, duration, tags like "Long Rally"/"Contains Smash"/"Ends in Foul").
 - Compiler bar (appears when ?1 rally selected): text field for output filename, "Compile Highlights" button.
- ****Loading overlay:**** full-screen spinner + dynamic title/subtitle (e.g., "Analyzing Video Stream?").

****What it calls:****

1. POST `/api/process` with video_path and player_pool.
2. Parses response; renders validation results.
3. On success, POST `/api/query` with empty filters to populate the initial rally list.
4. On filter button, POST `/api/query` with server/receiver/duration/event_type.
5. On compile, POST `/api/compile` with segment_ids (from selected rally cards) and output_filename.

****Rough edges:**** No persistent UI state (alert dialogs on error; alerts on success with file p? [/external_agent_tool_result])

62. assistant (2026-07-02T06:11:34.067Z)

[external_agent_tool_result]

Summary

I've completed a comprehensive synthesis of the product and UI documentation for the badminton-highlight-indexer project. The output covers all eight required sections in dense, specific language grounded in the docs:

****Key findings for UI/product planning:****

1. ****Personas:**** India badminton coaching academies (B2B primary), with B2C enthusiasts as a funnel. Static-camera constraint is a hard filter but creates alignment: coaches already record this way.
2. ****Pricing:**** B2C USD 60?180/yr; B2B academy USD 300?1,500/yr. India market needs INR-tuned, per-match pricing. Per-run API cost ~\$0.05 today (Gemini).
3. ****V1 scope:**** Rally detection + highlights + match history + human-correction loop + per-rally stats. Minors excluded; gait/biomechanics deferred.
4. ****Workflow:**** Record with guidelines ? upload with validator ? 2-stage processing ?

review/correct ? clip gallery + stats ? share. Golden-set labeling feeds training data.

5. ****Guardrails:**** No minors; MIT/Apache-2.0/BSD licensing hard rule; training-data opt-in per-user/match; Gemini paid-tier required for privacy; WASB weights provenance is a blocker.

6. ****UI ideas:**** Minimal mockups exist. Observability reports (quality verdict), input validator with footgun flags, and structured event-store query layer (SQL, not chatbot) are the main surfaces. Conversational QA deferred.

7. ****Roadmap:**** Phase 3 (eval, in progress), Phase 4 (learned segmenter, next). Platform scaling deferred. Owned-model Gen-0 ? Gen-1 ? Gen-2 (VLM), gated on corpus growth + cost benchmarking.

8. ****Landmines:**** Shuttle-tracking weights C3 blocker; audio weaker than expected; B2C TAM risk if amateurs won't tripod; Gen-2 cost unquantified; minors exclude most academy footage; repo name needs urgent change; ByteTrack deprecation trap; H.264 output needs AV1 migration.

All claims cite specific filenames and sections inline. No dumped file contents ? only load-bearing extracts.

[/external_agent_tool_result]

63. assistant (2026-07-02T06:11:34.068Z)

Research is in ? both repos studied and pricing verified. Now writing the deliverables.

64. assistant (2026-07-02T06:11:34.068Z)

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"1","status":"completed"}
[/external_agent_tool_call]

65. assistant (2026-07-02T06:11:34.068Z)

[external_agent_tool_result]
Updated task #1 status
[/external_agent_tool_result]

66. assistant (2026-07-02T06:11:34.068Z)

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"2","status":"completed"}
[/external_agent_tool_call]

67. assistant (2026-07-02T06:11:34.068Z)

[external_agent_tool_result]
Updated task #2 status
[/external_agent_tool_result]

68. assistant (2026-07-02T06:11:34.068Z)

[external_agent_tool_result]
Updated task #4 status
[/external_agent_tool_result]

69. assistant (2026-07-02T06:11:34.068Z)

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"3","status":"completed"}
[/external_agent_tool_call]

70. assistant (2026-07-02T06:11:34.068Z)

[external_agent_tool_result]

Updated task #3 status
[/external_agent_tool_result]

71. assistant (2026-07-02T06:11:34.068Z)

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"4","status":"in_progress"}
[/external_agent_tool_call]

72. assistant (2026-07-02T06:11:34.068Z)

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md
[/external_agent_tool_call]

73. assistant (2026-07-02T06:11:34.068Z)

[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

74. assistant (2026-07-02T06:11:34.068Z)

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"4","status":"completed"}
[/external_agent_tool_call]

75. assistant (2026-07-02T06:11:34.068Z)

[external_agent_tool_result]
Updated task #4 status
[/external_agent_tool_result]

76. assistant (2026-07-02T06:11:34.068Z)

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"5","status":"in_progress"}
[/external_agent_tool_call]

77. assistant (2026-07-02T06:11:34.068Z)

[external_agent_tool_result]
Updated task #5 status
[/external_agent_tool_result]

78. assistant (2026-07-02T06:11:34.068Z)

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\HOSTING_PLAN.md
[/external_agent_tool_call]

79. assistant (2026-07-02T06:11:34.068Z)

[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\HOSTING_PLAN.md (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

80. assistant (2026-07-02T06:11:34.068Z)

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"5","status":"completed"}
[/external_agent_tool_call]

```
[/external_agent_tool_call]
```

81. assistant (2026-07-02T06:11:34.068Z)

```
[external_agent_tool_result]
Updated task #5 status
[/external_agent_tool_result]
```

82. assistant (2026-07-02T06:11:34.068Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"6","status":"in_progress"}
[/external_agent_tool_call]
```

83. assistant (2026-07-02T06:11:34.068Z)

```
[external_agent_tool_result]
Updated task #6 status
[/external_agent_tool_result]
```

84. assistant (2026-07-02T06:11:34.069Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\mockups\ui_alpha_mockup.html
[/external_agent_tool_call]
```

85. assistant (2026-07-02T06:11:34.069Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\mockups\ui_alpha_mockup.html (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

86. assistant (2026-07-02T06:11:34.069Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"6","status":"completed"}
[/external_agent_tool_call]
```

87. assistant (2026-07-02T06:11:34.069Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"7","status":"in_progress"}
[/external_agent_tool_call]
```

88. assistant (2026-07-02T06:11:34.069Z)

```
[external_agent_tool_result]
Updated task #6 status
[/external_agent_tool_result]
```

89. assistant (2026-07-02T06:11:34.069Z)

```
[external_agent_tool_result]
Updated task #7 status
[/external_agent_tool_result]
```

90. assistant (2026-07-02T06:11:34.069Z)

```
[external_agent_tool_call: mcp_cowork__present_files]
input: {"files":[{"file_path":"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\UI_PROPOSAL.md"}, {"file_path":"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\UI_PROPOSAL.md"}]}
[/external_agent_tool_call]
```

```
indexer\\docs\\HOSTING_PLAN.md"}, {"file_path": "C:\\Users\\avidu\\Projects\\badminton-highlight-  
indexer\\docs\\mockups\\ui_alpha_mockup.html"}]]  
[/external_agent_tool_call]
```

91. assistant (2026-07-02T06:11:34.069Z)

```
[external_agent_tool_result]  
C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\UI_PROPOSAL.md  
C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\HOSTING_PLAN.md  
C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\mockups\\ui_alpha_mockup.html  
[/external_agent_tool_result]
```

92. assistant (2026-07-02T06:11:34.069Z)

```
[external_agent_tool_call: TaskUpdate]  
input: {"taskId": "7", "status": "completed"}  
[/external_agent_tool_call]
```

93. assistant (2026-07-02T06:11:34.069Z)

```
[external_agent_tool_result]  
Updated task #7 status  
[/external_agent_tool_result]
```

94. assistant (2026-07-02T06:11:34.069Z)

Done ? three files in the repo (uncommitted, for your review; no PR per convention):

****UI proposal**** ? the alpha is 6 screens (Matches, Upload, Job Progress, Rally Review, Highlights, Report) built as a small React SPA over your existing FastAPI. The key framing: the Rally Review screen *is* the corpus-growth engine ? every tester correction lands in `candidate\_segments.human\_label`, feeding the exact n you need for sign-test significance. Honest dependency: the alpha requires greenlighting the async job slice (#16) plus ~5 small backend additions (chunked upload, download endpoint, per-user scoping, correction endpoints). ~8?10 sessions to first tester. Beta maps 1:1 onto PLATFORM P1?P3 in four shippable drops, with consent/minors guardrails built into Drop 4.

****Hosting plan**** ? on your GCP/DO preference: I recommend a hybrid. Alpha runs at ****\$0/mo**** on Cloudflare's free edge (DNS + Pages + email-allowlist auth for ?50 users + tunnel to your box) because GCP has no free equivalent of the tunnel+auth combo (IAP needs a ~\$18/mo load balancer) ? this *saves* your credits for beta, where GCP shines: Cloud Run control plane (scale-to-zero), GCS uploads via signed URLs, and your GPU box as the first pull-based `byo\_worker`. ~\$15?20/mo post-credits, inside budget. All-DigitalOcean (~\$11?17/mo, Spaces' included 1 TiB egress is the best video-download deal) is written up as the fallback if credits expire. GoDaddy nameserver steps and the exact access grants I need are in \$5?6.

****Clickable mockup**** ? open `ui\_alpha\_mockup.html` in a browser to walk all six alpha screens with fake data.

Both docs end with explicit decision lists for you (8 total) ? the big ones: greenlight alpha scope (= greenlights #16), product name (****Khelstura**** vs ****KhelAcharya**** ? domain says one, project says the other), and DNS-to-Cloudflare approval. Once you ratify, I'll record them in `DECISIONS.md` and can start with the GoDaddy nameserver move.

Sources: [Cloud Run pricing](<https://cloud.google.com/run/pricing>), [GCP free tier](<https://docs.cloud.google.com/free/docs/free-cloud-features>), [Cloudflare Zero Trust free plan](<https://costbench.com/software/business-vpn/cloudflare-zero-trust/free-plan/>), [Cloudflare plans](<https://www.cloudflare.com/plans/>), [DO Spaces pricing](<https://docs.digitalocean.com/products/spaces/details/pricing/>), [DO droplets](<https://www.digitalocean.com/pricing/droplets>)

95. user (2026-07-02T06:11:34.069Z)

Really impressive for an alpha UI! And I can clarify all the questions you have! The name of the

product will be "KhelSutra Guru" essentially "teacher or the art of sports". I will be reviewing the docs more thoroughly but I browsed the UI and it looks impeccable!
Can you chalk out a plan of execution for this? And also should we isolate this work in a new github repo?

96. assistant (2026-07-02T06:11:34.069Z)

Two quick confirmations before I bake the name in everywhere ? one of them affects DNS setup directly:

97. assistant (2026-07-02T06:11:34.069Z)

```
[external_agent_tool_call: AskUserQuestion]
input: {"questions":[{"question":"Exact registered domain spelling? Your first message said \"khelstura.guru\" but the product name KhelSutra suggests \"khelsutra.guru\" ? DNS setup and every doc reference depend on the exact string.", "header":"Domain", "multiSelect":false, "options":[{"label":"khelsutra.guru", "description":"khel-SUTRA, matching the product name \"KhelSutra Guru\" ? the earlier message had a typo."}, {"label":"khelstura.guru", "description":"The spelling as originally typed (s-t-u-r-a) is what's actually registered at GoDaddy."}]}], {"question":"Confirm the repo split? My recommendation: SPA in a new repo, backend changes stay in the indexer repo.", "header":"Repo split", "multiSelect":false, "options":[{"label":"New repo: khelsutra-ui (Recommended)", "description":"SPA gets its own repo ? clean Pages auto-deploy, npm dependency world stays out of the indexer's MIT/Apache/BSD licensing audit, separate CI. Backend PRs (B1?B7) remain in the indexer repo."}, {"label":"Folder in indexer repo", "description":"Single repo, e.g. frontend/ directory. Fewer repos to manage but mixes toolchains, complicates Pages deploys and the licensing gate."}]}]}
[/external_agent_tool_call]
```

98. assistant (2026-07-02T06:11:34.070Z)

```
[external_agent_tool_result: error]
Tool permission request failed: Error: Tool permission stream closed before response received
[/external_agent_tool_result]
```

99. assistant (2026-07-02T06:11:34.070Z)

No response requested.

100. user (2026-07-02T06:11:34.070Z)

here is the godaddy receipt!

101. assistant (2026-07-02T06:11:34.070Z)

The receipt settles it ? **khelsutra.guru** (the first message's "khelstura" was indeed a typo). Domain's registered for 3 years with protection. Updating everything to the confirmed name + domain, writing the execution plan, and persisting the mockup as an artifact so it can't get lost.

102. assistant (2026-07-02T06:11:34.070Z)

```
[external_agent_tool_call: TaskCreate]
description: Global spelling fix (khelstura?khelsutra) in UI_PROPOSAL.md and HOSTING_PLAN.md; resolve the name decision; rebrand mockup logo/title.
[/external_agent_tool_call]
```

103. assistant (2026-07-02T06:11:34.071Z)

```
[external_agent_tool_result]
Task #8 created successfully: Rebrand docs + mockup to KhelSutra Guru / khelsutra.guru
[/external_agent_tool_result]
```

104. assistant (2026-07-02T06:11:34.071Z)

```
[external_agent_tool_call: TaskCreate]
description: Gated execution plan: G0 owner setup, Phase 1 backend PRs, Phase 2 SPA milestones,
hardening, tester rollout, risks, session map.
[/external_agent_tool_call]
```

105. assistant (2026-07-02T06:11:34.071Z)

```
[external_agent_tool_result]
Task #9 created successfully: Write docs/UI_ALPHA_EXECUTION_PLAN.md
[/external_agent_tool_result]
```

106. assistant (2026-07-02T06:11:34.071Z)

```
[external_agent_tool_call: TaskCreate]
description: Append product name + domain decision to docs/archives/decisions/DECISIONS.md
following existing ADR format; repo-split noted as recommended pending owner ack.
[/external_agent_tool_call]
```

107. assistant (2026-07-02T06:11:34.071Z)

```
[external_agent_tool_result]
Task #10 created successfully: Record ratified decisions (ADR-009)
[/external_agent_tool_result]
```

108. assistant (2026-07-02T06:11:34.071Z)

```
[external_agent_tool_call: TaskCreate]
description: Create a Cowork artifact from the final rebranded mockup HTML so it persists across
sessions per Avi's request.
[/external_agent_tool_call]
```

109. assistant (2026-07-02T06:11:34.071Z)

```
[external_agent_tool_result]
Task #11 created successfully: Save UI mockup as persistent artifact
[/external_agent_tool_result]
```

110. assistant (2026-07-02T06:11:34.071Z)

```
[external_agent_tool_call: mcp_workspace_bash]
command: tail -n 50 "/sessions/great-sharp-gauss/mnt/badminton-highlight-
indexer/docs/archives/decisions/DECISIONS.md"
[/external_agent_tool_call]
```

111. assistant (2026-07-02T06:11:34.071Z)

```
[external_agent_tool_call: ToolSearch]
input: {"query":"select:mcp_cowork_create_artifact","max_results":3}
[/external_agent_tool_call]
```

112. assistant (2026-07-02T06:11:34.071Z)

```
[external_agent_tool_result]
4. **The ship-gate is owned by the product side** (extends `backend/eval/regression.py` +
`metrics.match_intervals`): the trainer runs it as pre-flight but never grades itself.
The gate refuses commercial ship while C3 is unresolved, regardless of F1.
5. **Pre-committed split trigger** ? extract `training/` into a separate repo (working name
`sports-temporal-models`, to be finalized with the pending repo rename) **when**:
(a) Gen-2 arrives (ms-swift / Qwen3-VL / cloud-GPU training ? an alien dependency tree), OR
(b) a second consumer of the trained artifacts appears (e.g. the khelacharya layer),
AND (c) the harness's only data input is the collector's M0.1 export. By then the artifact
```

contract is proven and extraction is a file move across a known seam.

****Prerequisite frozen first: the golden-label/corpus contract.**** One canonical GT loader (the repo carried two incompatible ones: `annotations.load\_annotations` `start,end` vs `calibrate\_wasb.load\_golden\_gts` `start\_time,end\_time`, the latter feeding 7 modules) + a round-trip test against the collector's `curate export` format. Every ship-gate quantity (gt_hash, LOVO numbers, the future consent fingerprint) flows through this boundary.

Rationale (the decisive, code-verified facts)

- ****The "just another runner like WASB/Gemini" analogy fails at the seam location.**** WASB's contract sits at the video boundary (video ? trajectory CSV); Gemini's at video ? segments. The Gen-0 head consumes ****features this repo computes**** from the trajectory ? the seam is mid-pipeline. Since the indexer is an app (not pip-installable), a separate repo today could share that code only by ****copying**** it ? this project's one empirically observed failure mode (the GT-header fork; the duplicated TUNED constants).
- ****The in-house separate-repo precedent is unproven for load-bearing deps****: the `obsreport` pin has never been live (commented in requirements.txt, repo unpublished, CI skips via importorskip). It works because its absence is a harmless no-op ? a property a featurizer can never have.
- ****Gen-0 has no environmental forcing function**** (trains in minutes, ~4.5 GB, MIT pip-able deps, on the owner's WSL/2070S box); Gen-2 genuinely does. Splitting then is a scheduled event, not an overruled instinct.
- The prerequisite work for a clean split (featurizer promotion, loader consolidation, manifest) is ****identical**** to the in-repo design ? doing it in place first makes the eventual extraction cheap and safe. This is "envision now, build later" applied to topology: ****the split is driven by productionization needs, not code-isolation aesthetics**** (owner's framing, ratified).

Consequences

- `backend/` may never import `training.\*` (CI-enforced). Training tests use ****synthetic fixtures only**** until the M0.1 consent ledger exists (no real trajectories/labels committed).
- Weights are never committed; artifact manifests are the reviewable/committable surface.
- The collector remains system-of-record for labels + consent; the indexer owns scoring and the ship/no-ship decision; manifests pin both by hash.
- Revisit triggers are explicit (above); if agent-driven development repeatedly breaches the import wall despite CI, that is evidence the social+test boundary failed and the split should happen early.

Panel artifacts: workflow `wf\_471a9cdd-d1d` (3 grounded position papers + cross-examination), 2026-06-10.

[/external_agent_tool_result]

113. assistant (2026-07-02T06:11:34.071Z)

[external_agent_tool_result]
[/external_agent_tool_result]

114. assistant (2026-07-02T06:11:34.071Z)

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"8","status":"in_progress"}
[/external_agent_tool_call]

115. assistant (2026-07-02T06:11:34.072Z)

[external_agent_tool_result]
Updated task #8 status
[/external_agent_tool_result]

116. assistant (2026-07-02T06:11:34.072Z)


```
[external_agent_tool_call: Grep]
input: {"pattern": "^#+ .*ADR-00[78]", "path": "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\archives\\decisions\\DECISIONS.md", "output_mode": "content", "-n": true}
[/external_agent_tool_call]
```

117. assistant (2026-07-02T06:11:34.072Z)

```
[external_agent_tool_result]
295:## ADR-007: Audio "thwack" hit-detection (fused with WASB) is the next rally-detection cue;
pose/stance parked
388:## ADR-008: Owned-model topology ? artifact boundary now, repo split at Gen-2
[/external_agent_tool_result]
```

118. assistant (2026-07-02T06:11:34.072Z)

```
[external_agent_tool_call: Edit]
file: C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\UI_PROPOSAL.md
[/external_agent_tool_call]
```

119. assistant (2026-07-02T06:11:34.072Z)

```
[external_agent_tool_result]
The file C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\UI_PROPOSAL.md has been
updated. All occurrences were successfully replaced. (file state is current in your context ? no
need to Read it back)
[/external_agent_tool_result]
```

120. assistant (2026-07-02T06:11:34.072Z)

```
[external_agent_tool_call: Edit]
file: C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\HOSTING_PLAN.md
[/external_agent_tool_call]
```

121. assistant (2026-07-02T06:11:34.072Z)

```
[external_agent_tool_result]
The file C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\HOSTING_PLAN.md has been
updated. All occurrences were successfully replaced. (file state is current in your context ? no
need to Read it back)
[/external_agent_tool_result]
```

122. assistant (2026-07-02T06:11:34.072Z)

```
[external_agent_tool_call: Edit]
file: C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\mockups\\ui_alpha_mockup.html
[/external_agent_tool_call]
```

123. assistant (2026-07-02T06:11:34.072Z)

```
[external_agent_tool_result]
The file C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\mockups\\ui_alpha_mockup.html
has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

124. assistant (2026-07-02T06:11:34.072Z)

```
[external_agent_tool_call: Edit]
file: C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\mockups\\ui_alpha_mockup.html
[/external_agent_tool_call]
```

125. assistant (2026-07-02T06:11:34.072Z)

```
[external_agent_tool_result]
```

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\mockups\ui_alpha_mockup.html has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

126. assistant (2026-07-02T06:11:34.072Z)

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"8","status":"completed"}
[/external_agent_tool_call]

127. assistant (2026-07-02T06:11:34.073Z)

[external_agent_tool_result]
Updated task #8 status
[/external_agent_tool_result]

128. assistant (2026-07-02T06:11:34.073Z)

[external_agent_tool_call: TaskCreate]
description: Request folder access, git init khelsutra (master branch), scaffold README/.gitignore/web placeholder, first commit, hand owner the private-GitHub push commands.
[/external_agent_tool_call]

129. assistant (2026-07-02T06:11:34.073Z)

[external_agent_tool_result]
Task #12 created successfully: Init private khelsutra repo in C:\Users\avidu\Projects
[/external_agent_tool_result]

130. assistant (2026-07-02T06:11:34.073Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md
[/external_agent_tool_call]

131. assistant (2026-07-02T06:11:34.073Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

132. assistant (2026-07-02T06:11:34.073Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md
[/external_agent_tool_call]

133. assistant (2026-07-02T06:11:34.073Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

134. assistant (2026-07-02T06:11:34.073Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\HOSTING_PLAN.md
[/external_agent_tool_call]

135. assistant (2026-07-02T06:11:34.073Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\HOSTING_PLAN.md has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

136. assistant (2026-07-02T06:11:34.073Z)

```
[external_agent_tool_call: ToolSearch]
input: {"query":"select:mcp__cowork__request_cowork_directory","max_results":3}
[/external_agent_tool_call]
```

137. assistant (2026-07-02T06:11:34.073Z)

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

138. assistant (2026-07-02T06:11:34.073Z)

```
[external_agent_tool_call: mcp__cowork__request_cowork_directory]
input: {"path":"C:\\Users\\avidu\\Projects"}
[/external_agent_tool_call]
```

139. assistant (2026-07-02T06:11:34.073Z)

```
[external_agent_tool_result]
Folder connected: C:\Users\avidu\Projects
```

Use this exact path with Read/Write/Edit/Grep/Glob.

For mcp__workspace__bash ONLY, this folder appears at /sessions/great-sharp-gauss/mnt/Projects. Do NOT pass that /sessions/? form to Read/Write/Edit/Grep/Glob ? those run on the host, where that path does not exist.

Read/Write/Edit/Grep/Glob can use this folder immediately.

```
[/external_agent_tool_result]
```

140. assistant (2026-07-02T06:11:34.073Z)

Folder connected. Initializing the repo, writing the local-first execution plan, and prepping the mockup for artifact persistence ? in parallel:

141. assistant (2026-07-02T06:11:34.073Z)

```
[external_agent_tool_call: mcp__workspace__bash]
command: set -e
cd /sessions/great-sharp-gauss/mnt/Projects
mkdir -p khelsutra/web khelsutra/docs
cd khelsutra
git init -b master -q
git config user.name "Avi Dullu"
git config user.email "avi.dullu@gmail.com"
cat > README.md <<'EOF'
```

KhelSutra Guru ?

****Private product repo**** ? the heart and brain of khelsutra.guru: the web app (alpha SPA) now, the hosted control plane at beta.

- ****Engine**** (pipeline, API, eval): sibling repo `badminton-highlight-indexer` (rename pending).
- ****Canonical planning docs**** (live in the engine repo's `docs/` for now, migrate here at beta):
`UI\_PROPOSAL.md` · `HOSTING\_PLAN.md` · `UI\_ALPHA\_EXECUTION\_PLAN.md`.
- ****Status**** pre-alpha scaffold. SPA lands in `web/` per execution-plan milestone M1.

Structure

- `web/` ? React SPA (alpha; Vite + React + TS + Tailwind; Cloudflare Pages build root)
- `docs/` ? product docs as they migrate from the engine repo
- `control-plane/` ? (beta) Cloud Run FastAPI control plane

License

All rights reserved (private; no OSS license).

EOF

```
cat > .gitignore <<'EOF'
```

```
node_modules/
```

```
dist/
```

```
.env
```

```
.env.*
```

```
*.local
```

```
.DS_Store
```

```
Thumbs.db
```

EOF

```
cat > web/README.md <<'EOF'
```

SPA scaffold lands here at execution-plan milestone M1 (Vite + React + TS + Tailwind).

Dev: talks to the engine at <http://localhost:8000> via Vite proxy; no auth locally

(DEV_USER_EMAIL).

EOF

```
cat > docs/README.md <<'EOF'
```

Canonical planning docs currently live in the engine repo (badminton-highlight-indexer/docs):

UI_PROPOSAL.md · HOSTING_PLAN.md · UI_ALPHA_EXECUTION_PLAN.md. They migrate here when the

control plane lands (per ADR-009 in the engine repo's DECISIONS.md).

EOF

```
git add -A
```

```
git commit -q -m "init: KhelSutra Guru product repo ? scaffold (web/, docs pointers, all-rights-reserved)"
```

```
git log --oneline && git status --short && echo OK
```

```
[/external_agent_tool_call]
```

142. assistant (2026-07-02T06:11:34.074Z)

```
[external_agent_tool_result: error]
```

```
Exit code 128
```

```
fatal: bad config line 1 in file /sessions/great-sharp-gauss/mnt/Projects/khelsutra/.git/config
```

```
warning: unable to unlink '/sessions/great-sharp-gauss/mnt/Projects/khelsutra/.git/config.lock':
```

```
Operation not permitted
```

```
[/external_agent_tool_result]
```

143. assistant (2026-07-02T06:11:34.074Z)

```
[external_agent_tool_call: Write]
```

```
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md
```

```
[/external_agent_tool_call]
```

144. assistant (2026-07-02T06:11:34.074Z)

```
[external_agent_tool_result]
```

```
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-
```

```
indexer\docs\UI_ALPHA_EXECUTION_PLAN.md (file state is current in your context ? no need to Read it back)
```

```
[/external_agent_tool_result]
```

145. assistant (2026-07-02T06:11:34.074Z)

```
[external_agent_tool_call: Edit]
```

```
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\mockups\ui_alpha_mockup.html
```

```
[/external_agent_tool_call]
```

146. assistant (2026-07-02T06:11:34.074Z)

```
[external_agent_tool_result]
```


150. assistant (2026-07-02T06:11:34.074Z)

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

151. assistant (2026-07-02T06:11:34.074Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"9","status":"completed"}
[/external_agent_tool_call]
```

152. assistant (2026-07-02T06:11:34.074Z)

```
[external_agent_tool_result]
Updated task #9 status
[/external_agent_tool_result]
```

153. assistant (2026-07-02T06:11:34.074Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"12","status":"in_progress"}
[/external_agent_tool_call]
```

154. assistant (2026-07-02T06:11:34.074Z)

```
[external_agent_tool_result]
Updated task #12 status
[/external_agent_tool_result]
```

155. assistant (2026-07-02T06:11:34.074Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\khelsutra\README.md
[/external_agent_tool_call]
```

156. assistant (2026-07-02T06:11:34.074Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\khelsutra\README.md (file state is current
in your context ? no need to Read it back)
[/external_agent_tool_result]
```

157. assistant (2026-07-02T06:11:34.075Z)

```
[external_agent_tool_result]
425      (a) Gen-2 arrives (ms-swift / Qwen3-VL / cloud-GPU training ? an alien dependency
tree), OR
426      (b) a second consumer of the trained artifacts appears (e.g. the khelacharya layer),
427      AND (c) the harness's only data input is the collector's M0.1 export. By then the
artifact
428      contract is proven and extraction is a file move across a known seam.
429
430      **Prerequisite frozen first: the golden-label/corpus contract.** One canonical GT loader
(the
431      repo carried two incompatible ones: `annotations.load_annotations` `start,end` vs
432      `calibrate_wasb.load_golden_gts` `start_time,end_time`, the latter feeding 7 modules) +
a
433      round-trip test against the collector's `curate export` format. Every ship-gate quantity
434      (gt_hash, LOVO numbers, the future consent fingerprint) flows through this boundary.
435
436      ### Rationale (the decisive, code-verified facts)
437
438      - **The "just another runner like WASB/Gemini" analogy fails at the seam location.**
WASB's
```

439 contract sits at the video boundary (video ? trajectory CSV); Gemini's at video ?
segments.
440 The Gen-0 head consumes **features this repo computes** from the trajectory ? the seam
is
441 mid-pipeline. Since the indexer is an app (not pip-installable), a separate repo today
could
442 share that code only by **copying** it ? this project's one empirically observed
failure
443 mode (the GT-header fork; the duplicated TUNED constants).
444 - **The in-house separate-repo precedent is unproven for load-bearing deps**: the
`obsreport`
445 pin has never been live (commented in requirements.txt, repo unpublished, CI skips via
446 importorskip). It works because its absence is a harmless no-op ? a property a
featurizer
447 can never have.
448 - **Gen-0 has no environmental forcing function** (trains in minutes, ~4.5 GB, MIT pip-
able
449 deps, on the owner's WSL/2070S box); Gen-2 genuinely does. Splitting then is a
scheduled
450 event, not an overruled instinct.
451 - The prerequisite work for a clean split (featurizer promotion, loader consolidation,
452 manifest) is **identical** to the in-repo design ? doing it in place first makes the
453 eventual extraction cheap and safe. This is "envision now, build later" applied to
topology:
454 **the split is driven by productionization needs, not code-isolation aesthetics**
(owner's
455 framing, ratified).
456
457 **### Consequences**
458
459 - **backend/` may never import `training.`** (CI-enforced). Training tests use **synthetic**
460 **fixtures only** until the M0.1 consent ledger exists (no real trajectories/labels
committed).
461 - Weights are never committed; artifact manifests are the reviewable/committable
surface.
462 - The collector remains system-of-record for labels + consent; the indexer owns scoring
and
463 the ship/no-ship decision; manifests pin both by hash.
464 - Revisit triggers are explicit (above); if agent-driven development repeatedly breaches
the
465 import wall despite CI, that is evidence the social+test boundary failed and the split
466 should happen early.
467
468 Panel artifacts: workflow `wf\_471a9cdd-d1d` (3 grounded position papers + cross-
examination),
469 2026-06-10.
470
[/external_agent_tool_result]

158. assistant (2026-07-02T06:11:34.075Z)

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\khelsutra\.gitignore
[/external_agent_tool_call]

159. assistant (2026-07-02T06:11:34.075Z)

[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\khelsutra\.gitignore (file state is
current in your context ? no need to Read it back)
[/external_agent_tool_result]

160. assistant (2026-07-02T06:11:34.075Z)

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\khelsutra\web\README.md
[/external_agent_tool_call]

161. assistant (2026-07-02T06:11:34.075Z)

[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\khelsutra\web\README.md (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

162. assistant (2026-07-02T06:11:34.075Z)

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\khelsutra\docs\README.md
[/external_agent_tool_call]

163. assistant (2026-07-02T06:11:34.075Z)

[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\khelsutra\docs\README.md (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

164. assistant (2026-07-02T06:11:34.075Z)

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\decisions\DECISIONS.md
[/external_agent_tool_call]

165. assistant (2026-07-02T06:11:34.075Z)

[external_agent_tool_call: mcp_cowork__allow_cowork_file_delete]
file: /sessions/great-sharp-gauss/mnt/Projects/khelsutra/.git
[/external_agent_tool_call]

166. assistant (2026-07-02T06:11:34.075Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\decisions\DECISIONS.md
[/external_agent_tool_call]

167. assistant (2026-07-02T06:11:34.075Z)

[external_agent_tool_call: mcp_cowork__create_artifact]
description: KhelSutra Guru alpha UI blueprint ? clickable static mockup of all six alpha screens (Matches, Upload & Process, Job Progress, Rally Review, Highlights, Quality Report) with fake data. Source of truth: docs/mockups/ui_alpha_mockup.html in the badminton-highlight-indexer repo.
[/external_agent_tool_call]

168. assistant (2026-07-02T06:11:34.075Z)

[external_agent_tool_result]
File deletion is now enabled for the "Projects" folder.
[/external_agent_tool_result]

169. assistant (2026-07-02T06:11:34.075Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\decisions\DECISIONS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

170. assistant (2026-07-02T06:11:34.075Z)

[external_agent_tool_result]

Artifact "khelsutra-alpha-ui-mockup" created.

[/external_agent_tool_result]

171. assistant (2026-07-02T06:11:34.075Z)

Delete permission granted ? retrying the full git init:

172. assistant (2026-07-02T06:11:34.075Z)

[external_agent_tool_call: mcp__workspace__bash]

command: cd /sessions/great-sharp-gauss/mnt/Projects/khelsutra

rm -rf .git && echo RM_OK

git init -b master -q && git config user.name "Avi Dullu" && git config user.email

"avi.dullu@gmail.com" && echo INIT_OK

git add -A && git commit -q -m "init: KhelSutra Guru product repo ? scaffold (web/, docs pointers, all-rights-reserved)" && echo COMMIT_OK

git log --oneline && git status --short && ls -A

[/external_agent_tool_call]

173. assistant (2026-07-02T06:11:34.076Z)

[external_agent_tool_result: error]

Exit code 128

RM_OK

fatal: bad config line 1 in file /sessions/great-sharp-gauss/mnt/Projects/khelsutra/.git/config

fatal: not a git repository (or any parent up to mount point /sessions/great-sharp-gauss/mnt)

Stopping at filesystem boundary (GIT_DISCOVERY_ACROSS_FILESYSTEM not set).

fatal: not a git repository (or any parent up to mount point /sessions/great-sharp-gauss/mnt)

Stopping at filesystem boundary (GIT_DISCOVERY_ACROSS_FILESYSTEM not set).

[/external_agent_tool_result]

174. assistant (2026-07-02T06:11:34.076Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md

[/external_agent_tool_call]

175. assistant (2026-07-02T06:11:34.076Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

176. assistant (2026-07-02T06:11:34.076Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\crlf-mount-artifact.md

[/external_agent_tool_call]

177. assistant (2026-07-02T06:11:34.076Z)

[external_agent_tool_result: error]

<tool_use_error>String to replace not found in file.

String: ---

name: crlf-mount-artifact

description: In this workspace, git status on badminton-highlight-indexer shows ALL tracked files modified ? it's a CRLF/LF line-ending mount artifact, not real changes

metadata:
type: project

`git status` in `~/sessions/.../mnt/badminton-highlight-indexer` (observed 2026-06-10) lists every tracked file as modified with symmetric insertions==deletions (whole-file rewrites). This is a line-ending normalization artifact of the Windows?Linux mount, not real edits. Check `git diff --stat` symmetry or a single-file diff before assuming uncommitted work; only untracked files reflect real local state.
(note: Edit also tried swapping \uXXXX escapes and their characters; neither form matched, so the mismatch is likely elsewhere in old_string. Re-read the file and copy the exact surrounding text.)</tool_use_error>
[/external_agent_tool_result]

178. assistant (2026-07-02T06:11:34.076Z)

[external_agent_tool_result]
Updated task #12 subject, status
[/external_agent_tool_result]

179. assistant (2026-07-02T06:11:34.076Z)

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"10","status":"completed"}
[/external_agent_tool_call]

180. assistant (2026-07-02T06:11:34.076Z)

[external_agent_tool_result]
Updated task #10 status
[/external_agent_tool_result]

181. assistant (2026-07-02T06:11:34.076Z)

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"11","status":"completed"}
[/external_agent_tool_call]

182. assistant (2026-07-02T06:11:34.076Z)

[external_agent_tool_result]
Updated task #11 status
[/external_agent_tool_result]

183. assistant (2026-07-02T06:11:34.076Z)

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"12","subject":"Scaffold private khelsutra repo (git init/push = owner, Windows-side)","status":"completed"}
[/external_agent_tool_call]

184. assistant (2026-07-02T06:11:34.076Z)

[external_agent_tool_call: Write]
file: C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\crlf-mount-artifact.md
[/external_agent_tool_call]

185. assistant (2026-07-02T06:11:34.076Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\crlf-mount-artifact.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

186. assistant (2026-07-02T06:11:34.076Z)

```
[external_agent_tool_call: mcp_cowork_present_files]
input: {"files":[{"file_path":"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\UI_ALPHA_EXECUTION_PLAN.md"}, {"file_path":"C:\\Users\\avidu\\Projects\\khelsutra\\README.md"}]}
[/external_agent_tool_call]
```

187. assistant (2026-07-02T06:11:34.076Z)

```
[external_agent_tool_result]
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md
C:\Users\avidu\Projects\khelsutra\README.md
[/external_agent_tool_result]
```

188. assistant (2026-07-02T06:11:34.076Z)

All wrapped up. On the task you pinged ("Verify proposals against guardrails") ? that one's already done, nothing needed there. Where you *can* help is the three G0 items that are yours by nature:

1. ****Init + push the khelsutra repo**** (git can't write through the Cowork mount ? verified, it corrupts `.git/config` ? so this is a Windows-side step; scaffold files are already in place):
...

```
cd C:\Users\avidu\Projects\khelsutra
git init -b master
git add -A
git commit -m "init: KhelSutra Guru product repo"
gh repo create khelsutra --private --source . --push
...
```

2. ****Say "go" for PR-1**** (async job model #16) whenever you've finished reviewing the docs ? that starts the local build phase.

3. ****Rotate the Gemini API key**** (pre-existing housekeeping; must happen before anything is exposed).

Everything else from your messages is done: the mockup is saved as a ****persistent artifact**** (``khelsutra-alpha-ui-mockup`` ? survives sessions, openable from the sidebar) *and* lives in the repo at ``docs/mockups/``; the execution plan is resequenced ****local-first**** exactly as you asked (Phase L: backend PRs 1?4 + SPA milestones M1?M4 + local E2E gate ? only then Phase D deployment ? Phase R rollout, ~11 sessions total); all docs and the mockup now say ****KhelSutra Guru / khelsutra.guru**** per the receipt; and the name, repo-split, and local-first decisions are recorded as ****ADR-009**** in DECISIONS.md.

189. user (2026-07-02T06:11:34.077Z)

I got -- "Pushed commits to <https://github.com/avidullu/khelsutra.git> "

190. assistant (2026-07-02T06:11:34.077Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md
[/external_agent_tool_call]
```

191. assistant (2026-07-02T06:11:34.077Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)
```

[/external_agent_tool_result]

192. assistant (2026-07-02T06:11:34.077Z)

Checked off in the plan ? `github.com/avidullu/khelsutra` is live and private. G0 now has just two items left, both yours: the explicit **"go"** for PR-1** (async job model, starts the local build phase) and the **Gemini key rotation**.

One optional convenience for later: when we reach the PR workflow, you can connect the GitHub connector in Cowork's connectors panel ? that would let me file branches/PRs on the private repo directly instead of handing you push commands each time. Not needed yet.

Say "go" whenever you're ready and S1 begins: PR-1, branch off master, built and tested locally per the plan.

193. user (2026-07-02T06:11:34.077Z)

How do I "Rotate the Gemini API key" ?

194. assistant (2026-07-02T06:11:34.077Z)

```
[external_agent_tool_call: WebSearch]
input: {"query":"Google AI Studio rotate Gemini API key delete create new key 2026"}
[/external_agent_tool_call]
```

195. assistant (2026-07-02T06:11:34.077Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"GEMINI_API_KEY|load_dotenv","path":"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer","output_mode":"content","-n":true,"glob":"*.py","head_limit":25}
[/external_agent_tool_call]
```

196. assistant (2026-07-02T06:11:34.077Z)

```
[external_agent_tool_result]
C:\Users\avidu\Projects\badminton-highlight-indexer\run_phase3_eval.py:14:Full corpus with the
two-stage detector (needs GEMINI_API_KEY; costs API calls):
C:\Users\avidu\Projects\badminton-highlight-indexer\run_phase3_eval.py:29:from dotenv import
load_dotenv
C:\Users\avidu\Projects\badminton-highlight-indexer\run_phase3_eval.py:30:load_dotenv() # make
GEMINI_API_KEY (etc.) available to the AI-backed segmenters
C:\Users\avidu\Projects\badminton-highlight-indexer\run_phase3_eval.py:55:
"yolo_hybrid/gemini need GEMINI_API_KEY and cost API calls.")
C:\Users\avidu\Projects\badminton-highlight-indexer\run_process_and_stitch.py:5:from dotenv
import load_dotenv
C:\Users\avidu\Projects\badminton-highlight-indexer\run_process_and_stitch.py:7:load_dotenv()
C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\list_models.py:2:from dotenv import
load_dotenv
C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\list_models.py:5:load_dotenv()
C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\list_models.py:6:api_key =
os.environ.get("GEMINI_API_KEY")
C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\list_models.py:8:    print("[ERROR]
GEMINI_API_KEY is not set.")
C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\gemini_video_debugger.py:32:
api_key = os.environ.get("GEMINI_API_KEY")
C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\gemini_video_debugger.py:34:
print("[ERROR] GEMINI_API_KEY environment variable is not set!")
C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\gemini_video_debugger.py:35:
print("Please set it in your PowerShell: $env:GEMINI_API_KEY=\"your_api_key\"")
C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\test_ai_segmenter.py:3:from dotenv
import load_dotenv
C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\test_ai_segmenter.py:5:load_dotenv()
C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\test_ai_segmenter.py:54:    api_key
= os.environ.get("GEMINI_API_KEY")
```

```

C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\test_ai_segmenter.py:56:
print("\n[SKIP] Skipping 'gemini' segmenter because GEMINI_API_KEY is not set.")
C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\test_ai_segmenter.py:58:
print(" $env:GEMINI_API_KEY=\"your_key\"")
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py:7:from dotenv import
load_dotenv
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py:9:load_dotenv()
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\gemini_refine.py:17:Run
(GEMINI_API_KEY in env):
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\gemini_refine.py:105:
api_key = os.environ.get("GEMINI_API_KEY")
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\gemini_refine.py:107:
raise SystemExit("GEMINI_API_KEY not set in env")
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\gemini_refine.py:156:
api_key = os.environ.get("GEMINI_API_KEY")
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\gemini_refine.py:158:
raise SystemExit("GEMINI_API_KEY not set in env")

```

```

[Showing results with pagination = limit: 25]
[/external_agent_tool_result]

```

197. assistant (2026-07-02T06:11:34.077Z)

```

[external_agent_tool_result]
Web search results for query: "Google AI Studio rotate Gemini API key delete create new key
2026"

```

```

Links: [{"title":"Using Gemini API keys | Google AI for
Developers","url":"https://ai.google.dev/gemini-api/docs/api-key"}, {"title":"Proposed Fix:
Resolving Rate Limiting by Removing AI Studio API Keys · google-gemini/gemini-cli · Discussion
#24430","url":"https://github.com/google-gemini/gemini-cli/discussions/24430"}, {"title":"Dynamic
Google Gemini API Keys with HashiCorp Vault | by Narish Samplay | Google Cloud - Community |
Apr, 2026 | Medium","url":"https://medium.com/google-cloud/dynamic-google-gemini-api-keys-with-
hashicorp-vault-c560ca92d85b"}, {"title":"Google Gemini API Key: AI Studio Setup Guide
(2026)","url":"https://www.chaterimo.com/en/blog/how-to-gemini-api-account/"}, {"title":"Gemini
CLI API Key Setup: Get and Configure Your Google AI API Key | Inventive
HQ","url":"https://inventivehq.com/knowledge-base/gemini/how-to-set-up-api-key"}, {"title":"Add
your own Gemini API key | Android Studio | Android
Developers","url":"https://developer.android.com/studio/gemini/add-api-key"}, {"title":"How to
Access Your Existing Gemini API Keys - salesso.com","url":"https://salesso.com/blog/how-to-
access-the-existing-gemini-api-keys-final/"}, {"title":"How to Get a Gemini API Key ? and Connect
It to Your Product | Unified.to","url":"https://unified.to/blog/how_to_get_a_gemini_api_key_and_
connect_it_to_your_product"}, {"title":"Securely Create & Manage Gemini API Keys in Google AI
Studio","url":"https://geminidocumentation.com/gemini-api/docs/api-key"}, {"title":"How to Get a
Free or Paid Google Gemini API Key - H2S Media","url":"https://www.how2shout.com/how-to/how-to-
get-google-gemini-api-key.html"}]

```

Based on the search results, here's information about managing Gemini API keys in Google AI Studio as of 2026:

Creating a New API Key

To create a new API key, click "Create API key" to generate a new key. All new keys created in AI Studio are automatically created as auth keys. After copying the new auth API key, you should update your application code, environment variables, and any deployment configurations to use the new auth API key.

Deleting/Revoking Old Keys

Once verified, delete or revoke your old key to prevent misuse. If a key is compromised, delete it immediately in AI Studio and generate a new one.

Key Rotation Best Practices

Keys should be rotated quarterly, not just when you suspect a leak. Periodically rotating keys every 90 days is a reasonable rhythm for production.

Important 2026 Security Changes

On June 19, 2026, the Gemini API will reject requests from unrestricted standard keys. In September 2026, the Gemini API will reject requests from Standard keys, and you must migrate to auth keys before this date to avoid service interruption.

The limitation is that AI Studio offers no programmatic SDK or API for key lifecycle management?you cannot create, rotate, restrict, or revoke keys through code.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

[/external_agent_tool_result]

198. assistant (2026-07-02T06:11:34.077Z)

Here's the safe rotation sequence (new key first, revoke old last). The key lives in ``.env`` at the indexer repo root on your box ? ``backend/main.py`` and the ``run_*.py`` scripts all ``load_dotenv()`` from there.

1. **Create the new key:**
aistudio.google.com/app/apikey (same Google account as your paid project) ? Create API key ? **attach it to the same Cloud project** so the paid tier/billing carries over ? copy it.
2. **Swap it in:** edit ``.env`` in the repo root ? ``GEMINI_API_KEY=<new key>``. Also check you don't have an overriding ``$env:GEMINI_API_KEY`` in your PowerShell profile or any Task Scheduler job.
3. **Restart** any running FastAPI server/long process ? the key is read once at startup.
4. **Verify cheaply:** ``python scratch\list_models.py`` (already in your repo; it just lists models with the key ? prints an error if the key is bad).
5. **Revoke the old key:** back in AI Studio (or GCP Console ? APIs & Services ? Credentials) ? delete it. Since the old one was pasted in chat, treat it as compromised ? delete, don't keep as backup.
6. **Optional hardening:** in GCP Console, restrict the new key to the ***Generative Language API*** only.

One timely flag from the docs: Google is migrating Gemini API key types in 2026 ? reportedly ****unrestricted "standard" keys start getting rejected June 19, 2026**** (next week), with full migration to the new "auth keys" by September. New keys created in AI Studio today are already auth keys ? so this rotation conveniently also future-proofs you. Worth doing step 4's verification carefully.

Sources: [Using Gemini API keys (official docs)](https://ai.google.dev/gemini-api/docs/api-key), [AI Studio key setup guide](https://www.chaterimo.com/en/blog/how-to-gemini-api-account/), [Key management guide](https://geminidocumentation.com/gemini-api/docs/api-key)

199. user (2026-07-02T06:11:34.077Z)

I have verified that the key is WAI. Can you validate as well?

200. assistant (2026-07-02T06:11:34.077Z)

I can try ? the ``.env`` sits in the mounted repo, so I can run your own ``scratch/list_models.py`` against it from the sandbox without ever printing the key. Checking the script and environment first:

201. assistant (2026-07-02T06:11:34.077Z)

```
[external_agent_tool_call: mcp_workspace_bash]
command: cd /sessions/great-sharp-gauss/mnt/badminton-highlight-indexer
[ -f .env ] && echo ".env present" || echo ".env MISSING"
```

```
grep -c "GEMINI_API_KEY" .env 2>/dev/null | sed 's/^/GEMINI_API_KEY lines: /'
echo "--- list_models.py ---"; cat scratch/list_models.py
[/external_agent_tool_call]
```

202. assistant (2026-07-02T06:11:34.077Z)

```
[external_agent_tool_result]
.env present
GEMINI_API_KEY lines: 1
--- list_models.py ---
import os
from dotenv import load_dotenv
from google import genai

load_dotenv()
api_key = os.environ.get("GEMINI_API_KEY")
if not api_key:
    print("[ERROR] GEMINI_API_KEY is not set.")
    exit(1)

client = genai.Client(api_key=api_key)
print("Listing all available models on your API Key:")
try:
    models = client.models.list()
    for m in models:
        print(f" * {m.name} (Display: {m.display_name})")
except Exception as e:
    print(f"[FAIL] Error listing models: {e}")

[/external_agent_tool_result]
```

203. assistant (2026-07-02T06:11:34.078Z)

Script's clean ? it only prints model names, never the key. Installing the SDK and running it against your ``.env``:

204. assistant (2026-07-02T06:11:34.078Z)

```
[external_agent_tool_call: mcp_workspace_bash]
command: pip install -q --break-system-packages google-genai python-dotenv 2>&1 | tail -1; echo
"--- install done ---"
[/external_agent_tool_call]
```

205. assistant (2026-07-02T06:11:34.078Z)

```
[external_agent_tool_result]
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-
warn-script-location.
--- install done ---

[/external_agent_tool_result]
```

206. assistant (2026-07-02T06:11:34.078Z)

```
[external_agent_tool_call: mcp_workspace_bash]
command: cd /sessions/great-sharp-gauss/mnt/badminton-highlight-indexer && python3
scratch/list_models.py 2>&1 | head -20
[/external_agent_tool_call]
```

207. assistant (2026-07-02T06:11:34.078Z)

```
[external_agent_tool_result]
Traceback (most recent call last):
  File "/sessions/great-sharp-gauss/mnt/badminton-highlight-indexer/scratch/list_models.py",
```

```

line 11, in <module>
    client = genai.Client(api_key=api_key)
File "/sessions/great-sharp-gauss/.local/lib/python3.10/site-packages/google/genai/client.py",
line 458, in __init__
    self._api_client = self._get_api_client(
File "/sessions/great-sharp-gauss/.local/lib/python3.10/site-packages/google/genai/client.py",
line 506, in _get_api_client
    return BaseApiClient(
File "/sessions/great-sharp-gauss/.local/lib/python3.10/site-
packages/google/genai/_api_client.py", line 816, in __init__
    self._httpx_client = SyncHttpxCliet(**client_args)
File "/sessions/great-sharp-gauss/.local/lib/python3.10/site-
packages/google/genai/_api_client.py", line 562, in __init__
    super().__init__(**kwargs)
File "/sessions/great-sharp-gauss/.local/lib/python3.10/site-packages/httpx/_client.py", line
697, in __init__
    self._mounts: dict[URLPattern, BaseTransport | None] = {
File "/sessions/great-sharp-gauss/.local/lib/python3.10/site-packages/httpx/_client.py", line
700, in <dictcomp>
    else self._init_proxy_transport(
File "/sessions/great-sharp-gauss/.local/lib/python3.10/site-packages/httpx/_client.py", line
750, in _init_proxy_transport
    return HTTPTransport(
File "/sessions/great-sharp-gauss/.local/lib/python3.10/site-
packages/httpx/_transports/default.py", line 191, in __init__
    raise ImportError(
ImportError: Using SOCKS proxy, but the 'socksio' package is not installed. Make sure to install
httpx using `pip install httpx[socks]`.

```

[/external_agent_tool_result]

208. assistant (2026-07-02T06:11:34.078Z)

Sandbox egress goes through a SOCKS proxy ? needs one more package:

209. assistant (2026-07-02T06:11:34.078Z)

```

[external_agent_tool_call: mcp__workspace__bash]
command: pip install -q --break-system-packages "httpx[socks]" socksio 2>&1 | tail -1
cd /sessions/great-sharp-gauss/mnt/badminton-highlight-indexer && python3 scratch/list_models.py
2>&1 | head -15
[/external_agent_tool_call]

```

210. assistant (2026-07-02T06:11:34.078Z)

```

[external_agent_tool_result]
Listing all available models on your API Key:
[FAIL] Error listing models: 403 Forbidden

[/external_agent_tool_result]

```

211. assistant (2026-07-02T06:11:34.078Z)

403 ? need to know if that's the sandbox proxy blocking the domain or Google rejecting the call.
Probing the endpoint with and without the key (key stays in a shell var, never printed):

212. assistant (2026-07-02T06:11:34.078Z)

```

[external_agent_tool_call: mcp__workspace__bash]
command: cd /sessions/great-sharp-gauss/mnt/badminton-highlight-indexer
echo "--- no key ---"
curl -s -m 15 -w "%{http_code}\n"
"https://generativelanguage.googleapis.com/v1beta/models" | head -12
echo "--- with key (from .env, not echoed) ---"

```



```
KEY=$(grep -oP '(?<=GEMINI_API_KEY=).*' .env | tr -d '"\r')
curl -s -m 15 -w "\nHTTP:%{http_code}\n" -H "x-goog-api-key: ${KEY}"
"https://generativelanguage.googleapis.com/v1beta/models" | head -12
[/external_agent_tool_call]
```

213. assistant (2026-07-02T06:11:34.078Z)

```
[external_agent_tool_result]
```

```
--- no key ---
```

```
HTTP:000
```

```
--- with key (from .env, not echoed) ---
```

```
HTTP:000
```

```
[/external_agent_tool_result]
```

214. assistant (2026-07-02T06:11:34.078Z)

I can't validate it from here, and the reason is clean: the sandbox's network egress doesn't allow `generativelanguage.googleapis.com` at all (the bare connection gets HTTP 000 ? never reaches Google; the SDK's earlier "403" was the egress proxy refusing, not your key). So your machine-side check is the authoritative one ? which matches our working split anyway: GPU-box/API checks are yours, and `list\_models.py` succeeding there is exactly the right verification.

What I *could* verify from here: your `.env` is well-formed ? exactly one `GEMINI\_API\_KEY` entry, no stale duplicate that could shadow the new key.

Two 30-second closers on your side and rotation is fully done: confirm the **old key is deleted** in AI Studio (not just unused), and while you're there check the new key shows as an **auth key** type (the June 19 migration cutoff). Then G0 has exactly one item left: your "go" for PR-1.

215. user (2026-07-02T06:11:34.078Z)

would you work better in WSL ? I have ubuntu and the repo can be made on it.

216. assistant (2026-07-02T06:11:34.078Z)

Short answer: it wouldn't help me much, and it might help *you* ? with one real catch for the engine repo.

My shell isn't running on your machine at all ? it's an isolated Linux sandbox, and your folders reach me through a mount bridge. That bridge is where this session's friction lives (git-metadata writes corrupt, deletes blocked) ? it's not a Windows-vs-Linux thing. Moving the repo into WSL's filesystem doesn't move *me* into WSL; I'd still talk to it through the same bridge, now with an extra hop (`\\wsl.localhost\...\` over 9P), assuming Cowork can mount a UNC path at all. If it can't, I'd lose file access entirely ? a regression from today, where everything except git-writes works fine.

Two things worth separating:

For you and Claude Code, a WSL-hosted repo is genuinely nicer: native git (no CRLF noise ? the phantom all-files-modified I see is exactly that), faster I/O, and the conda/GPU stack is already there. Claude Code running inside WSL gets all of that natively. That combo ? Claude Code in WSL for execution-heavy work, me here for research/docs/hosting/cross-repo planning ? plays to both our strengths, and git ops were already yours per our working agreement.

For the engine repo specifically, there's a migration cost: the WASB/TrackNet runners assume **Windows host invoking WSL** (`wsl bash -c`, `C:\ ? /mnt/c` path staging). Move the repo inside WSL and those invocation contracts need rework. Not huge, but not free ? and it'd deserve an ADR if we go there.

If you're curious whether the bridge behaves better with a WSL path, there's a 2-minute experiment: add `\\wsl.localhost\Ubuntu\home\<you>\...` as a folder in Cowork and I'll test write + git behavior on it. Happy to run that ? otherwise my recommendation is keep the repos where they are, and we lose nothing.

217. user (2026-07-02T06:11:34.078Z)

Sure. And can you elaborate as to who will execute the ": the explicit "go" for PR-1 (async job model, starts the local build phase)"

218. assistant (2026-07-02T06:11:34.079Z)

Kicking off the WSL experiment first ? a folder picker should appear: navigate to your Ubuntu home via `\\wsl.localhost\Ubuntu\home\<your-user>` (or any scratch folder inside WSL) and select it.

219. assistant (2026-07-02T06:11:34.079Z)

```
[external_agent_tool_call: mcp__cowork__request_cowork_directory]
input: {}
[/external_agent_tool_call]
```

220. assistant (2026-07-02T06:11:34.079Z)

```
[external_agent_tool_result]
Directory selection was cancelled by the user.
[/external_agent_tool_result]
```

221. assistant (2026-07-02T06:11:34.079Z)

No problem ? picker cancelled; whenever you want to run that test, just say so (or tell me your WSL username and I'll request `\\wsl.localhost\Ubuntu\home\<user>` directly so you only have to click Approve).

On who does what for PR-1 ? the "go" is ****yours****, and it's literally one word: reply "go" (or "go for PR-1") in chat. That's the owner-gating rule from your own CLAUDE.md ? I don't start implementation without explicit approval, so the approval itself is the only thing you execute there.

After that, the division for the whole PR cycle:

****Me:**** branch off `master`, implement B1 (jobs table migration, `202 + job\_id`, `GET /api/jobs/{id}`, threadpool worker, crash/restart handling), write the tests, run the full offline suite (`run\_all\_tests.py` is fully mocked, so it runs in my sandbox).

****The seam ? getting the branch onto GitHub:**** three options, best first. (1) You connect the ****GitHub connector**** in Cowork's connectors panel ? then I push the branch and open the PR via API, no local git involved, mount problem irrelevant. (2) You or Claude Code run a three-line commit/push handoff I prepare. (3) If the WSL-mount experiment pans out, I get native git and this seam disappears entirely.

****You:**** review the PR, run the machine-side checks I can't (real video on the GPU box, WSL suite), and merge. Per the plan, nothing lands on `master` without your review.

So in practice: you say "go" ? I disappear into PR-1 ? you get back a reviewable branch with a verification checklist for the parts only your box can test.

222. user (2026-07-02T06:11:34.079Z)

btw my wsl - `\\wsl.localhost\Ubuntu\home\avidullu\projects`

223. assistant (2026-07-02T06:11:34.079Z)

```
[external_agent_tool_call: mcp__cowork__request_cowork_directory]
```

```
input: {"path":"\\\\wsl.localhost\\Ubuntu\\home\\avidullu\\projects"}
[/external_agent_tool_call]
```

224. assistant (2026-07-02T06:11:34.079Z)

```
[external_agent_tool_result: error]
UNC paths are not supported: \\wsl.localhost\ubuntu\home\avidullu\projects
[/external_agent_tool_result]
```

225. assistant (2026-07-02T06:11:34.079Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\crlf-mount-artifact.md
[/external_agent_tool_call]
```

226. assistant (2026-07-02T06:11:34.079Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\crlf-mount-artifact.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

227. assistant (2026-07-02T06:11:34.079Z)

That settles it definitively ? Cowork can't mount UNC paths at all (\\wsl.localhost\...` is rejected outright), so a WSL-hosted repo would be invisible to me. Experiment over, result recorded.

The practical takeaway stands as before: repos stay on the Windows filesystem where I have full file access; for git autonomy on my side, the ****GitHub connector**** is the clean answer (API-based branch push + PR creation, no local git involved). WSL remains a fine place for **your** interactive work via Claude Code if you ever want it ? just with the caveat that moving the engine repo there means reworking the Windows?WSL detector-runner contracts, so I wouldn't.

So the board is: G0 done except your "go" for PR-1 ? and optionally, connect GitHub in the connectors panel whenever convenient so I can hand you PRs instead of patch files.

228. user (2026-07-02T06:11:34.079Z)

Ok. So can you point me to the doc from which thsi PR1 is coming ?

229. assistant (2026-07-02T06:11:34.079Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"async job|#16|^## 16|^### 16","path":"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\DEFERRED_HARDENING_PLAN.md","output_mode":"content","-n":true,"-i":true,"head_limit":15}
[/external_agent_tool_call]
```

230. assistant (2026-07-02T06:11:34.079Z)

```
[external_agent_tool_result]
1:# Deferred Hardening Plan ? Train/Eval Guardrails, Async Jobs, Packaging
13:#13 Train/eval guardrails (correctness of the data flywheel) ? #16 Async jobs
90:## 3. Async job model (debt #16)
[/external_agent_tool_result]
```

231. assistant (2026-07-02T06:11:34.080Z)

The lineage runs through three docs, from origin to execution:

1. ****Origin / canonical design:**** `docs/DEFERRED\_HARDENING\_PLAN.md` ? section 3, ****"Async job model (debt #16)"**** (line 90). That's where the design lives: `POST /api/process` returns 202 + `job\_id`, a `jobs` table via the `user\_version` migration ladder, ThreadPoolExecutor worker, `GET /api/jobs/{id}` polling. It's also flagged in `CLAUDE.md` and `NEXT\_STEPS.md` as the ***committed next PLATFORM slice*** (and `PLATFORM\_ARCHITECTURE.md` lists it as P0) ? it predates the UI work; the UI just made it urgent.

2. ****Why the UI needs it:**** `docs/UI\_PROPOSAL.md` §3, backend dependency table, row ****B1**** ? a 5?60 minute blocking HTTP call can't sit behind a web UI, so #16 is the alpha's first prerequisite.

3. ****Where "PR-1" gets its name:**** `docs/UI\_ALPHA\_EXECUTION\_PLAN.md` ? Phase L ? L1 ? ****PR-1 . B1 async job model (#16)****, first item in the local build phase.

So when you say "go", what I implement is the design from DEFERRED_HARDENING_PLAN §3, scoped per UI_PROPOSAL B1, sequenced as EXECUTION_PLAN PR-1.

232. user (2026-07-02T06:11:34.080Z)

Got it. But will you be able to ramp up using CODE_MAP and session handoffs?

233. assistant (2026-07-02T06:11:34.080Z)

Yes ? that's literally what this session opened with. My session-start routine reads `CLAUDE.md` ? `CODE\_MAP.md` ? `NEXT\_STEPS.md` ? recent git log, and that got me from cold to current in about three reads today. For PR-1 specifically, CODE_MAP already marks every touchpoint: the `process\_video` endpoint and its module-globals gotcha in `backend/main.py`, the `PRAGMA user\_version` migration ladder in `database.py` (currently 1 ? PR-1 bumps it to 2), the SQLite WAL/busy-timeout posture, and the fact that segmenters already thread internally ? which is exactly the executor model #16 prescribes.

Two honest caveats. First, CODE_MAP carries its own staleness banner (the 2026-06-10 sweep changed behaviors it documents), so while coding I verify any contract I touch against the actual code, not the map ? and PR-1 itself triggers a CODE_MAP update per its regen rule, plus a NEXT_STEPS refresh, so the next session inherits a current map. Second, the "session handoff" memories referenced in NEXT_STEPS (`current-state.md`, `session-handoff.md`) live in your ***Claude Code*** setup ? I can't see those. My equivalents are the repo docs plus my own persistent memory here (which already carries this workspace's quirks ? the mount/git findings, the WSL result).

The practical implication: the repo docs are the shared brain between me and Claude Code. As long as decisions and state land in `NEXT\_STEPS.md` / `DECISIONS.md` / the plan docs ? which is already your convention, and ADR-009 + the three new docs follow it ? either of us can pick up exactly where the other left off. So yes: say "go" in any future session and the ramp-up cost is a few file reads, not an archaeology dig.

234. user (2026-07-02T06:11:34.080Z)

Well I did not know that you and Claude Code were besties, in that case .. Go for PR1 ! Please make sure to make all the changes necessary and I will be happy to review in the morning. (With Claude Code I can tell it to keep merging as well, but I doubt you can do that)

235. assistant (2026-07-02T06:11:34.080Z)

Go received! ? And yes ? merging stays with you (or Claude Code post-review); my lane ends at "complete, tested, reviewable." You'll wake up to a ready changeset. Setting up the build:

236. assistant (2026-07-02T06:11:34.080Z)

[external_agent_tool_call: TaskCreate]
description: Read DEFERRED_HARDENING_PLAN §3 exact design, backend/main.py, backend/infrastructure/database.py, tests/test_api_endpoints.py, run_all_tests.py.

[/external_agent_tool_call]

237. assistant (2026-07-02T06:11:34.080Z)

[external_agent_tool_result]

Task #13 created successfully: PR-1: Read #16 spec + current code (main.py, database.py, tests)

[/external_agent_tool_result]

238. assistant (2026-07-02T06:11:34.080Z)

[external_agent_tool_call: TaskCreate]

description: user_version 1?2, jobs table, add/update/get job methods, restart-recovery sweep.

[/external_agent_tool_call]

239. assistant (2026-07-02T06:11:34.080Z)

[external_agent_tool_result]

Task #14 created successfully: PR-1: Implement jobs migration + job store (database.py)

[/external_agent_tool_result]

240. assistant (2026-07-02T06:11:34.080Z)

[external_agent_tool_call: TaskCreate]

description: POST /api/process ? 202+job_id; GET /api/jobs/{id}; ThreadPoolExecutor(1); pipeline body moved to worker with finally-report preserved.

[/external_agent_tool_call]

241. assistant (2026-07-02T06:11:34.080Z)

[external_agent_tool_result]

Task #15 created successfully: PR-1: Async endpoints + worker (main.py)

[/external_agent_tool_result]

242. assistant (2026-07-02T06:11:34.080Z)

[external_agent_tool_result]

Task #17 created successfully: PR-1: Docs sync + reviewable handoff

[/external_agent_tool_result]

243. assistant (2026-07-02T06:11:34.080Z)

[external_agent_tool_call: TaskCreate]

description: tests/test_async_jobs.py (202, transitions, failure, 404, recovery, migration), update test_api_endpoints.py, run run_all_tests.py green.

[/external_agent_tool_call]

244. assistant (2026-07-02T06:11:34.081Z)

[external_agent_tool_result]

Task #16 created successfully: PR-1: Tests (new async suite + update api tests) + full offline run

[/external_agent_tool_result]

245. assistant (2026-07-02T06:11:34.081Z)

[external_agent_tool_call: TaskCreate]

description: CODE_MAP/NEXT_STEPS/plan touch-ups; scratch/PR1_HANDOFF.md with branch commands, commit message, PR description, machine-side checklist.

[/external_agent_tool_call]

246. assistant (2026-07-02T06:11:34.081Z)

[external_agent_tool_call: TaskUpdate]

```
input: {"taskId":"13","status":"in_progress"}
[/external_agent_tool_call]
```

247. assistant (2026-07-02T06:11:34.081Z)

```
[external_agent_tool_result]
Updated task #13 status
[/external_agent_tool_result]
```

248. assistant (2026-07-02T06:11:34.081Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DEFERRED_HARDENING_PLAN.md
[/external_agent_tool_call]
```

249. assistant (2026-07-02T06:11:34.081Z)

```
[external_agent_tool_result]
85     - Unit test: overlapping train/eval ids ? `assert_disjoint` raises.
86     - Integration: a deliberately-leaked manifest fails the training endpoint.
87
88     ---
89
90     ## 3. Async job model (debt #16)
91
92     ### Problem
93     `POST /api/process` runs validate ? segment ? report **inline**, blocking the
94     request for the full duration of a long video. Unworkable as a hosted/product API.
95
96     ### Approach (minimal first slice ? single self-hosted box)
97     - `POST /api/process` returns **202 + `job_id`**; add `GET /api/jobs/{job_id}`
98     returning `{status, progress, result, reports}`.
99     - Run the pipeline on a background worker ? start with a `ThreadPoolExecutor`
100     (segmenters already use threads internally for AI handoff, so this fits); leave a
101     real queue (RQ/Celery) for later if multi-process is needed.
102     - Persist job state in the existing SQLite DB via the migration ladder
103     (`backend/infrastructure/database.py::_init_db`, bump `PRAGMA user_version` 1 ? 2,
104     add a `jobs` table). The observability report becomes the job's artifact.
105     - Update the static UI to poll `GET /api/jobs/{id}` instead of blocking.
106
107     ### Risks
108     - **SQLite thread-safety:** verify `check_same_thread`/connection-per-thread;
109     `_get_connection` already re-issues `PRAGMA foreign_keys=ON` per connection ?
110     confirm it's safe under the executor.
111     - Cancellation/timeout and crash recovery (a job in RUNNING after a restart).
112     - In-process store is not horizontally scalable ? acceptable for the current
113     single-box self-hosted target; call it out so it isn't mistaken for production-HA.
114
115     ### Verification
116     - Submit a job ? poll ? DONE with the same segments the sync path produced.
117     - Server restart mid-job leaves a recoverable/clearly-FAILED job, not a hang.
118
119     ---
120
121     ## Out of this doc
122     - **Detector/WSL abstraction (#6):** in flight (PR #49).
123     - **Unify hybrid segmenters (#7 in LOW_REGRET):** owner-deferred.
124     - **ByteTrack `supervision<0.30.0` migration + real-footage validation (#14/#15):**
125     need a GPU/WSL box and real video ? owner-machine work, tracked separately.
126
[/external_agent_tool_result]
```

250. assistant (2026-07-02T06:11:34.081Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py
[/external_agent_tool_call]
```

251. assistant (2026-07-02T06:11:34.081Z)

```
[external_agent_tool_result]
1     import os
2     import sys
3     import argparse
4     import json
5     import logging
6     import uvicorn
7     from dotenv import load_dotenv
8
9     load_dotenv()
10
11     # Centralized logging setup for the indexer (hot paths use logger, CLI output uses print
for direct user-facing messages).
12     logging.basicConfig(
13         level=logging.INFO,
14         format='%(asctime)s [%(levelname)s] %(name)s: %(message)s',
15         datefmt='%H:%M:%S'
16     )
17     logger = logging.getLogger(__name__)
18     from fastapi import FastAPI, HTTPException
19     from fastapi.middleware.cors import CORSMiddleware
20     from pydantic import BaseModel
21     from typing import List, Dict, Any, Optional
22
23     from fastapi.staticfiles import StaticFiles
24     from fastapi.responses import HTMLResponse
25     from backend.pipeline.validators.base import registry
26     from backend.infrastructure.database import Database
27     from backend.pipeline.segmenters.base import segmenter_registry, BasePlaySegmenter
28     from backend.pipeline.stitcher.compiler import VideoCompiler
29     from backend.utils.hashing import compute_video_id # canonical file-identity hashing
30     from backend.utils.paths import safe_output_filename, validate_input_path
31     from backend.config import load_config as load_app_config, AppConfig # new typed config
32     from backend.results import Result, Failure # standardized error/result contract
33     from backend.reporting import emit_indexer_report
34
35     app = FastAPI(title="Sports Highlight Indexer API")
36
37     # Enable CORS for the local web interface. allow_origins='' with allow_credentials=True
is
38     # rejected by browsers per the fetch spec and is an unsafe default to carry into the
39     # auth/tenancy work; this tool uses no cookies, so credentials stay off.
40     app.add_middleware(
41         CORSMiddleware,
42         allow_origins=["*"],
43         allow_credentials=False,
44         allow_methods=["*"],
45         allow_headers=["*"],
46     )
47
48     # Serves static frontend files directly (now under api/ per approved structure)
49     os.makedirs("backend/api/static", exist_ok=True)
50     app.mount("/static", StaticFiles(directory="backend/api/static"), name="static")
51
52     @app.get("/", response_class=HTMLResponse)
53     def serve_index():
54         index_path = "backend/api/static/index.html"
55         if os.path.exists(index_path):
56             with open(index_path, "r", encoding="utf-8") as f:
```

```

57         return f.read()
58     return "<h3>Sports Highlight Indexer Server is Running. Please create index.html in
backend/api/static/</h3>"
59
60     # Global variables initialized at startup
61     db_instance: Optional[Database] = None
62     global_config: Optional[AppConfig] = None # now typed (Pydantic model)
63
64     # Legacy thin wrapper for any remaining direct calls during transition.
65     # New code should import load_app_config / AppConfig from backend.config directly.
66     def load_config(config_path: str = "config.json") -> Dict[str, Any]:
67         cfg = load_app_config(config_path)
68         return cfg.model_dump(mode="json")
69
70     # --- API Models ---
71     class ProcessRequest(BaseModel):
72         video_path: str
73         player_pool: Optional[List[str]] = None
74         max_frames: Optional[int] = None
75         segmenter: Optional[str] = None
76
77     class QueryRequest(BaseModel):
78         video_id: str
79         server: Optional[str] = None
80         receiver: Optional[str] = None
81         duration_min: Optional[float] = None
82         event_type: Optional[str] = None
83
84     class CompileRequest(BaseModel):
85         video_id: str
86         segment_ids: List[int]
87         output_filename: str
88
89     def _finalize_reingest(video_id: str, segments, play_wm: int, cand_wm: int) -> None:
90         """Commit a (re)segmentation with destroy-AFTER-confirm semantics.
91
92         On a non-empty run, drop the PRIOR rows (id <= the pre-run watermark) so only the
93         fresh segments remain. On an empty run, drop the NEW partial rows (id > watermark)
94         and
95         keep the prior good results intact ? a failed/empty re-run never destroys prior
96         data.
97         """
98         if segments:
99
100     [/external_agent_tool_result]

```

252. assistant (2026-07-02T06:11:34.081Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\infrastructure\database.py
[/external_agent_tool_call]

```

253. assistant (2026-07-02T06:11:34.081Z)

```

[external_agent_tool_result]
1     import sqlite3
2     import json
3     import os
4     import logging
5     from typing import List, Dict, Any, Optional, Tuple
6
7     logger = logging.getLogger(__name__)
8
9     class Database:
10         def __init__(self, db_path: str):

```



```

11         self.db_path = db_path
12         self._init_db()
13
14     def _get_connection(self):
15         # busy_timeout: wait (not fail) on a locked DB ? needed once the async job model
16         # adds concurrent writers (a swallowed 'database is locked' silently drops
segments).
17         conn = sqlite3.connect(self.db_path, timeout=30.0)
18         conn.row_factory = sqlite3.Row
19         # Enforce ON DELETE CASCADE / SET NULL ? SQLite leaves FKs off per-connection.
20         conn.execute("PRAGMA foreign_keys = ON")
21         # WAL allows concurrent readers during a write; busy_timeout bounds lock waits.
22         conn.execute("PRAGMA busy_timeout = 30000")
23         try:
24             conn.execute("PRAGMA journal_mode = WAL")
25         except sqlite3.OperationalError:
26             pass # e.g. a read-only/networked FS that can't do WAL ? degrade gracefully
27         return conn
28
29     def _init_db(self):
30         """Initializes tables for videos, play_segments, and events."""
31         with self._get_connection() as conn:
32             cursor = conn.cursor()
33
34             # Videos Table
35             cursor.execute("""
36                 CREATE TABLE IF NOT EXISTS videos (
37                     id TEXT PRIMARY KEY,
38                     filepath TEXT NOT NULL,
39                     processed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
40                     status TEXT NOT NULL,
41                     validation_results TEXT
42                 )
43             """)
44
45             # Play Segments Table (Extensible metadata JSON)
46             cursor.execute("""
47                 CREATE TABLE IF NOT EXISTS play_segments (
48                     id INTEGER PRIMARY KEY AUTOINCREMENT,
49                     video_id TEXT NOT NULL,
50                     start_time REAL NOT NULL,
51                     end_time REAL NOT NULL,
52                     duration REAL NOT NULL,
53                     server TEXT,
54                     receiver TEXT,
55                     metadata TEXT,
56                     FOREIGN KEY (video_id) REFERENCES videos(id) ON DELETE CASCADE
57                 )
58             """)
59
60             # Events Table
61             cursor.execute("""
62                 CREATE TABLE IF NOT EXISTS events (
63                     id INTEGER PRIMARY KEY AUTOINCREMENT,
64                     segment_id INTEGER NOT NULL,
65                     timestamp REAL NOT NULL,
66                     type TEXT NOT NULL,
67                     player TEXT,
68                     details TEXT,
69                     FOREIGN KEY (segment_id) REFERENCES play_segments(id) ON DELETE
CASCADE
70                 )
71             """)
72
73             # Versioned migrations live here. PRAGMA user_version is the schema

```

```

74         # contract ? bump it for every change and add an ordered `if version < N`
75         # block below. Tests assert the expected version, so accidental drift fails
CI.
76         version = cursor.execute("PRAGMA user_version").fetchone()[0]
77
78         if version < 1:
79             # v1: raw candidate detections (pre-confirmation), detector-agnostic.
80             # Common fields are columns; detector-specific metrics go in `stats`
JSON,
81             # so a new segmenter reuses this table by setting its own
`detector`/`stats`.
82             cursor.execute("""
83                 CREATE TABLE IF NOT EXISTS candidate_segments (
84                     id INTEGER PRIMARY KEY AUTOINCREMENT,
85                     video_id TEXT NOT NULL,
86                     detector TEXT NOT NULL,
87                     chunk_index INTEGER,
88                     start_time REAL NOT NULL,
89                     ?
[/external_agent_tool_result]

```

254. assistant (2026-07-02T06:11:34.081Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py
[/external_agent_tool_call]

```

255. assistant (2026-07-02T06:11:34.081Z)

```

[external_agent_tool_result]
1     """TestClient coverage for the FastAPI seam (previously zero ? the compiler SyntaxError
2     that broke `import backend.main` shipped green because nothing imported it)."""
3     import pytest
4
5     pytest.importorskip("httpx") # TestClient needs httpx; skip cleanly where absent
6     from fastapi.testclient import TestClient
7
8     import backend.main as m
9     from backend.infrastructure.database import Database
10    from backend.config.models import AppConfig
11    from backend.pipeline.validators.base import ValidationResult
12
13
14    @pytest.fixture
15    def env(tmp_path, monkeypatch):
16        db = Database(str(tmp_path / "t.db"))
17        cfg = AppConfig()
18        monkeypatch.setattr(m, "db_instance", db)
19        monkeypatch.setattr(m, "global_config", cfg)
20        client = TestClient(m.app)
21        return client, db, tmp_path, monkeypatch
22
23
24    def _video(tmp_path, name="m.mp4"):
25        p = tmp_path / name
26        p.write_bytes(b"not-a-real-video-but-hashable")
27        return p
28
29
30    def _pass():
31        return [ValidationResult(validator_name="x", passed=True, message="ok", details={})]
32
33
34    def _fail():
35        return [ValidationResult(validator_name="blur", passed=False, message="blurry",

```

```

details={}])
36
37
38     class _FakeSeg:
39         def __init__(self, db, cfg):
40             self.db = db
41
42         def process_video(self, video_id, video_path, *a, **k):
43             sid = self.db.add_play_segment(video_id, 0.0, 5.0)
44             return [{"id": sid, "start_time": 0.0, "end_time": 5.0, "duration": 5.0,
"metadata": {}}]
45
46
47     def test_api_config_ok(env):
48         client = env[0]
49         r = client.get("/api/config")
50         assert r.status_code == 200
51         assert r.json()["sport"] == "badminton"
52
53
54     def test_process_success_with_mocked_segmenter(env):
55         client, db, tmp_path, mp = env
56         vid_file = _video(tmp_path)
57         mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {}))
58         mp.setattr(m.segmenter_registry, "get", lambda name: _FakeSeg)
59         r = client.post("/api/process", json={"video_path": str(vid_file), "segmenter":
"x"})
60         assert r.status_code == 200
61         body = r.json()
62         assert body["status"] == "success"
63         assert len(body["play_segments"]) == 1
64
65
66     def test_process_validation_failure_preserves_prior_segments(env):
67         client, db, tmp_path, mp = env
68         vid_file = _video(tmp_path)
69         vid = m.compute_video_id(str(vid_file))
70         db.add_video(vid, str(vid_file), "processed", [])
71         db.add_play_segment(vid, 1.0, 2.0) # prior good result
72         mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_fail(), {}))
73         r = client.post("/api/process", json={"video_path": str(vid_file)})
74         assert r.status_code == 200 and r.json()["status"] == "failed"
75         # PR-C: a failed re-validation must NOT cascade-delete prior segments.
76         assert len(db.query_play_segments(vid, {})) == 1
77
78
79     def test_process_rejects_null_byte_path(env):
80         client = env[0]
81         r = client.post("/api/process", json={"video_path": "a\x00b.mp4"})
82         assert r.status_code == 400
83
84
85     def test_query_returns_segments(env):
86         client, db, tmp_path, mp = env
87         db.add_video("vid", "/x.mp4", "processed", [])
88         db.add_play_segment("vid", 1.0, 4.0)
89         r = client.post("/api/query", json={"video_id": "vid"})
90         assert r.status_code == 200
91         assert len(r.json()["play_segments"]) == 1
92
93
94     def test_compile_rejects_traversal_output_filename(env):
95         client, db, tmp_path, mp = env
96         db.add_video("vid", str(_video(tmp_path)), "processed", [])
97         sid = db.add_play_segment("vid", 1.0, 4.0)

```

```

98         r = client.post("/api/compile", json={
99             "video_id": "vid", "segment_ids": [sid], "output_filename": "../evil.mp4"})
100         assert r.status_code == 400
101
102
103     def test_compile_success_with_mocked_compiler(env):
104         client, db, tmp_path, mp = env
105         db.add_video("vid", str(_video(tmp_path)), "?
[/external_agent_tool_result]

```

256. assistant (2026-07-02T06:11:34.081Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\run_all_tests.py
[/external_agent_tool_call]

```

257. assistant (2026-07-02T06:11:34.081Z)

```

[external_agent_tool_result]
1     #!/usr/bin/env python
2     """Run the FULL test suite for the badminton-highlight-indexer.
3
4     A single, discoverable entry point so you (or a reviewer / CI) can verify the
5     whole suite before opening a PR:
6
7         python run_all_tests.py          # full suite, quiet
8         python run_all_tests.py -v       # verbose (passthrough to pytest)
9         python run_all_tests.py -k config # filter by keyword (passthrough)
10        python run_all_tests.py --cov=backend --cov-report=term-missing # coverage
11
12    Anything after the script name is passed straight through to pytest.
13
14    Notes
15    -----
16    - The suite is fully mocked (no GPU, no network, no WSL) and runs offline.
17    - The observability **reporting** tests skip cleanly if the optional ``obsreport``
18      package isn't installed (they use ``pytest.importorskip``). To exercise them too:
19      pip install -e ../sports-obsreport --no-build-isolation
20    - Exit code is pytest's own, so this is CI / pre-PR friendly (0 = all green).
21    """
22    from __future__ import annotations
23
24    import importlib.util
25    import subprocess
26    import sys
27
28
29    def _installed(module: str) -> bool:
30        try:
31            return importlib.util.find_spec(module) is not None
32        except (ImportError, ValueError):
33            return False
34
35
36    def main() -> int:
37        args = list(sys.argv[1:])
38        # Default to quiet unless the caller chose a verbosity.
39        if not any(a in ("-q", "-v", "-vv", "-vvv", "--quiet", "--verbose") for a in args):
40            args = ["-q", *args]
41        cmd = [sys.executable, "-m", "pytest", *args]
42
43        obs = "available" if _installed("obsreport") else "NOT installed ? reporting tests
will SKIP"
44        bar = "=" * 72
45        print(bar)

```

```

46         print(" Full test suite: " + " ".join(cmd))
47         print(f" obsreport (optional reporting dep): {obs}")
48         print(bar, flush=True)
49
50         rc = subprocess.run(cmd).returncode
51
52         print(bar)
53         print(" RESULT: " + ("PASS - all tests green" if rc == 0 else f"FAIL - pytest exit
{rc}"))
54         print(bar)
55         return rc
56
57
58     if __name__ == "__main__":
59         raise SystemExit(main())
60
[/external_agent_tool_result]

```

258. assistant (2026-07-02T06:11:34.081Z)

```

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"13","status":"completed"}
[/external_agent_tool_call]

```

259. assistant (2026-07-02T06:11:34.082Z)

```

[external_agent_tool_result]
Updated task #13 status
[/external_agent_tool_result]

```

260. assistant (2026-07-02T06:11:34.082Z)

```

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"14","status":"in_progress"}
[/external_agent_tool_call]

```

261. assistant (2026-07-02T06:11:34.082Z)

```

[external_agent_tool_result]
Updated task #14 status
[/external_agent_tool_result]

```

262. assistant (2026-07-02T06:11:34.082Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\infrastructure\database.py
[/external_agent_tool_call]

```

263. assistant (2026-07-02T06:11:34.082Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\infrastructure\database.py
has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

264. assistant (2026-07-02T06:11:34.082Z)

Now the job-store methods, appended after `clear\_video\_data`:

265. assistant (2026-07-02T06:11:34.082Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\infrastructure\database.py

```

[/external_agent_tool_call]

266. assistant (2026-07-02T06:11:34.082Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\infrastructure\database.py has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

267. assistant (2026-07-02T06:11:34.082Z)

Database layer done. Now the main.py surgery ? imports, executor, the endpoint swap, and startup recovery:

268. assistant (2026-07-02T06:11:34.082Z)

[external_agent_tool_call: TaskUpdate]

input: {"taskId":"14","status":"completed"}

[/external_agent_tool_call]

269. assistant (2026-07-02T06:11:34.082Z)

[external_agent_tool_result]

Updated task #14 status

[/external_agent_tool_result]

270. assistant (2026-07-02T06:11:34.082Z)

[external_agent_tool_call: TaskUpdate]

input: {"taskId":"15","status":"in_progress"}

[/external_agent_tool_call]

271. assistant (2026-07-02T06:11:34.082Z)

[external_agent_tool_result]

Updated task #15 status

[/external_agent_tool_result]

272. assistant (2026-07-02T06:11:34.082Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py

[/external_agent_tool_call]

273. assistant (2026-07-02T06:11:34.082Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

274. assistant (2026-07-02T06:11:34.082Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py

[/external_agent_tool_call]

275. assistant (2026-07-02T06:11:34.082Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

276. assistant (2026-07-02T06:11:34.082Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py
[/external_agent_tool_call]
```

277. assistant (2026-07-02T06:11:34.083Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

278. assistant (2026-07-02T06:11:34.083Z)

Now the core swap ? the blocking endpoint becomes submit + worker + poll:

279. assistant (2026-07-02T06:11:34.083Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py
[/external_agent_tool_call]
```

280. assistant (2026-07-02T06:11:34.083Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

281. assistant (2026-07-02T06:11:34.083Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py
[/external_agent_tool_call]
```

282. assistant (2026-07-02T06:11:34.083Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

283. assistant (2026-07-02T06:11:34.083Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py
[/external_agent_tool_call]
```

284. assistant (2026-07-02T06:11:34.083Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

285. assistant (2026-07-02T06:11:34.083Z)

Backend done. Now the static UI poll update (#16 explicitly requires it) ? locating the submit handler:

286. assistant (2026-07-02T06:11:34.083Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"api/process","path":"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend\\api\\static\\app.js","output_mode":"content","-n":true,"-B":8,"-A":40}
[/external_agent_tool_call]
```

287. assistant (2026-07-02T06:11:34.083Z)

```
[external_agent_tool_result]
79-     if (rawPlayers) {
80-         playerPool = rawPlayers.split(",").map(p => p.trim()).filter(p => p.length > 0);
81-         populatePlayerSelects(playerPool);
82-     }
83-
84-     showLoader("Analyzing Video Stream", "Running structural diagnostics, video focus
checks, and relevance validators...");
85-
86-     try {
87-         const response = await fetch("/api/process", {
88-             method: "POST",
89-             headers: { "Content-Type": "application/json" },
90-             body: JSON.stringify({
91-                 video_path: videoPath,
92-                 player_pool: playerPool
93-             })
94-         });
95-
96-         if (!response.ok) {
97-             const err = await response.json();
98-             throw new Error(err.detail || "Server failed to process the request.");
99-         }
100-
101-         const data = await response.json();
102-         currentVideoId = data.video_id;
103-
104-         // Render validation list
105-         renderValidationResults(data.results);
106-
107-         if (data.status === "failed") {
108-             hideLoader();
109-             alert("Validation Failed! This video does not meet our quality or structural
criteria. Please review the checklist.");
110-             renderEmptyRallies("Validation failed. Please load a valid badminton raw
video.");
111-             compileBoard.style.display = "none";
112-             return;
113-         }
114-
115-         // Successfully processed
116-         indexedRallies = data.rallies || [];
117-         selectedRallyIds.clear();
118-         renderRallyCards(indexedRallies);
119-
120-         hideLoader();
121-         alert(`Success! Successfully processed and indexed ${indexedRallies.length}
badminton rallies.`);
122-
123-     } catch (error) {
124-         hideLoader();
125-         alert(`Error: ${error.message}`);
126-         renderEmptyRallies("Error processing file.");
127-     }
[/external_agent_tool_result]
```

288. assistant (2026-07-02T06:11:34.083Z)


```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\api\static\app.js
[/external_agent_tool_call]
```

289. assistant (2026-07-02T06:11:34.083Z)

```
[external_agent_tool_result: error]
<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>
[/external_agent_tool_result]
```

290. assistant (2026-07-02T06:11:34.083Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\api\static\app.js
[/external_agent_tool_call]
```

291. assistant (2026-07-02T06:11:34.083Z)

```
[external_agent_tool_result]
60         opt2.value = player;
61         opt2.textContent = player;
62         queryReceiver.appendChild(opt2);
63     });
64 };
65
66 // Initialize player selects at start
67 populatePlayerSelects(playerPool);
68
69 // --- 1. RUN VIDEO PROCESSING & VALIDATIONS ---
70 btnProcess.addEventListener("click", async () => {
71     const videoPath = videoPathInput.value.trim();
72     if (!videoPath) {
73         alert("Please enter a valid local MP4 video file path.");
74         return;
75     }
76
77     // Parse custom players
78     const rawPlayers = playerPoolInput.value.trim();
79     if (rawPlayers) {
80         playerPool = rawPlayers.split(",").map(p => p.trim()).filter(p => p.length >
0);
81         populatePlayerSelects(playerPool);
82     }
83
84     showLoader("Analyzing Video Stream", "Running structural diagnostics, video
focus checks, and relevance validators...");
85
86     try {
87         const response = await fetch("/api/process", {
88             method: "POST",
89             headers: { "Content-Type": "application/json" },
90             body: JSON.stringify({
91                 video_path: videoPath,
92                 player_pool: playerPool
93             })
94         });
95
96         if (!response.ok) {
97             const err = await response.json();
98             throw new Error(err.detail || "Server failed to process the request.");
99         }
100
101         const data = await response.json();
102         currentVideoId = data.video_id;
```

```

103
104         // Render validation list
105         renderValidationResults(data.results);
106
107         if (data.status === "failed") {
108             hideLoader();
109             alert("Validation Failed! This video does not meet our quality or
structural criteria. Please review the checklist.");
110             renderEmptyRallies("Validation failed. Please load a valid badminton raw
video.");
111             compileBoard.style.display = "none";
112             return;
113         }
114
115         // Successfully processed
116         indexedRallies = data.rallies || [];
117         selectedRallyIds.clear();
118         renderRallyCards(indexedRallies);
119
120         hideLoader();
121         alert(`Success! Successfully processed and indexed ${indexedRallies.length}
badminton rallies.`);
122
123     } catch (error) {
124         hideLoader();
125         alert(`Error: ${error.message}`);
126         renderEmptyRallies("Error processing file.");
127     }
128 });
129
130 // --- 2. RENDER VALIDATION CHECKLIST ---
131 const renderValidationResults = (results) => {
132     validationList.innerHTML = "";
133
134     results.forEach(res => {
[/external_agent_tool_result]

```

292. assistant (2026-07-02T06:11:34.083Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\api\static\app.js
[/external_agent_tool_call]

```

293. assistant (2026-07-02T06:11:34.084Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\api\static\app.js has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

294. assistant (2026-07-02T06:11:34.084Z)

```

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"15","status":"completed"}
[/external_agent_tool_call]

```

295. assistant (2026-07-02T06:11:34.084Z)

```

[external_agent_tool_result]
Updated task #15 status
[/external_agent_tool_result]

```

296. assistant (2026-07-02T06:11:34.084Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"16","status":"in_progress"}
[/external_agent_tool_call]
```

297. assistant (2026-07-02T06:11:34.084Z)

```
[external_agent_tool_result]
Updated task #16 status
[/external_agent_tool_result]
```

298. assistant (2026-07-02T06:11:34.084Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_async_jobs.py
[/external_agent_tool_call]
```

299. assistant (2026-07-02T06:11:34.084Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_async_jobs.py (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

300. assistant (2026-07-02T06:11:34.084Z)

Now updating the existing API tests to the async contract:

301. assistant (2026-07-02T06:11:34.084Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py
[/external_agent_tool_call]
```

302. assistant (2026-07-02T06:11:34.084Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

303. assistant (2026-07-02T06:11:34.084Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py
[/external_agent_tool_call]
```

304. assistant (2026-07-02T06:11:34.084Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

305. assistant (2026-07-02T06:11:34.084Z)

Now installing the offline deps and running the new suite first, then the full suite:

306. assistant (2026-07-02T06:11:34.084Z)

```
[external_agent_tool_call: mcp__workspace__bash]
command: pip install -q --break-system-packages fastapi httpx pytest python-dotenv pydantic
uvicorn pyyaml 2>&1 | tail -1; echo READY
```

[/external_agent_tool_call]

307. assistant (2026-07-02T06:11:34.084Z)

[external_agent_tool_result: error]

UNC paths are not supported: \\wsl.localhost\ubuntu\home\avidullu\projects

[/external_agent_tool_result]

308. assistant (2026-07-02T06:11:34.084Z)

[external_agent_tool_call: mcp__workspace__bash]

command: echo ping && python3 --version

[/external_agent_tool_call]

309. assistant (2026-07-02T06:11:34.084Z)

[external_agent_tool_result: error]

UNC paths are not supported: \\wsl.localhost\ubuntu\home\avidullu\projects

[/external_agent_tool_result]

310. assistant (2026-07-02T06:11:34.085Z)

[external_agent_tool_call: mcp__cowork__request_cowork_directory]

input: {"path":"C:\\Users\\avidu\\Projects"}

[/external_agent_tool_call]

311. assistant (2026-07-02T06:11:34.085Z)

[external_agent_tool_result: error]

UNC paths are not supported: \\wsl.localhost\ubuntu\home\avidullu\projects

[/external_agent_tool_result]

312. assistant (2026-07-02T06:11:34.085Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DEFERRED_HARDENING_PLAN.md

[/external_agent_tool_call]

313. assistant (2026-07-02T06:11:34.085Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DEFERRED_HARDENING_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

314. assistant (2026-07-02T06:11:34.085Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md

[/external_agent_tool_call]

315. assistant (2026-07-02T06:11:34.085Z)

[external_agent_tool_result: error]

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

[/external_agent_tool_result]

316. assistant (2026-07-02T06:11:34.085Z)

[external_agent_tool_result: error]

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

[/external_agent_tool_result]

317. assistant (2026-07-02T06:11:34.085Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md
[/external_agent_tool_call]
```

318. assistant (2026-07-02T06:11:34.085Z)

CODE_MAP needs a fresh read in this context ? grabbing both anchor regions:

319. assistant (2026-07-02T06:11:34.085Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md
[/external_agent_tool_call]
```

320. assistant (2026-07-02T06:11:34.085Z)

```
[external_agent_tool_result]
83     - `@backend/main.py` ? Thin CLI + FastAPI bootstrap (orchestrator; modes via FLAGS).
Static UI mounts `/static` from `backend/api/static`. Logging is initialized with `basicConfig`
(INFO, timestamped format); backend modules use `logger = logging.getLogger(__name__)` (replaces
stray prints in hot paths like motion, stitcher, wasb_infer, config loader). User-facing CLI
summaries may still use `print` for direct output.
84     - `::app` FastAPI; `::db_instance`, `::global_config` (`Optional[AppConfig]`) module
globals set ONLY in `main()`. ? importing `app` for ASGI without `main()` -> both `None` ->
every endpoint NPEs.
85     - `::process_video` `@POST /api/process` -> validate -> add_video -> segment ->
`finally:` always `emit_indexer_report` (never raises). `::compile_highlights` `@POST
/api/compile`. `::query_play_segments` `@POST /api/query`. `::get_config` `@GET /api/config`.
`::run_cli_diagnose_clip` `--debug-clip`: lazy `import cv2`, samples ±3s vs
`indexing.motion_threshold` (dual model/dict read).
86     - `::load_config` ? ? legacy thin wrapper returning a **dict**
(`load_app_config(path).model_dump(...)`); kept for transition ? new code imports
`load_app_config`/`AppConfig` directly. `::ProcessRequest`/`::QueryRequest`/`::CompileRequest`
pydantic bodies.
87     - `main()` sets `global_config.output.output_dir` = args.output_dir` (typed field on
`OutputConfig`); `db_path` = os.path.join(global_config.output.output_dir, output.db_name)`.
`compile_highlights` reads the SAME `global_config.output.output_dir`, so `--output-dir` is
honored end-to-end (the old write-only `runtime_overrides` hack is gone).
88     - ? `--segmenter` argparse `choices` frozen at build time from
`segmenter_registry.list_available()` ? a lazily-registered segmenter wouldn't appear. CLI
default_segmenter fallback `motion`.
89     - `@run_process_and_stitch.py` ? one-shot demo; ? hardcoded paths, interactive `input()`
(blocks automation), skips validation. Config via `backend.config.load_config` (typed);
segmenter, padding, and db path all read from the model (no literal fallbacks).
90     - `@run_eval.py::main` ? score one video's DB preds vs GT CSV.
`--stage{candidates,final,both}`. Exit 2=arg/IO. ? pure scorer; errors if video never processed.
`::default_db_path` resolves via `backend.config.load_config`
(`output.output_dir`/`output.db_name`). `get_file_md5` is an alias for `compute_video_id`.
[/external_agent_tool_result]
```

321. assistant (2026-07-02T06:11:34.085Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md
[/external_agent_tool_call]
```

322. assistant (2026-07-02T06:11:34.085Z)

```
[external_agent_tool_result]
173
174     ### 2.7 Infrastructure (DB) ?
175     - `@backend/infrastructure/database.py::Database` ? SQLite, 4 tables
```

```
(`videos/play_segments/events/candidate_segments`), `PRAGMA user_version` migration ladder in
`:::_init_db` (currently =1, `version<1` creates `candidate_segments`). `:::_get_connection` (?
re-issues `PRAGMA foreign_keys=ON` per connection ? else CASCADE/SET NULL silently skipped).
Methods: `add_video`, `add_play_segment`, `add_event`, `add_candidate_segment`, `link_candidate_to_s
egment`, `query_candidate_segments(detector=,only_unlabeled=)`, `query_play_segments`, `get_video`,
`clear_video_data`. ? candidate_segments is the detector-agnostic fine-tuning extension point
(common fields columns, detector-specifics in `stats`/`detection_params` JSON; index
`idx_candidates_video_detector`). ? `add_video` is `INSERT OR REPLACE` -> cascade-DELETES
dependent segments on re-ingest; events column is `type` not `event_type`; `query_play_segments`
filters use truthiness so `duration_min=0`/empty-string = "no filter"; write helpers swallow
exceptions.
```

```
176
```

```
177     ### 2.8 Reporting / observability ? (per-video, soft-dep `obsreport`)
178     - `@backend/reporting/__init__.py` ? `::emit_indexer_report(*, db, video_id, video_path,
config, val_results, val_context=None, segmenter_requested=, segmenter_actual=,
was_reingest=False, segmentation_ran=True)` ? single entrypoint, keyword-only. Normalizes typed
config (`config.model_dump(...)` if `hasattr model_dump`); short-circuits to None if
unavailable/disabled; pulls play_segments/candidates from `db` (read-only
`query_play_segments(vid,{})`/`query_candidate_segments(vid)`); delegates to
`harvest.record_run`; writes via `rec.write(output_dir=config["report"]["output_dir"])`. ? wraps
everything in bare `except Exception: return None` ? **NEVER raises** (designed for `main.py`'s
`finally:`); writes `<video>.report.json`/`.report.md`/`.summary.md` next to video.
`::OBSREPORT_AVAILABLE` ? soft-dep gate (True only if `import obsreport` AND `from . import
harvest` both succeed). `::report_enabled(config)` reads `config["report"]["enabled"]` default
True.
```

```
[/external_agent_tool_result]
```

323. assistant (2026-07-02T06:11:34.085Z)

```
[external_agent_tool_call: Edit]
```

```
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md
```

```
[/external_agent_tool_call]
```

324. assistant (2026-07-02T06:11:34.085Z)

```
[external_agent_tool_result]
```

```
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
```

```
[/external_agent_tool_result]
```

325. assistant (2026-07-02T06:11:34.086Z)

```
[external_agent_tool_call: Edit]
```

```
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md
```

```
[/external_agent_tool_call]
```

326. assistant (2026-07-02T06:11:34.086Z)

```
[external_agent_tool_result]
```

```
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
```

```
[/external_agent_tool_result]
```

327. assistant (2026-07-02T06:11:34.086Z)

```
[external_agent_tool_call: Read]
```

```
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md
```

```
[/external_agent_tool_call]
```

328. assistant (2026-07-02T06:11:34.086Z)

```
[external_agent_tool_result]
```

```
1     # Next steps (live) ? owned-model + multisport
```

```
2
```

```
3      **Updated:** 2026-06-10 (end of the audit?M0?Gemini-battery session). **Authoritative
roadmap:**
4      `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-
blow:** memory
5      `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
6
7      ## Where we are (one paragraph)
8      ADR-008 ratified ("repo split driven by productionization, not isolation"): training
lives in-repo (`training/`)
[/external_agent_tool_result]
```

329. assistant (2026-07-02T06:11:34.086Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md
[/external_agent_tool_call]
```

330. assistant (2026-07-02T06:11:34.086Z)

```
[external_agent_tool_result]
53      (script, ground rules, Phase-2 consent rules; owner fills ?/contact placeholders).
54      6. **Housekeeping:** rotate the Gemini API key (was shared in chat); **rename repo off
"badminton" (owner action,
55      in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the
battery completed.
56
[/external_agent_tool_result]
```

331. assistant (2026-07-02T06:11:34.086Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md
[/external_agent_tool_call]
```

332. assistant (2026-07-02T06:11:34.086Z)

```
[external_agent_tool_result]
197
198      $ **play_segment dict** (`db.add_play_segment` / `query_play_segments`):
199      `{id:int, video_id, start_time:float, end_time:float, duration:float(end-start), server,
receiver, metadata:dict, events:[...]}`. Hybrid P4 metadata `{shot_count,
shot_count_confidence:'Unknown', detector:f'{family}-hybrid-{model}'}`; gemini detector = bare
`model_name`; motion adds random `intensity/avg_speed_kmh`.
200
201      $ **AI segment dict** (provider output, consumed by hybrid P3 / gemini): `{start_time,
end_time, shuttle_exchange_count?, shot_count_confidence?}` (offset by chunk/window start;
`_candidate_id` stamped internally).
202
[/external_agent_tool_result]
```

333. assistant (2026-07-02T06:11:34.086Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md
[/external_agent_tool_call]
```

334. assistant (2026-07-02T06:11:34.086Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

335. assistant (2026-07-02T06:11:34.086Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md
[/external_agent_tool_call]
```

336. assistant (2026-07-02T06:11:34.086Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

337. assistant (2026-07-02T06:11:34.086Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md
[/external_agent_tool_call]
```

338. assistant (2026-07-02T06:11:34.086Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

339. assistant (2026-07-02T06:11:34.087Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md
[/external_agent_tool_call]
```

340. assistant (2026-07-02T06:11:34.087Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md has
been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

341. assistant (2026-07-02T06:11:34.087Z)

One more try on the sandbox before I write the handoff ? if it healed, I run the suite:

342. assistant (2026-07-02T06:11:34.087Z)

```
[external_agent_tool_call: mcp__workspace__bash]
command: echo ping
[/external_agent_tool_call]
```

343. assistant (2026-07-02T06:11:34.087Z)

```
[external_agent_tool_result: error]
UNC paths are not supported: \\wsl.localhost\ubuntu\home\avidullu\projects
[/external_agent_tool_result]
```

344. assistant (2026-07-02T06:11:34.087Z)

Still wedged ? the handoff gets the honest test status then. Writing it now:

345. assistant (2026-07-02T06:11:34.087Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\PR1_HANDOFF.md
```



```
[/external_agent_tool_call]
```

346. assistant (2026-07-02T06:11:34.087Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\PR1_HANDOFF.md (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

347. assistant (2026-07-02T06:11:34.087Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"16","subject":"PR-1: Tests written (suite run blocked by sandbox failure ? handed to owner)","status":"completed"}
[/external_agent_tool_call]
```

348. assistant (2026-07-02T06:11:34.087Z)

```
[external_agent_tool_result]
Updated task #16 subject, status
[/external_agent_tool_result]
```

349. assistant (2026-07-02T06:11:34.087Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"17","status":"completed"}
[/external_agent_tool_call]
```

350. assistant (2026-07-02T06:11:34.088Z)

```
[external_agent_tool_result]
Updated task #17 status
[/external_agent_tool_result]
```

351. assistant (2026-07-02T06:11:34.088Z)

```
[external_agent_tool_call: mcp_cowork__present_files]
input: {"files":[{"file_path":"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\scratch\\PR1_HANDOFF.md"}, {"file_path":"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\tests\\test_async_jobs.py"}, {"file_path":"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend\\main.py"}]}
[/external_agent_tool_call]
```

352. assistant (2026-07-02T06:11:34.088Z)

```
[external_agent_tool_result]
C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\PR1_HANDOFF.md
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_async_jobs.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py
[/external_agent_tool_result]
```

353. assistant (2026-07-02T06:11:34.088Z)

PR-1 is built ? see you in the morning. The short version for your coffee:

****Shipped:**** the async job model (#16) exactly per spec ? `POST /api/process` now fast-fails 4xx then returns ****202 + job_id****, `GET /api/jobs/{id}` exposes `{status, progress, result, reports}`, the pipeline runs on a single-worker executor, job state lives in a new `jobs` table (`user\_version` 2), orphaned jobs get swept to `failed` on restart, and the static UI polls with live stage labels. Two bonus fixes that fell out naturally: raising segmenters now restore pre-run segments (destroy-after-confirm held only for `Failure` returns before), and the static UI's `rallies` field bug is dead ? rallies will actually render for the first time.

****The one caveat:**** my Linux sandbox died mid-session (that failed WSL mount wedged it permanently), so the 11 new tests + updated API tests are written but ****unexecuted**** ? ``python run_all_tests.py`` is step one in ``scratch/PR1_HANDOFF.md``, before you read any code. If anything's red, leave it for me.

The handoff doc has everything else: the 7-file change list, design decisions to review (job-status semantics being the main one), paste-ready branch/commit commands (code and planning-docs as two separate PRs per convention), the PR description, and the machine-side checklist ? including the spec's kill-the-server-mid-job test. Good night! ?

354. user (2026-07-02T06:11:34.088Z)

have you sent a PR for the same?

355. assistant (2026-07-02T06:11:34.088Z)

No ? there's no PR yet, and transparently, I couldn't send one: git can't write through my mount (the corruption issue from earlier), and the GitHub connector isn't authenticated, so I have no API path either. Right now PR-1 exists as uncommitted changes in your working tree, with the branch/commit/PR commands ready to paste from ``scratch/PR1_HANDOFF.md``.

Two ways to get the PR opened:

1. ****You (or Claude Code) run the two command blocks**** in the handoff ? ~1 minute, creates ``feat/async-job-model`` + the docs branch, then ``gh pr create`` (or push and click).
2. ****Authenticate the GitHub connector here**** (Connectors panel ? GitHub) ? then I can push the branch and open the PR myself via API, description and all, and that seam disappears for every future PR too.

Worth noting either way: the repo has CI (``github/workflows/ci.yml``), so the moment the PR opens, the offline suite runs there ? which neatly covers the "tests written but unexecuted" gap from last night even before you pull the branch locally.

356. user (2026-07-02T06:11:34.088Z)

I am happy to authenticate the github. Please ask!

357. assistant (2026-07-02T06:11:34.088Z)

```
[external_agent_tool_call: ToolSearch]
input: {"query":"select:mcp__plugin_engineering_github__authenticate,mcp__plugin_engineering_github__complete_authentication","max_results":3}
[/external_agent_tool_call]
```

358. assistant (2026-07-02T06:11:34.088Z)

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

359. assistant (2026-07-02T06:11:34.088Z)

```
[external_agent_tool_call: mcp__plugin_engineering_github__authenticate]
input: {}
[/external_agent_tool_call]
```

360. assistant (2026-07-02T06:11:34.088Z)

```
[external_agent_tool_result]
Failed to start OAuth flow for plugin:engineering:github: SDK auth failed: Incompatible auth server: does not support dynamic client registration. Ask the user to run /mcp and authenticate manually.
[/external_agent_tool_result]
```

361. assistant (2026-07-02T06:11:34.089Z)

The automatic OAuth flow isn't supported for this server ? it needs the manual route, which is quick:

Type `**`/mcp`**` in the chat input here, find `**github**` in the server list, and authenticate it from there (it'll open GitHub's authorization page in your browser).

Once you've done that, the GitHub tools light up on my side and I'll push the ``feat/async-job-model`` branch and open the PR with the prepared description right away ? just say "done" when you've authorized.

362. user (2026-07-02T06:11:34.089Z)

says unknown skill mcp!

363. assistant (2026-07-02T06:11:34.089Z)

Ah, my bad ? ``/mcp`` is a Claude Code command, not a Cowork one. In Cowork the equivalent lives in the app UI: `**Settings ? Connectors**` (or the connectors/plugin icon near the chat box) ? find `**GitHub**` ? Connect/Authorize. If you installed GitHub as part of the engineering plugin, it may show an "authenticate" button under the plugin's entry instead.

But honestly, the fastest path to a PR this morning doesn't need any of that ? let me make it copy-paste trivial:

364. assistant (2026-07-02T06:11:34.089Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\PR1_BODY.md
[/external_agent_tool_call]
```

365. assistant (2026-07-02T06:11:34.089Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\PR1_BODY.md (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

366. assistant (2026-07-02T06:11:34.089Z)

PR body written to a file. Here's the complete copy-paste block ? PR opened in under a minute:

```
...
cd C:\Users\avidu\Projects\badminton-highlight-indexer
git checkout -b feat/async-job-model
git add backend/infrastructure/database.py backend/main.py backend/api/static/app.js
tests/test_async_jobs.py tests/test_api_endpoints.py docs/DEFERRED_HARDENING_PLAN.md
docs/CODE_MAP.md
git commit -m "feat(api): async job model - 202+job_id, jobs table (uv2), worker, crash sweep (#16)"
git push -u origin feat/async-job-model
gh pr create --title "feat(api): async job model (#16) - 202+job_id, jobs table, worker, crash sweep" --body-file scratch/PR1_BODY.md
...
```

CI will run the full suite on the PR automatically ? that's the unexecuted-tests gap closed.

The planning-docs branch is a separate item (commands in ``scratch/PR1_HANDOFF.md`` `$"Separate docs commit"`) ? run it whenever, no rush.

And for the connector: if you find GitHub in Settings ? Connectors and authorize it, future PRs won't need your terminal at all. If it's not obvious where, send me a screenshot of your Settings and I'll point at the right spot.

367. user (2026-07-02T06:11:34.089Z)

it is authorized! But maybe I need to disconnect and reconnect. Would be great if Claude Code could take over! But let me to this right now

368. assistant (2026-07-02T06:11:34.089Z)

```
[external_agent_tool_call: ToolSearch]
input: {"query":"+github pull request create branch","max_results":10}
[/external_agent_tool_call]
```

369. assistant (2026-07-02T06:11:34.089Z)

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

370. assistant (2026-07-02T06:11:34.090Z)

On my side the GitHub server still shows only the auth stub ? your instinct is right: a disconnect/reconnect (or simply our next session) should make the real tools appear, since the authorization is done. Nothing blocked meanwhile ? you're pushing it yourself, which is faster anyway.

And yes, Claude Code taking over the git/merge flow fits the working agreement perfectly ? review and merge were always on your side of the seam; whether your fingers or Claude Code's do the typing is your call.

Once the PR is up, paste the CI result (or just the URL) if anything's red ? the suite ran written-but-unexecuted, so that's the one genuine unknown. If it's green, PR-1 is done pending your review, and PR-2 (chunked upload + download endpoint) is queued for my next "go".

371. user (2026-07-02T06:11:34.090Z)

<https://github.com/avidullu/badminton-highlight-indexer/pull/87> -- CI seems to be failing

372. assistant (2026-07-02T06:11:34.090Z)

Let me find it from here ? most likely suspect: other test files hitting `/api/process` that I didn't know about. Checking, plus what CI actually runs:

373. assistant (2026-07-02T06:11:34.090Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"api/process|api/jobs","path":"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\tests","output_mode":"content","-n":true,"head_limit":40}
[/external_agent_tool_call]
```

374. assistant (2026-07-02T06:11:34.090Z)

```
[external_agent_tool_result]
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py:69:      r =
client.post("/api/process", json={"video_path": str(vid_file), "segmenter": "x"})
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py:71:      job =
client.get(f"/api/jobs/{r.json()['job_id']}").json()
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py:84:      r =
client.post("/api/process", json={"video_path": str(vid_file)})
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py:86:      job =
client.get(f"/api/jobs/{r.json()['job_id']}").json()
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py:95:      r =
client.post("/api/process", json={"video_path": "a\x00b.mp4"})
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_async_jobs.py:4:- POST
/api/process ? 202 + job_id (fast-fail 4xx still happens at submit time)
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_async_jobs.py:5:- GET
/api/jobs/{id} ? {status, progress, result, reports, ...}
```

```

C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_async_jobs.py:111: r =
client.post("/api/process", json={"video_path": str(vid_file), "segmenter": "x"})
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_async_jobs.py:121: r =
client.post("/api/process", json={"video_path": "a\x00b.mp4"})
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_async_jobs.py:127: r =
client.post("/api/process", json={"video_path": str(tmp_path / "nope.mp4")})
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_async_jobs.py:136: r =
client.post("/api/process", json={"video_path": str(vid_file), "segmenter": "nope"})
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_async_jobs.py:152:
"/api/process", json={"video_path": str(vid_file), "segmenter": "x"}).json()["job_id"]
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_async_jobs.py:153: r =
client.get(f"/api/jobs/{job_id}")
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_async_jobs.py:177: job_id =
client.post("/api/process", json={"video_path": str(vid_file)}).json()["job_id"]
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_async_jobs.py:178: body =
client.get(f"/api/jobs/{job_id}").json()
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_async_jobs.py:196:
"/api/process", json={"video_path": str(vid_file), "segmenter": "x"}).json()["job_id"]
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_async_jobs.py:197: body =
client.get(f"/api/jobs/{job_id}").json()
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_async_jobs.py:220:
client.post("/api/process", json={"video_path": str(vid_file), "segmenter": "x"})
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_async_jobs.py:229: assert
client.get("/api/jobs/deadbeef").status_code == 404
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_async_jobs.py:261:
"/api/process", json={"video_path": str(vid_file), "segmenter": "x"}).json()["job_id"]
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_async_jobs.py:265:
body = client.get(f"/api/jobs/{job_id}").json()
[/external_agent_tool_result]

```

375. assistant (2026-07-02T06:11:34.090Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.github\workflows\ci.yml
[/external_agent_tool_call]

```

376. assistant (2026-07-02T06:11:34.090Z)

```

[external_agent_tool_result]
1     name: CI
2
3     on:
4         push:
5             branches: [master]
6         pull_request:
7
8     permissions:
9         contents: read
10
11    concurrency:
12        group: ci-${{ github.ref }}
13        cancel-in-progress: true
14
15    jobs:
16        # Fast guard against the 7fbbeb0 class of breakage: a SyntaxError or
17        # module-level import error shipping while the suite stays green.
18        # Deliberately does NOT install cv2/torch/numpy ? the smoke tests must hold
19        # on machines without the GPU stack (test_import_smoke stubs what's absent).
20        import-smoke:
21            name: Syntax sweep + import smoke (no GPU stack)
22            runs-on: ubuntu-latest
23            steps:
24                - uses: actions/checkout@v4
25                - uses: actions/setup-python@v5
26                with:

```

```

27         python-version: "3.13"
28     - name: Install minimal deps
29       run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-
genai
30     - name: Run import smoke tests
31       run: python -m pytest tests/test_import_smoke.py -v
32
33     full-suite:
34       name: Full test suite (offline, mocked)
35       runs-on: ubuntu-latest
36       steps:
37         - uses: actions/checkout@v4
38         - uses: actions/setup-python@v5
39         with:
40           python-version: "3.13"
41           cache: pip
42         - name: System libs for opencv-python
43           run: |
44             sudo apt-get update
45             sudo apt-get install -y libgl1 libglib2.0-0t64 || sudo apt-get install -y
libgl1 libglib2.0-0
46         - name: Install CPU-only torch (no CUDA wheels on a CPU runner)
47           run: python -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
48         - name: Install requirements
49           run: python -m pip install -r requirements.txt pytest httpx # httpx: fastapi
TestClient
50         # obsreport (private, soft dep) is absent here: reporting tests skip via
51         # pytest.importorskip ? run_all_tests.py prints which.
52         - name: Run full suite
53           run: python run_all_tests.py
54
[/external_agent_tool_result]

```

377. assistant (2026-07-02T06:11:34.090Z)

```

[external_agent_tool_call: ToolSearch]
input: {"query":"chrome","max_results":20}
[/external_agent_tool_call]

```

378. assistant (2026-07-02T06:11:34.091Z)

```

[external_agent_tool_result]
[/external_agent_tool_result]

```

379. assistant (2026-07-02T06:11:34.091Z)

```

[external_agent_tool_call: mcp_Claude_in_Chrome__tabs_context_mcp]
input: {"createIfEmpty":true}
[/external_agent_tool_call]

```

380. assistant (2026-07-02T06:11:34.091Z)

```

[external_agent_tool_result]
{"availableTabs":[{"tabId":2039717955,"title":"New
Tab","url":"chrome://newtab/"}],"tabGroupId":543115371}

```

Tab Context:

- Available tabs:

? tabId 2039717955: "New Tab" (chrome://newtab/)

<system-reminder>You used a single tool call this turn. Prefer browser_batch to execute multiple actions in one call ? it is significantly faster. Batch your next sequence of clicks, types, navigations, and screenshots together.</system-reminder>

[/external_agent_tool_result]

381. assistant (2026-07-02T06:11:34.091Z)

```
[external_agent_tool_call: mcp__Claude_in_Chrome__browser_batch]
input: {"actions":[{"name":"navigate","input":{"url":"https://github.com/avidullu/badminton-highlight-indexer/pull/87/checks","tabId":2039717955}},{"name":"get_page_text","input":{"tabId":2039717955}}]}
[/external_agent_tool_call]
```

382. assistant (2026-07-02T06:11:34.091Z)

```
[external_agent_tool_result]
[navigate] Navigated to https://github.com/avidullu/badminton-highlight-indexer/pull/87/checks
[get_page_text] Title: feat(api): async job model (#16) - 202+job_id, jobs table, worker, crash sweep by avidullu · Pull Request #87 · avidullu/badminton-highlight-indexer
URL: https://github.com/avidullu/badminton-highlight-indexer/pull/87/checks
Source element: <main>
---
feat(api): async job model (#16) - 202+job_id, jobs table, worker, crash sweep
#87
Able to merge
Code
Open
avidullu
wants to merge 1 commit into
master
from
feat/async-job-model
+603
-31
Lines changed: 603 additions & 31 deletions
Conversation0
(0)
Commits1
(1)
Checks2
(2)
Files changed7
(7)
feat(api): async job model - 202+job_id, jobs table (uv2), worker, cr?
2d27dd8
CI
on: pull_request
3
Select a check to view from the sidebar
```

Tab Context:

```
- Executed on tabId: 2039717955
- Available tabs:
  ? tabId 2039717955: "feat(api): async job model (#16) - 202+job_id, jobs table, worker, crash sweep by avidullu · Pull Request #87 · avidullu/badminton-highlight-indexer"
(https://github.com/avidullu/badminton-highlight-indexer/pull/87/checks)
[/external_agent_tool_result]
```

383. assistant (2026-07-02T06:11:34.091Z)

```
[external_agent_tool_call: mcp__Claude_in_Chrome__find]
input: {"query":"failing check job in the left sidebar (red X icon)","tabId":2039717955}
[/external_agent_tool_call]
```

384. assistant (2026-07-02T06:11:34.091Z)

```
[external_agent_tool_result]
Found 1 matching element
```

- ref_195: img ""This job failed"" (image) - This is the red X icon (failed status) in the left sidebar under "Check suites" section, indicating a failing check job. It's associated with the "Full test suite (offline, mocked)" job link [ref_196].

Tab Context:

```
- Executed on tabId: 2039717955
- Available tabs:
  ? tabId 2039717955: "feat(api): async job model (#16) - 202+job_id, jobs table, worker, crash
sweep by avidullu · Pull Request #87 · avidullu/badminton-highlight-indexer"
(https://github.com/avidullu/badminton-highlight-indexer/pull/87/checks)
[/external_agent_tool_result]
```

385. assistant (2026-07-02T06:11:34.091Z)

Import smoke passed; **full-suite** is the red one. Opening its log:

386. assistant (2026-07-02T06:11:34.091Z)

```
[external_agent_tool_call: mcp_Claude_in_Chrome_browser_batch]
input: {"actions":[{"name":"computer","input":{"action":"left_click","ref":"ref_196","tabId":203
9717955}},{"name":"computer","input":{"action":"wait","duration":3,"tabId":2039717955}},{"name":
"get_page_text","input":{"tabId":2039717955}}]}
[/external_agent_tool_call]
```

387. assistant (2026-07-02T06:11:34.091Z)

```
[external_agent_tool_result]
[computer:left_click] Clicked on element ref_196
[computer:wait] Waited for 3 seconds
[get_page_text] Title: feat(api): async job model (#16) - 202+job_id, jobs table, worker, crash
sweep by avidullu · Pull Request #87 · avidullu/badminton-highlight-indexer
URL: https://github.com/avidullu/badminton-highlight-indexer/pull/87/checks
Source element: <main>
---
feat(api): async job model (#16) - 202+job_id, jobs table, worker, crash sweep
#87
Able to merge
Code
Open
avidullu
wants to merge 1 commit into
master
from
feat/async-job-model
+603
-31
Lines changed: 603 additions & 31 deletions
Conversation0
(0)
Commits1
(1)
Checks2
(2)
Files changed7
(7)
feat(api): async job model - 202+job_id, jobs table (uv2), worker, cr?
2d27dd8
CI
on: pull_request
3
```


Select a check to view from the sidebar

Tab Context:

```
- Executed on tabId: 2039717955
- Available tabs:
  ? tabId 2039717955: "feat(api): async job model (#16) - 202+job_id, jobs table, worker, crash
sweep by avidullu · Pull Request #87 · avidullu/badminton-highlight-indexer"
(https://github.com/avidullu/badminton-highlight-indexer/pull/87/checks)
[/external_agent_tool_result]
```

388. assistant (2026-07-02T06:11:34.091Z)

```
[external_agent_tool_call: mcp__Claude_in_Chrome__javascript_tool]
input: {"action":"javascript_exec","tabId":2039717955,"text":"Array.from(document.querySelectorAll('a')).filter(a => a.textContent.includes('Full test suite')).map(a => a.href).join('\n')}
[/external_agent_tool_call]
```

389. assistant (2026-07-02T06:11:34.091Z)

```
[external_agent_tool_result]
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27297748855/job/80634917956?pr=87
```

Tab Context:

```
- Executed on tabId: 2039717955
- Available tabs:
  ? tabId 2039717955: "feat(api): async job model (#16) - 202+job_id, jobs table, worker, crash
sweep by avidullu · Pull Request #87 · avidullu/badminton-highlight-indexer"
(https://github.com/avidullu/badminton-highlight-indexer/pull/87/checks)
[/external_agent_tool_result]
```

390. assistant (2026-07-02T06:11:34.092Z)

```
[external_agent_tool_call: mcp__Claude_in_Chrome__browser_batch]
input: {"actions":[{"name":"navigate","input":{"url":"https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27297748855/job/80634917956?pr=87","tabId":2039717955}},{"name":"computer","input":{"action":"wait","duration":4,"tabId":2039717955}},{"name":"get_page_text","input":{"tabId":2039717955}}]}
[/external_agent_tool_call]
```

391. assistant (2026-07-02T06:11:34.092Z)

```
[external_agent_tool_result]
[navigate] Navigated to https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27297748855/job/80634917956?pr=87
[computer:wait] Waited for 4 seconds
[get_page_text] Title: feat(api): async job model (#16) - 202+job_id, jobs table, worker, crash sweep · avidullu/badminton-highlight-indexer@2d27dd8
URL: https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27297748855/job/80634917956?pr=87
Source element: <main>
---
Back to pull request #87
feat(api): async job model (#16) - 202+job_id, jobs table, worker, crash sweep #49
Re-run jobs
Annotations
1 error and 1 warning
Full test suite (offline, mocked)
failed in 1m 25s
Explain error
Set up job
1s
```

```
Run actions/checkout@v4
1s
Run actions/setup-python@v5
4s
System libs for opencv-python
12s
Install CPU-only torch (no CUDA wheels on a CPU runner)
26s
Install requirements
16s
Run full suite
21s
1
Run python run_all_tests.py

11
=====

12
Full test suite: /opt/hostedtoolcache/Python/3.13.13/x64/bin/python -m pytest -q

13
obsreport (optional reporting dep): NOT installed ? reporting tests will SKIP

14
=====

15
..... [ 21%]

16
FF..... [ 43%]

17
..... [ 64%]

18
..... [ 86%]

19
..... [100%]

20
===== FAILURES =====

21
_____ test_schema_version_is_current _____

22

23
tmp_path = PosixPath('/tmp/pytest-of-runner/pytest-0/test_schema_version_is_current0')

24

25
def test_schema_version_is_current(tmp_path):

26
    db_path = tmp_path / "test.db"

27
    Database(str(db_path))

28
```

```
conn = sqlite3.connect(str(db_path))
```

```
29
version = conn.execute("PRAGMA user_version").fetchone()[0]
```

```
30
conn.close()
```

```
31
> assert version == 1
```

```
32
E assert 2 == 1
```

```
33
```

```
34
tests/test_database.py:87: AssertionError
```

```
35
_____ test_init_db_idempotent_and_upgrades_old_db _____
```

```
36
```

```
37
tmp_path = PosixPath('/tmp/pytest-of-runner/pytest-0/test_init_db_idempotent_and_up0')
```

```
38
```

```
39
def test_init_db_idempotent_and_upgrades_old_db(tmp_path):
```

```
40
    db_path = tmp_path / "legacy.db"
```

```
41
```

Simulate a pre-v1 database that only has the old tables and no version.

```
42
conn = sqlite3.connect(str(db_path))
```

```
43
conn.execute("CREATE TABLE videos (id TEXT PRIMARY KEY, filepath TEXT, status TEXT)")
```

```
44
conn.commit()
```

```
45
conn.close()
```

```
46
```

```
47
```

Opening with the new Database should add candidate_segments without error...

```
48
Database(str(db_path))
```

```
49
```

...and running init again must be a no-op (idempotent).

```
50
Database(str(db_path))
```

```

51
52
conn = sqlite3.connect(str(db_path))

53
tables = [r[0] for r in conn.execute(

54
"SELECT name FROM sqlite_master WHERE type='table'").fetchall()]

55
version = conn.execute("PRAGMA user_version").fetchone()[0]

56
conn.close()

57
assert "candidate_segments" in tables

58
> assert version == 1

59
E assert 2 == 1

60

61
tests/test_database.py:109: AssertionError

62
===== warnings summary =====

63
../../../../opt/hostedtoolcache/Python/3.13.13/x64/lib/python3.13/site-
packages/fastapi/testclient.py:1

64
/opt/hostedtoolcache/Python/3.13.13/x64/lib/python3.13/site-packages/fastapi/testclient.py:1:
StarletteDeprecationWarning: Using `httpx` with `starlette.testclient` is deprecated; install
`httpx2` instead.

65
from starlette.testclient import TestClient as TestClient # noqa

66

67
tests/test_api_endpoints.py::test_process_success_with_mocked_segmenter

68
tests/test_async_jobs.py::test_submit_returns_202_with_job_id

69
tests/test_async_jobs.py::test_job_done_with_se?
[/external_agent_tool_result]

```

392. assistant (2026-07-02T06:11:34.092Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_database.py
[/external_agent_tool_call]

```

393. assistant (2026-07-02T06:11:34.092Z)

```

[external_agent_tool_result]
78         assert by_name[col][3] == 1, f"{col} should be NOT NULL"
79
80
81     def test_schema_version_is_current(tmp_path):
82         db_path = tmp_path / "test.db"
83         Database(str(db_path))
84         conn = sqlite3.connect(str(db_path))
85         version = conn.execute("PRAGMA user_version").fetchone()[0]
86         conn.close()
87         assert version == 1
88
89
90     def test_init_db_idempotent_and_upgrades_old_db(tmp_path):
91         db_path = tmp_path / "legacy.db"
92         # Simulate a pre-v1 database that only has the old tables and no version.
93         conn = sqlite3.connect(str(db_path))
94         conn.execute("CREATE TABLE videos (id TEXT PRIMARY KEY, filepath TEXT, status
TEXT)")
95         conn.commit()
96         conn.close()
97
98         # Opening with the new Database should add candidate_segments without error...
99         Database(str(db_path))
100        # ...and running init again must be a no-op (idempotent).
101        Database(str(db_path))
102
103        conn = sqlite3.connect(str(db_path))
104        tables = [r[0] for r in conn.execute(
105            "SELECT name FROM sqlite_master WHERE type='table'").fetchall()]
106        version = conn.execute("PRAGMA user_version").fetchone()[0]
107        conn.close()
108        assert "candidate_segments" in tables
109        assert version == 1
110
111
112    def test_candidate_round_trip(tmp_path):
[/external_agent_tool_result]

```

394. assistant (2026-07-02T06:11:34.092Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_database.py
[/external_agent_tool_call]

```

395. assistant (2026-07-02T06:11:34.092Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_database.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

396. assistant (2026-07-02T06:11:34.092Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_database.py
[/external_agent_tool_call]

```

397. assistant (2026-07-02T06:11:34.092Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_database.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

398. assistant (2026-07-02T06:11:34.092Z)

That was the whole failure ? the repo's intentional migration tripwire (`database.py`'s comment says it outright: "Tests assert the expected version, so accidental drift fails CI"). It caught my deliberate bump, as designed. The happy news from the same log: ****all 331 other tests passed****, including every one of the 11 new async-job tests and the cross-thread executor test ? so the "written but unexecuted" risk is fully retired.

I've updated the two pins (and strengthened the upgrade test to also assert the `jobs` table exists). You're one push away from green ? on the `feat/async-job-model` branch:

```
...  
cd C:\Users\avidu\Projects\badminton-highlight-indexer  
git add tests/test_database.py  
git commit -m "test(db): bump schema-version pin to v2 (jobs table, #16)"  
git push  
...
```

CI will re-run on the push. Once it's green, PR-1 is review-ready ? design notes are in the PR body and `scratch/PR1\_HANDOFF.md`, and the machine-side checklist (real video, kill-mid-job restart test) is the part that still wants your GPU box before merge.

399. user (2026-07-02T06:11:34.092Z)

PR has been merged. Can I follow a task list for all this?

400. assistant (2026-07-02T06:11:34.092Z)

Merged ? PR-1 is in! ? And yes, there's exactly one list to follow: the master checklist in `docs/UI\_ALPHA\_EXECUTION\_PLAN.md`. Syncing it to reality first (merge, the given "go", the done key rotation):

401. assistant (2026-07-02T06:11:34.092Z)

```
[external_agent_tool_call: Edit]  
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md  
[/external_agent_tool_call]
```

402. assistant (2026-07-02T06:11:34.092Z)

```
[external_agent_tool_result]  
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md has  
been updated successfully. (file state is current in your context ? no need to Read it back)  
[/external_agent_tool_result]
```

403. assistant (2026-07-02T06:11:34.092Z)

```
[external_agent_tool_call: Edit]  
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md  
[/external_agent_tool_call]
```

404. assistant (2026-07-02T06:11:34.092Z)

```
[external_agent_tool_result]  
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md has  
been updated successfully. (file state is current in your context ? no need to Read it back)  
[/external_agent_tool_result]
```

405. assistant (2026-07-02T06:11:34.092Z)

```
[external_agent_tool_call: Edit]  
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DEFERRED_HARDENING_PLAN.md  
[/external_agent_tool_call]
```

406. assistant (2026-07-02T06:11:34.093Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DEFERRED_HARDENING_PLAN.md has
been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

407. assistant (2026-07-02T06:11:34.093Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md
[/external_agent_tool_call]
```

408. assistant (2026-07-02T06:11:34.093Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

409. assistant (2026-07-02T06:11:34.093Z)

```
[external_agent_tool_call: mcp_cowork__present_files]
input: {"files":[{"file_path":"C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\docs\\UI_ALPHA_EXECUTION_PLAN.md"}]}
[/external_agent_tool_call]
```

410. assistant (2026-07-02T06:11:34.093Z)

```
[external_agent_tool_result]
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md
[/external_agent_tool_result]
```

411. assistant (2026-07-02T06:11:34.093Z)

One housekeeping note: the planning docs (proposal, hosting plan, execution plan, ADR-009, mockup, NEXT_STEPS) are still uncommitted in your working tree ? PR #87 only carried the 7 code files. Whenever convenient:

```
...
git checkout master && git pull
git checkout -b docs/khelsutra-ui-alpha-plan
git add docs/UI_PROPOSAL.md docs/HOSTING_PLAN.md docs/UI_ALPHA_EXECUTION_PLAN.md docs/mockups
docs/archives/decisions/DECISIONS.md docs/NEXT_STEPS.md docs/DEFERRED_HARDENING_PLAN.md
docs/CODE_MAP.md
git commit -m "docs: KhelSutra UI alpha - proposal, hosting plan, execution plan, ADR-009 + PR-1
status sync"
git push -u origin docs/khelsutra-ui-alpha-plan && gh pr create --fill
...
```

And that's the rhythm from here: the doc is the map, the session widget is the live ticker. Say "go for PR-2" whenever you're ready.

412. user (2026-07-02T06:11:34.093Z)

```
I got "\badminton-highlight-indexer> git push -u origin docs/khelsutra-ui-alpha-plan && gh pr
create --fill
At line:1 char:49
+ git push -u origin docs/khelsutra-ui-alpha-plan && gh pr create --fil ...
+
The token '&&' is not a valid statement separator in this version.
+ CategoryInfo          : ParserError: (:) [], ParentContainsErrorRecordException
+ FullyQualifiedErrorId : InvalidEndOfLine"
```

413. assistant (2026-07-02T06:11:34.093Z)

Ah, that's Windows PowerShell 5.x ? it doesn't understand `&&` (only PowerShell 7+ does). Just run them as separate lines:

```
```\ngit push -u origin docs/khelsutra-ui-alpha-plan\ngit pr create --fill\n```\n
```

(Or join with `;` instead of `&&` in PowerShell.) I'll keep future command blocks `&&`-free for you.

**414. assistant (2026-07-02T06:11:34.093Z)**

```
[external_agent_tool_call: Write]\nfile: C:\\Users\\avidu\\AppData\\Roaming\\Claude\\local-agent-mode-sessions\\87f748ce-e72b-44f3-8ef0-78e3212a99c3\\b4180250-5146-4b86-9001-c9f3dd74258c\\spaces\\313cf772-aa7b-42d3-8233-50773dd063e6\\memory\\avi-windows-powershell.md\n[/external_agent_tool_call]\n
```

**415. assistant (2026-07-02T06:11:34.093Z)**

```
[external_agent_tool_result]\nFile created successfully at: C:\\Users\\avidu\\AppData\\Roaming\\Claude\\local-agent-mode-sessions\\87f748ce-e72b-44f3-8ef0-78e3212a99c3\\b4180250-5146-4b86-9001-c9f3dd74258c\\spaces\\313cf772-aa7b-42d3-8233-50773dd063e6\\memory\\avi-windows-powershell.md (file state is current in your context ? no need to Read it back)\n[/external_agent_tool_result]\n
```

**416. assistant (2026-07-02T06:11:34.093Z)**

```
[external_agent_tool_call: Edit]\nfile: C:\\Users\\avidu\\AppData\\Roaming\\Claude\\local-agent-mode-sessions\\87f748ce-e72b-44f3-8ef0-78e3212a99c3\\b4180250-5146-4b86-9001-c9f3dd74258c\\spaces\\313cf772-aa7b-42d3-8233-50773dd063e6\\memory\\MEMORY.md\n[/external_agent_tool_call]\n
```

**417. assistant (2026-07-02T06:11:34.093Z)**

```
[external_agent_tool_result]\nThe file C:\\Users\\avidu\\AppData\\Roaming\\Claude\\local-agent-mode-sessions\\87f748ce-e72b-44f3-8ef0-78e3212a99c3\\b4180250-5146-4b86-9001-c9f3dd74258c\\spaces\\313cf772-aa7b-42d3-8233-50773dd063e6\\memory\\MEMORY.md has been updated successfully. (file state is current in your context ? no need to Read it back)\n[/external_agent_tool_result]\n
```

**418. assistant (2026-07-02T06:11:34.093Z)**

Noted permanently ? all future command blocks will be PowerShell-friendly. Once that push + `git pr create --fill` goes through, the docs PR is up and the whole paper trail (proposal ? hosting ? execution plan ? ADR-009) is in the repo where it belongs. Then we're fully caught up: next move is yours ? "go for PR-2" starts the upload/download slice.

**419. user (2026-07-02T06:11:34.093Z)**

Go for Pr2

**420. assistant (2026-07-02T06:11:34.094Z)**

No response requested.

**421. user (2026-07-02T06:11:34.094Z)**



Igot merrge conflict in it  
<https://github.com/avidullu/badminton-highlight-indexer/pull/88>

**422. assistant (2026-07-02T06:11:34.094Z)**

Let me look at #88 first to confirm what's conflicting:

**423. assistant (2026-07-02T06:11:34.094Z)**

```
[external_agent_tool_call: mcp__Claude_in_Chrome__browser_batch]
input: {"actions":[{"name":"navigate","input":{"url":"https://github.com/avidullu/badminton-highlight-indexer/pull/88","tabId":2039717955}},{"name":"computer","input":{"action":"wait","duration":3,"tabId":2039717955}},{"name":"get_page_text","input":{"tabId":2039717955}}]}
[/external_agent_tool_call]
```

**424. assistant (2026-07-02T06:11:34.094Z)**

```
[external_agent_tool_result: error]
Tab 2039717955 no longer exists. Call tabs_context_mcp to get current tabs.
[/external_agent_tool_result]
```

**425. assistant (2026-07-02T06:11:34.094Z)**

```
[external_agent_tool_call: ToolSearch]
input: {"query":"+github","max_results":30}
[/external_agent_tool_call]
```

**426. assistant (2026-07-02T06:11:34.094Z)**

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

**427. assistant (2026-07-02T06:11:34.094Z)**

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\PR88_FIX.md
[/external_agent_tool_call]
```

**428. assistant (2026-07-02T06:11:34.094Z)**

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\PR88_FIX.md (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

**429. assistant (2026-07-02T06:11:34.094Z)**

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md
[/external_agent_tool_call]
```

**430. assistant (2026-07-02T06:11:34.094Z)**

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

**431. assistant (2026-07-02T06:11:34.094Z)**

Done. Here's the new world, no copy-paste in it:

**\*\*Right now:\*\*** tell Claude Code ? **\*\*"run scratch/PR88\_FIX.md"**. That recreates the docs branch

conflict-free off master (the conflict was my sequencing miss: two files overlapped #87, so they're dropped from the docs PR and will ride with PR-2), opens the new PR, and closes #88.

**\*\*Going forward:\*\*** the protocol is now written into the execution plan's working agreement ? I design/build/test and leave a `PRn\_HANDOFF.md`; Claude Code does branch/commit/push/PR/merge from it. Your role shrinks to reviewing PRs and saying "go".

**\*\*The connector:\*\*** if you want me holding the git seam directly someday, the likely fix is Settings ? disconnect GitHub ? reconnect ? start a fresh chat with me; I'll check at session start and take over if the tools light up. But the Claude Code split works fine indefinitely ? no pressure.

**\*\*PR-2:\*\*** your "go" is logged. This session is carrying a lot of context, so I'll build it at the start of our next one (cold-start is cheap ? that's what CODE\_MAP and the plan are for): chunked upload + streaming download, tests, docs, and `scratch/PR2\_HANDOFF.md` waiting for Claude Code. Deal?

**432. user (2026-07-02T06:11:34.094Z)**

good call on the context! See you next session!

**433. assistant (2026-07-02T06:11:34.094Z)**

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\session-handoff-2026-06-11.md
[/external_agent_tool_call]
```

**434. assistant (2026-07-02T06:11:34.095Z)**

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\session-handoff-2026-06-11.md (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

**435. assistant (2026-07-02T06:11:34.095Z)**

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\MEMORY.md
[/external_agent_tool_call]
```

**436. assistant (2026-07-02T06:11:34.095Z)**

```
[external_agent_tool_result]
The file C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\MEMORY.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

**437. assistant (2026-07-02T06:11:34.095Z)**

It was a fun one ? KhelSutra Guru went from a domain receipt to a named product with a face: mockup, plans, ADR, its own private repo, and the first real engine change merged underneath it. Handoff state is saved on my side, so next session opens with the connector check and goes straight into PR-2.

Rest well ? the face gets its upload hands next time. ?

**438. user (2026-07-02T06:11:34.095Z)**

hello. Can we resume the work today about the UI? As per our last chat, is it possible that Claude Code and you can collaborate rather than me copy pasting? PR-2 was where we left off last time and maybe we can resume from there? I am going for a amazing badminton session tomorrow morning and would love to use the UI to run through the CUIs! No rush though! I prefer slow and steady rather than fast and rashy!

**439. assistant (2026-07-02T06:11:34.095Z)**

```
[external_agent_tool_call: ToolSearch]
input: {"query":"+github","max_results":30}
[/external_agent_tool_call]
```

**440. assistant (2026-07-02T06:11:34.095Z)**

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

**441. assistant (2026-07-02T06:11:34.095Z)**

```
[external_agent_tool_call: mcp__workspace__bash]
command: cd /sessions/great-sharp-gauss/mnt/badminton-highlight-indexer && git log --oneline -8
&& echo "---UNTRACKED+MODIFIED (real state via diff-sym check)---" && git status --short | grep
-v "^ M" | head -20 && echo "---sandbox alive---" && python3 --version
[/external_agent_tool_call]
```

**442. assistant (2026-07-02T06:11:34.095Z)**

```
[external_agent_tool_result: error]
UNC paths are not supported: \\wsl.localhost\ubuntu\home\avidullu\projects
[/external_agent_tool_result]
```

**443. assistant (2026-07-02T06:11:34.095Z)**

```
[external_agent_tool_call: TaskCreate]
description: Read SecurityConfig + utils/paths.py; design chunked upload (init/part/complete) +
download endpoint.
[/external_agent_tool_call]
```

**444. assistant (2026-07-02T06:11:34.095Z)**

```
[external_agent_tool_result]
Task #18 created successfully: PR-2: Read config/paths code + design upload protocol
[/external_agent_tool_result]
```

**445. assistant (2026-07-02T06:11:34.095Z)**

```
[external_agent_tool_call: TaskCreate]
description: POST /api/upload/init, PUT part, POST complete ? assemble, MD5, register under
upload dir; caps + extension allowlist; typed config fields.
[/external_agent_tool_call]
```

**446. assistant (2026-07-02T06:11:34.095Z)**

```
[external_agent_tool_result]
Task #19 created successfully: PR-2: Implement upload endpoints + config (B2)
[/external_agent_tool_result]
```

**447. assistant (2026-07-02T06:11:34.095Z)**

```
[external_agent_tool_call: TaskCreate]
description: GET /api/highlights/{video_id}/{filename} streaming with safe-name validation;
```

```
app.js compile success ? download link.
[/external_agent_tool_call]
```

#### 448. assistant (2026-07-02T06:11:34.095Z)

```
[external_agent_tool_result]
Task #20 created successfully: PR-2: Implement download endpoint (B3) + static UI download link
[/external_agent_tool_result]
```

#### 449. assistant (2026-07-02T06:11:34.095Z)

```
[external_agent_tool_call: TaskCreate]
description: tests/test_upload_download.py covering init/part/complete, caps, ext allowlist,
order, traversal, 404s, stream success; run suite if sandbox heals.
[/external_agent_tool_call]
```

#### 450. assistant (2026-07-02T06:11:34.096Z)

```
[external_agent_tool_result]
Task #21 created successfully: PR-2: Tests + suite run (or CI fallback)
[/external_agent_tool_result]
```

#### 451. assistant (2026-07-02T06:11:34.096Z)

```
[external_agent_tool_call: TaskCreate]
description: CODE_MAP + contracts update; handoff with branch/PR steps (includes folding in last
session's uncommitted doc status-sync files); CUJ-for-tomorrow notes.
[/external_agent_tool_call]
```

#### 452. assistant (2026-07-02T06:11:34.096Z)

```
[external_agent_tool_result]
Task #22 created successfully: PR-2: Docs sync + PR2_HANDOFF.md for Claude Code
[/external_agent_tool_result]
```

#### 453. assistant (2026-07-02T06:11:34.096Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\utils\paths.py
[/external_agent_tool_call]
```

#### 454. assistant (2026-07-02T06:11:34.096Z)

```
[external_agent_tool_result]
1 """Path-safety helpers for the API boundary (untrusted request fields)."""
2 import os
3 from typing import Optional
4
5
6 def safe_output_filename(name: str) -> str:
7 """Validate an output filename is a BARE name (no dir part, no traversal, not
absolute).
8
9 `os.path.join(output_dir, name)` silently discards `output_dir` when `name` is
absolute,
10 and `..` segments escape it ? so an unchecked `output_filename` is an arbitrary-
write
11 primitive. Returns the name unchanged when safe; raises ValueError otherwise.
12 """
13 if not name or not name.strip():
14 raise ValueError("output_filename must not be empty")
15 if "\x00" in name:
16 raise ValueError("output_filename must not contain a null byte")
17 if "/" in name or "\\" in name:
```

```

18 raise ValueError(f"output_filename must not contain a path separator: {name!r}")
19 if os.path.isabs(name) or name != os.path.basename(name) or name in (".", ".."):
20 raise ValueError(f"output_filename must be a bare filename, got: {name!r}")
21 return name
22
23
24 def resolve_under_root(path: str, root: str) -> str:
25 """Return the real path, asserting it is contained under `root` (raises ValueError
26 if not)."""
27 rp = os.path.realpath(path)
28 rr = os.path.realpath(root)
29 try:
30 contained = os.path.commonpath([rp, rr]) == rr
31 except ValueError: # different drives on Windows
32 contained = False
33 if not contained:
34 raise ValueError(f"path {path!r} escapes the allowed root {root!r}")
35 return rp
36
37 def validate_input_path(path: str, allowed_root: Optional[str] = None) -> None:
38 """Reject null bytes and (when an allowed_root is configured) out-of-root inputs.
39
40 With no allowed_root (default, single-user local mode) only the null-byte check
41 applies,
42 so the local 'point it at any of my videos' workflow is preserved. Setting an
43 allowed_root (a future hosted/tenant deployment) enforces containment.
44 """
45 if "\x00" in path:
46 raise ValueError("video_path must not contain a null byte")
47 if allowed_root:
48 resolve_under_root(path, allowed_root)
49
50 [/external_agent_tool_result]

```

#### 455. assistant (2026-07-02T06:11:34.096Z)

```

[external_agent_tool_call: Grep]
input: {"pattern": "class
SecurityConfig|security|allowed_video_root", "path": "C:\\Users\\avidu\\Projects\\badminton-
highlight-indexer\\backend\\config\\models.py", "output_mode": "content", "-n": true, "-B": 2, "-A": 12}
[/external_agent_tool_call]

```

#### 456. assistant (2026-07-02T06:11:34.096Z)

```

[external_agent_tool_result]
129-
130-
131: class SecurityConfig(BaseModel):
132- # When set, /api/process video_path must resolve under this directory (hosted/tenant
133- # mode). Default None preserves the single-user local 'any local file' workflow.
134: allowed_video_root: Optional[str] = None
135-
136-
137: class AppConfig(BaseModel):
138- """Root configuration object.
139-
140- All runtime configuration should be accessed via an instance of this class
141- (or the loader) rather than raw dict lookups.
142- """
143-
144- sport: Sport = "badminton"
145- ai_provider: str = "gemini"
146- # Auto-tracking alias: 2.5-flash is legacy now that 3.5-flash is the stable flagship;
147-
148-

```

```

152- output: OutputConfig = Field(default_factory=OutputConfig)
153- stitching: StitchingConfig = Field(default_factory=StitchingConfig)
154- security: SecurityConfig = Field(default_factory=SecurityConfig)
155-
156- # Allow extra keys during transition so existing config.json files do not break.
157- model_config = {
158- "extra": "allow",
159- "validate_assignment": True,
160- }
161-
162- # Temporary migration shim for legacy dict-style access (config.get("section",
163- {}).get(...)).
164- # Will be removed once all call sites (validators, segmenters, etc.) move to typed
165- # attribute access / the typed config. A nested *section* is returned as a plain dict
166- # via model_dump, so legacy chained access keeps working ? returning the Pydantic
167- # section model instead breaks it (a model has no .get), which bit every validator on
168- [/external_agent_tool_result]

```

#### 457. assistant (2026-07-02T06:11:34.096Z)

Design locked (init ? sequential parts ? complete, with caps from typed config). Reading the compile handler in app.js before wiring the download link:

#### 458. assistant (2026-07-02T06:11:34.096Z)

```

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"18","status":"completed"}
[/external_agent_tool_call]

```

#### 459. assistant (2026-07-02T06:11:34.096Z)

```

[external_agent_tool_result]
Updated task #18 status
[/external_agent_tool_result]

```

#### 460. assistant (2026-07-02T06:11:34.096Z)

```

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"19","status":"in_progress"}
[/external_agent_tool_call]

```

#### 461. assistant (2026-07-02T06:11:34.096Z)

```

[external_agent_tool_result]
Updated task #19 status
[/external_agent_tool_result]

```

#### 462. assistant (2026-07-02T06:11:34.096Z)

```

[external_agent_tool_call: Grep]
input: {"pattern":"api/compile","path":"C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\backend\\api\\static\\app.js","output_mode":"content","-n":true,"-B":10,"-A":30}
[/external_agent_tool_call]

```

#### 463. assistant (2026-07-02T06:11:34.096Z)

```

[external_agent_tool_result]
363- btnCompile.addEventListener("click", async () => {
364- if (!currentVideoId || selectedRallyIds.size === 0) {
365- alert("No rallies selected for compiling highlights.");
366- return;
367- }
368-
369- const outputName = highlightOutputName.value.trim() || "highlights_compiled.mp4";

```

```

370- showLoader("Compiling Highlight Reels", "Triggering localized high-speed FFmpeg
stitching demuxers...");
371-
372- try {
373- const res = await fetch("/api/compile", {
374- method: "POST",
375- headers: { "Content-Type": "application/json" },
376- body: JSON.stringify({
377- video_id: currentVideoId,
378- rally_ids: Array.from(selectedRallyIds),
379- output_filename: outputName
380- })
381- });
382-
383- if (!res.ok) {
384- const err = await res.json();
385- throw new Error(err.detail || "Concatenation failed.");
386- }
387-
388- const data = await res.json();
389- alert(`Highlight compilation successfully completed! Output saved
to:\n${data.filepath}`);
390-
391- } catch (error) {
392- alert(`Stitching compilation failed: ${error.message}`);
393- } finally {
394- hideLoader();
395- }
396- });
397-});
[/external_agent_tool_result]

```

#### 464. assistant (2026-07-02T06:11:34.096Z)

```

[external_agent_tool_call: Grep]
input: {"pattern":"^from typing|^import","path":"C:\\Users\\avidu\\Projects\\badminton-
highlight-indexer\\backend\\config\\models.py","output_mode":"content","-n":true}
[/external_agent_tool_call]

```

#### 465. assistant (2026-07-02T06:11:34.097Z)

```

[external_agent_tool_result]
11:from typing import Any, Literal, Optional
[/external_agent_tool_result]

```

#### 466. assistant (2026-07-02T06:11:34.097Z)

```

[external_agent_tool_call: Read]
file: C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend\\config\\models.py
[/external_agent_tool_call]

```

#### 467. assistant (2026-07-02T06:11:34.097Z)

```

[external_agent_tool_result]
8
9 from __future__ import annotations
10
11 from typing import Any, Literal, Optional
12
13 from pydantic import BaseModel, Field, field_validator, model_validator
[/external_agent_tool_result]

```

#### 468. assistant (2026-07-02T06:11:34.097Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py
[/external_agent_tool_call]
```

**469. assistant (2026-07-02T06:11:34.097Z)**

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

**470. assistant (2026-07-02T06:11:34.097Z)**

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py
[/external_agent_tool_call]
```

**471. assistant (2026-07-02T06:11:34.097Z)**

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

**472. assistant (2026-07-02T06:11:34.097Z)**

Now main.py ? imports, then the upload + download endpoints in one block after the jobs endpoint:

**473. assistant (2026-07-02T06:11:34.097Z)**

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py
[/external_agent_tool_call]
```

**474. assistant (2026-07-02T06:11:34.097Z)**

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

**475. assistant (2026-07-02T06:11:34.097Z)**

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py
[/external_agent_tool_call]
```

**476. assistant (2026-07-02T06:11:34.097Z)**

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

**477. assistant (2026-07-02T06:11:34.097Z)**

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py
[/external_agent_tool_call]
```

**478. assistant (2026-07-02T06:11:34.097Z)**



```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

**479. assistant (2026-07-02T06:11:34.098Z)**

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py
[/external_agent_tool_call]
```

**480. assistant (2026-07-02T06:11:34.098Z)**

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

**481. assistant (2026-07-02T06:11:34.098Z)**

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"19","status":"completed"}
[/external_agent_tool_call]
```

**482. assistant (2026-07-02T06:11:34.098Z)**

```
[external_agent_tool_result]
Updated task #19 status
[/external_agent_tool_result]
```

**483. assistant (2026-07-02T06:11:34.098Z)**

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"20","status":"in_progress"}
[/external_agent_tool_call]
```

**484. assistant (2026-07-02T06:11:34.098Z)**

```
[external_agent_tool_result]
Updated task #20 status
[/external_agent_tool_result]
```

**485. assistant (2026-07-02T06:11:34.098Z)**

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\api\static\app.js
[/external_agent_tool_call]
```

**486. assistant (2026-07-02T06:11:34.098Z)**

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\api\static\app.js has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

**487. assistant (2026-07-02T06:11:34.098Z)**

Now the test suite for both endpoints:

**488. assistant (2026-07-02T06:11:34.098Z)**

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"20","status":"completed"}
```

[/external\_agent\_tool\_call]

**489. assistant (2026-07-02T06:11:34.098Z)**

[external\_agent\_tool\_result]  
Updated task #20 status  
[/external\_agent\_tool\_result]

**490. assistant (2026-07-02T06:11:34.098Z)**

[external\_agent\_tool\_call: TaskUpdate]  
input: {"taskId":"21","status":"in\_progress"}  
[/external\_agent\_tool\_call]

**491. assistant (2026-07-02T06:11:34.098Z)**

[external\_agent\_tool\_result]  
Updated task #21 status  
[/external\_agent\_tool\_result]

**492. assistant (2026-07-02T06:11:34.098Z)**

[external\_agent\_tool\_call: Write]  
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test\_upload\_download.py  
[/external\_agent\_tool\_call]

**493. assistant (2026-07-02T06:11:34.098Z)**

[external\_agent\_tool\_result]  
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test\_upload\_download.py (file state is current in your context ? no need to Read it back)  
[/external\_agent\_tool\_result]

**494. assistant (2026-07-02T06:11:34.098Z)**

Tests written. Checking if the sandbox healed (did you get a chance to remove the WSL folder?):

**495. assistant (2026-07-02T06:11:34.099Z)**

[external\_agent\_tool\_call: mcp\_\_workspace\_\_bash]  
command: echo ping  
[/external\_agent\_tool\_call]

**496. assistant (2026-07-02T06:11:34.099Z)**

[external\_agent\_tool\_result: error]  
UNC paths are not supported: \\wsl.localhost\ubuntu\home\avidullu\projects  
[/external\_agent\_tool\_result]

**497. assistant (2026-07-02T06:11:34.099Z)**

Still wedged ? CI stays the verifier (it did its job perfectly on PR-1). Final lap: docs sync + the handoff for Claude Code:

**498. assistant (2026-07-02T06:11:34.099Z)**

[external\_agent\_tool\_call: Edit]  
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE\_MAP.md  
[/external\_agent\_tool\_call]

**499. assistant (2026-07-02T06:11:34.099Z)**

[external\_agent\_tool\_result]  
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE\_MAP.md has been updated successfully. (file state is current in your context ? no need to Read it back)  
[/external\_agent\_tool\_result]

**500. assistant (2026-07-02T06:11:34.099Z)**

[external\_agent\_tool\_call: Edit]  
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI\_ALPHA\_EXECUTION\_PLAN.md  
[/external\_agent\_tool\_call]

**501. assistant (2026-07-02T06:11:34.099Z)**

[external\_agent\_tool\_result]  
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI\_ALPHA\_EXECUTION\_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)  
[/external\_agent\_tool\_result]

**502. assistant (2026-07-02T06:11:34.099Z)**

[external\_agent\_tool\_call: Edit]  
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT\_STEPS.md  
[/external\_agent\_tool\_call]

**503. assistant (2026-07-02T06:11:34.099Z)**

[external\_agent\_tool\_result]  
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT\_STEPS.md has been updated successfully. (file state is current in your context ? no need to Read it back)  
[/external\_agent\_tool\_result]

**504. assistant (2026-07-02T06:11:34.099Z)**

[external\_agent\_tool\_call: Write]  
file: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\PR2\_HANDOFF.md  
[/external\_agent\_tool\_call]

**505. assistant (2026-07-02T06:11:34.099Z)**

[external\_agent\_tool\_result]  
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\PR2\_HANDOFF.md (file state is current in your context ? no need to Read it back)  
[/external\_agent\_tool\_result]

**506. assistant (2026-07-02T06:11:34.099Z)**

[external\_agent\_tool\_call: TaskUpdate]  
input: {"taskId":"21","subject":"PR-2: Tests written (suite run ? Claude Code step 0, sandbox still down)","status":"completed"}  
[/external\_agent\_tool\_call]

**507. assistant (2026-07-02T06:11:34.099Z)**

[external\_agent\_tool\_result]  
Updated task #21 subject, status  
[/external\_agent\_tool\_result]

**508. assistant (2026-07-02T06:11:34.099Z)**

[external\_agent\_tool\_call: TaskUpdate]  
input: {"taskId":"22","status":"completed"}  
[/external\_agent\_tool\_call]

**509. assistant (2026-07-02T06:11:34.099Z)**

```
[external_agent_tool_result]
Updated task #22 status
[/external_agent_tool_result]
```

**510. assistant (2026-07-02T06:11:34.100Z)**

```
[external_agent_tool_call: mcp_cowork_present_files]
input: {"files":[{"file_path":"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\scratch\\PR2_HANDOFF.md"}]}
[/external_agent_tool_call]
```

**511. assistant (2026-07-02T06:11:34.100Z)**

```
[external_agent_tool_result]
C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\scratch\\PR2_HANDOFF.md
[/external_agent_tool_result]
```

**512. assistant (2026-07-02T06:11:34.100Z)**

PR-2 is built ? slow and steady, with two stowaway fixes that matter for tomorrow. Your entire to-do is one sentence to Claude Code:

```
> ***Run scratch/PR2_HANDOFF.md*** *(and if #88 is still open: "run scratch/PR88_FIX.md first")*
```

It runs the tests first (sandbox's still down here ? please do remove that `\\wsl.localhost\\...` folder from this session's folder list when you get a moment, it'll fix test-running for all future sessions), then branches, commits, opens the PR. You review and merge. That's the collaboration model working as ratified: I build and document, Claude Code does git, you decide.

What you get for tomorrow once it merges, in CUJ terms: record on a tripod in the morning ? `--server` ? paste the path in the browser ? watch job stages tick ? rally cards (which would have *\*never rendered\** before today ? the static UI had two fatal field-name bugs, both now fixed) ? select ? compile ? click the new *\*\*\*Download\*\** link. The full handoff has the step-by-step runbook at the bottom.

Honest scope note, since you prefer steady: tomorrow's CUJ is detect ? compile ? download. Accept/reject/nudge-corrections is PR-4 + the real KhelSutra SPA ? still ahead of us. Enjoy the badminton session ? and if the tripod's rolling, that footage feeds the corpus too. ?

**513. user (2026-07-02T06:11:34.100Z)**

Claude is working and just curious if you can communicate with it and check on its status? Additionally, when the work items are done will I have a nice UI in the morning ? :) I see some manual tasks for me which I am happy to do, but if its directory creation and permissions then I guess Claude Code has all those access as well.

**514. assistant (2026-07-02T06:11:34.100Z)**

Let me answer the status question the only way I genuinely can ? I have no direct line to Claude Code (we're separate processes; our "collaboration" is through files and git), but I *\*can\** watch its work land on GitHub through your browser. Checking now:

**515. assistant (2026-07-02T06:11:34.100Z)**

```
[external_agent_tool_call: mcp_Claude_in_Chrome_tabs_context_mcp]
input: {"createIfEmpty":true}
[/external_agent_tool_call]
```

**516. assistant (2026-07-02T06:11:34.100Z)**

```
[external_agent_tool_result]
{"availableTabs":[{"tabId":2039718366,"title":"New
```

```
Tab", "url": "chrome://newtab/"}], "tabGroupId": 2144914490}
```

Tab Context:

- Available tabs:

```
? tabId 2039718366: "New Tab" (chrome://newtab/)
<system-reminder>You used a single tool call this turn. Prefer browser_batch to execute multiple
actions in one call ? it is significantly faster. Batch your next sequence of clicks, types,
navigations, and screenshots together.</system-reminder>
[/external_agent_tool_result]
```

#### 517. assistant (2026-07-02T06:11:34.100Z)

```
[external_agent_tool_call: mcp__Claude_in_Chrome__browser_batch]
input: {"actions":[{"name":"navigate","input":{"url":"https://github.com/avidullu/badminton-
highlight-indexer/pulls?q=is%3Apr+sort%3Aupdated-desc","tabId":2039718366}},{"name":"computer","
input":{"action":"wait","duration":3,"tabId":2039718366}},{"name":"get_page_text","input":{"tabI
d":2039718366}}]}
[/external_agent_tool_call]
```

#### 518. assistant (2026-07-02T06:11:34.100Z)

```
[external_agent_tool_result]
[navigate] Navigated to https://github.com/avidullu/badminton-highlight-
indexer/pulls?q=is%3Apr+sort%3Aupdated-desc
[computer:wait] Waited for 3 seconds
[get_page_text] Title: Pull requests · avidullu/badminton-highlight-indexer
URL: https://github.com/avidullu/badminton-highlight-indexer/pulls?q=is%3Apr+sort%3Aupdated-desc
Source element: <main>

Pull requests: avidullu/badminton-highlight-indexer
Filters
Labels 9
Milestones 0
New pull request
Clear current search query, filters, and sorts
0 Open 100 Closed
Author
Label
Projects Milestones Reviews
Assignee
Sort
Pull requests list
docs: status sync - PR-2 merged (live-E2E-verified), quality sweep recorded
#100 by avidullu was merged
updated
feat(api): chunked upload + highlight download (B2+B3, PR-2)
#99 by avidullu was merged
updated
fix(detectors): keep ~ expandable in WSL staging/inference commands
#97 by avidullu was merged
updated
ci: bump checkout and setup-python to Node-24 majors (v6)
#98 by avidullu was merged
updated
feat(config): warn loudly on unknown config.json keys (dotted paths)
#96 by avidullu was merged
updated
ci: coverage floor on the full-suite lane (ratchet-only)
#95 by avidullu was merged
updated
fix(db): actually close SQLite connections (transaction scope != close)
#94 by avidullu was merged
updated
ci: add mypy lane (lenient green baseline + explicit quarantine list)
```

#93 by avidullu was merged  
updated  
ci: ruff lint lane + fix all violations; remove never-compiled flyer stub  
#92 by avidullu was merged  
updated  
fix: replace deprecated pydantic .dict() with model\_dump()  
#91 by avidullu was merged  
updated  
chore: repo hygiene - fix stale .coveragerc paths, archive TEST\_RUN\_SUMMARY  
#90 by avidullu was merged  
updated  
docs: KhelSutra UI alpha - proposal, hosting plan, execution plan, ADR-009  
#89 by avidullu was merged  
updated  
docs/khelsutra ui alpha plan  
#88 by avidullu was closed  
updated  
1  
feat(api): async job model (#16) - 202+job\_id, jobs table, worker, crash sweep  
#87 by avidullu was merged  
updated  
Docs: field-research recruiter flyer (C13 + Phase-2 rules)  
#86 by avidullu was merged  
updated  
Docs: NEXT\_STEPS session wrap (quality table + ship-target 0.66 + corpus priorities)  
#85 by avidullu was merged  
updated  
Eval: record the Gemini-wrapper baselines (moat reference)  
#84 by avidullu was merged  
updated  
Provider: 6 generate attempts with exponential backoff (503 waves)  
#83 by avidullu was merged  
updated  
test(reporting): CI-runnable coverage for the oversize-skip run\_signals channel  
#82 by avidullu was merged  
updated  
Tests: pin the CI-visible half of the oversize-signal channel (rescued from #80's branch)  
#81 by avidullu was merged  
updated  
fix(gemini): re-encode AI chunks so 4K sources don't silently lose every chunk to the upload cap  
#80 by avidullu was merged  
updated  
Gemini segmenter: honor ai\_max\_workers (burst-429 + shared-client fix path)  
#79 by avidullu was merged  
updated  
Config: ai\_model ? gemini-flash-latest (auto-tracking alias)  
#78 by avidullu was merged  
updated  
Gen-0 TAL harness: train ? honest LOVO ? ship-gate ? artifact bundle (M0.4 scaffold)  
#77 by avidullu was merged  
updated  
Fix: featurizer purity test skips on hosts without numpy/scipy  
#76 by avidullu was merged  
updated  
Previous 1 2 3 4 Next  
ProTip! Follow long discussions with comments:>50.

#### Tab Context:

- Executed on tabId: 2039718366  
- Available tabs:  
  ? tabId 2039718366: "Pull requests · avidullu/badminton-highlight-indexer"  
(<https://github.com/avidullu/badminton-highlight-indexer/pulls?q=is%3Apr+sort%3Aupdated-desc>)  
[/external\_agent\_tool\_result]

#### 519. assistant (2026-07-02T06:11:34.100Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"PR-2 · B2\\+B3|PR-2.*MERGED|CODE
COMPLETE", "path":"C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\docs\\UI_ALPHA_EXECUTION_PLAN.md", "output_mode":"content", "-n":true}
[/external_agent_tool_call]
```

#### 520. assistant (2026-07-02T06:11:34.100Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"PR-2|Updated:", "path":"C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\docs\\NEXT_STEPS.md", "output_mode":"content", "-n":true, "head_limit":8}
[/external_agent_tool_call]
```

#### 521. assistant (2026-07-02T06:11:34.100Z)

```
[external_agent_tool_result]
31:[Omitted long matching line]
[/external_agent_tool_result]
```

#### 522. assistant (2026-07-02T06:11:34.101Z)

```
[external_agent_tool_result]
3:**Updated:** 2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track
opened**:
5:`docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**; **PR-2
[/external_agent_tool_result]
```

#### 523. assistant (2026-07-02T06:11:34.101Z)

```
[external_agent_tool_call: Read]
file: C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\UI_ALPHA_EXECUTION_PLAN.md
[/external_agent_tool_call]
```

#### 524. assistant (2026-07-02T06:11:34.101Z)

```
[external_agent_tool_result]
28
29 **L1 ? backend enablers** (indexer repo · branch off master · ONE PR each, no stacking ·
offline tests in every PR · GPU/real-video checks = owner):
30 - [x] **PR-1 · B1 async job model (#16):** `jobs` table (user_version 1?2), `POST
/api/process` ? 202 + `job_id`, `GET /api/jobs/{job_id}`, ThreadPoolExecutor worker,
restart/crash state handling ? **MERGED 2026-06-11, PR #87** (CI green: 333 passed; machine-side
real-video checks fold into L3)
31 - [x] **PR-2 · B2+B3 upload & download:** chunked upload (init ? sequential ?95 MB parts
? complete ? MD5; caps + mp4/mov allowlist; per-user dirs deferred to PR-3) + streaming `GET
/api/highlights/{video_id}/{name}` ? **MERGED 2026-06-11, PR #99** (also fixes static-UI
`segment_ids` compile bug + adds browser download link). **Live E2E verified at merge:** 86 MB
real video uploaded in 3 sequential parts and reassembled to MD5 byte-identity; out-of-order 409
/ traversal 400 (handler) + 404 (router) / unknown-id 404 confirmed; compile of a real confirmed
rally ? streamed ? download (3.9 MB reel).
32 - [] **PR-3 · B4+B7 identity & exposure:** `owner_email` column + scoping on every
query; Access-JWT/header read with `DEV_USER_EMAIL` env fallback for local dev; CORS narrowed;
rate limit; per-user quota
33 - [] **PR-4 · B6 correction loop:** `PATCH /api/segments/{id}` accept/reject/boundary-
nudge ? `candidate_segments.human_label`/`notes` + provenance (who/when); retire
`backend/api/static` (B5)
34
35 **L2 ? SPA** (khelsutra repo, `web/`; dev server proxies to `http://localhost:8000`):
36 - [] **M1 scaffold:** Vite + React + TS + Tailwind, router, typed API client, branding
(KhelSutra Guru)
37 - [] **M2:** A2 Upload (chunked, guidelines checklist) + A3 Job Progress (poll)
[/external_agent_tool_result]
```

### 525. assistant (2026-07-02T06:11:34.101Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md
[/external_agent_tool_call]
```

### 526. assistant (2026-07-02T06:11:34.101Z)

```
[external_agent_tool_result]
3 **Updated:** 2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha
track opened**:
4 ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see
5 `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**;
**PR-2
6 upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +
7 compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep
#90?#98 landed** ? ruff/mypy/
8 coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into
[/external_agent_tool_result]
```

### 527. assistant (2026-07-02T06:11:34.101Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-
78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-
50773dd063e6\memory\session-handoff-2026-06-11.md
[/external_agent_tool_call]
```

### 528. assistant (2026-07-02T06:11:34.101Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-
8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-
50773dd063e6\memory\session-handoff-2026-06-11.md has been updated successfully. (file state is
current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

### 529. assistant (2026-07-02T06:11:34.101Z)

Here's the full status report, fresh from GitHub via your browser:

**Claude Code's status: done, and then some.** PR-2 is **merged (#99)** ? and the merge note says it was **\*live-E2E-verified\***: a real 86 MB video chunk-uploaded in 3 sequential parts, reassembled to MD5 byte-identity, a real rally compiled, and the 3.9 MB reel downloaded through the browser. The docs PR landed too (#89, old #88 closed). Then it kept going: a nine-PR quality sweep (#90?#98) I never asked for ? ruff and mypy CI lanes, a coverage floor, the pydantic deprecations from PR-1's logs cleaned up, a real SQLite connection-close fix, config unknown-key warnings. Zero open PRs. And here's the part I find elegant: it already updated the planning docs with all this (#100) ? so while I can't **\*talk\*** to Claude Code, we're genuinely collaborating: I write handoffs it executes, it writes status I read. The loop closed without you in the middle.

**Will the morning UI be "nice"?** It'll be **\*working\***, verified today with real footage ? record on the tripod, paste the path, watch job stages tick, rally cards, compile, click ? download. But honest answer: it's the functional developer UI, not the KhelSutra Guru face from the mockup ? that's the SPA (M1?M4), still ahead. One decision for you, no rush, maybe after tomorrow's CUJ run: next chunk could be **\*PR-3+PR-4\*** (per-user identity + the accept/reject correction loop ? the corpus engine) or **\*SPA M1+M2 first\*** (the pretty face sooner). Your morning experience will probably tell you which itch is bigger.

**Manual tasks:** you're right, and it's by design ? nothing filesystem-ish is yours. The upload



directory creates itself; dirs/permissions/git are all Claude Code's lane. The only things that are genuinely yours: PR review decisions, recording the footage, the Phase-D account-ownership steps (Cloudflare/GoDaddy) much later ? and one 10-second click only you can do: \*\*removing the stale `\\wsl.localhost\...\` folder chip from this Cowork session's folder list\*\*, which un-wedges my test sandbox for good.

Enjoy the badminton tomorrow ? first real match through the pipeline! ?

**530. user (2026-07-02T06:11:34.101Z)**

I tried to right click on that projects folder to find a "remove from workspace" option but I can't find it. Tell me how to relieve you of this WSL burden

**531. assistant (2026-07-02T06:11:34.101Z)**

Time to stop guessing about UI I can't see ? let me look at your screen and find the exact affordance:

**532. assistant (2026-07-02T06:11:34.101Z)**

```
[external_agent_tool_call: ToolSearch]
input: {"query":"computer-use","max_results":30}
[/external_agent_tool_call]
```

**533. assistant (2026-07-02T06:11:34.101Z)**

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

**534. assistant (2026-07-02T06:11:34.102Z)**

```
[external_agent_tool_call: mcp__computer-use__request_access]
input: {"apps":["Claude"],"reason":"Look at the Cowork folder list in the Claude app to find where to remove the stale WSL folder that's blocking the session's shell."}
[/external_agent_tool_call]
```

**535. assistant (2026-07-02T06:11:34.102Z)**

```
[external_agent_tool_result]
{"granted":[{"bundleId":"Claude_pzs8srxjxfjgc!Claude","displayName":"Claude","grantedAt":1781193578749,"tier":"full"}],"denied":[],"screenshotFiltering":"mask"}
[/external_agent_tool_result]
```

**536. assistant (2026-07-02T06:11:34.102Z)**

Access granted ? taking a look at your Claude app to find that folder control:

**537. assistant (2026-07-02T06:11:34.102Z)**

```
[external_agent_tool_call: mcp__computer-use__screenshot]
input: {}
[/external_agent_tool_call]
```

**538. assistant (2026-07-02T06:11:34.102Z)**

```
[external_agent_tool_result]
"Google Chrome", "Zoom Workplace" were open and got hidden before this screenshot (not in the session allowlist). If a previous action was meant to open one of them, that's why you don't see it ? call request_access to add them. "textinputhost.exe", "lockapp.exe", "nvidia overlay.exe", "omapsvcbroker.exe", "systemsettings.exe" were also hidden. These processes own the visible windows but aren't in the installed-apps list ? likely a worker process spawned by a launcher you already granted (e.g. LibreOffice's simpress.exe launches soffice.bin, which owns the actual window). Pass the exact basenames above to request_access.
```

```
[/external_agent_tool_result]
```

**539. assistant (2026-07-02T06:11:34.102Z)**

File Explorer's in front (nice view of the golden corpus, by the way). Bringing the Claude app forward:

**540. assistant (2026-07-02T06:11:34.102Z)**

```
[external_agent_tool_call: mcp__computer-use__open_application]
input: {"app":"Claude"}
[/external_agent_tool_call]
```

**541. assistant (2026-07-02T06:11:34.102Z)**

```
[external_agent_tool_result]
Opened "Claude".
[/external_agent_tool_result]
```

**542. assistant (2026-07-02T06:11:34.102Z)**

```
[external_agent_tool_call: mcp__computer-use__screenshot]
input: {}
[/external_agent_tool_call]
```

**543. assistant (2026-07-02T06:11:34.102Z)**

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

**544. assistant (2026-07-02T06:11:34.102Z)**

```
[external_agent_tool_call: mcp__Claude_in_Chrome__navigate]
input: {"url":"http://localhost:8000","tabId":2039718366}
[/external_agent_tool_call]
```

**545. assistant (2026-07-02T06:11:34.102Z)**

```
[external_agent_tool_result: error]
Tab 2039718366 no longer exists. Call tabs_context_mcp to get current tabs.
[/external_agent_tool_result]
```

**546. assistant (2026-07-02T06:11:34.102Z)**

```
[external_agent_tool_call: mcp__Claude_in_Chrome__tabs_context_mcp]
input: {"createIfEmpty":true}
[/external_agent_tool_call]
```

**547. assistant (2026-07-02T06:11:34.102Z)**

```
[external_agent_tool_result]
{"availableTabs":[{"tabId":2039718371,"title":"New
Tab","url":"chrome://newtab/"}],"tabGroupId":1526259416}
```

Tab Context:

- Available tabs:

? tabId 2039718371: "New Tab" (chrome://newtab/)

<system-reminder>You used a single tool call this turn. Prefer browser\_batch to execute multiple actions in one call ? it is significantly faster. Batch your next sequence of clicks, types, navigations, and screenshots together.</system-reminder>

```
[/external_agent_tool_result]
```

#### 548. assistant (2026-07-02T06:11:34.102Z)

```
[external_agent_tool_call: mcp_Claude_in_Chrome_browser_batch]
input: {"actions":[{"name":"navigate","input":{"url":"http://localhost:8000","tabId":2039718371}}, {"name":"computer","input":{"action":"wait","duration":2,"tabId":2039718371}}, {"name":"get_page_text","input":{"tabId":2039718371}}]}
[/external_agent_tool_call]
```

#### 549. assistant (2026-07-02T06:11:34.103Z)

```
[external_agent_tool_result: error]
[navigate] Navigated to http://localhost:8000
[computer:wait] Waited for 2 seconds

actions[2] (get_page_text) failed: Failed to extract page text: Frame with ID 0 is showing error
page (2 completed, 0 remaining)
[/external_agent_tool_result]
```

#### 550. assistant (2026-07-02T06:11:34.103Z)

Verified through your own Chrome: `localhost:8000` is refusing connections ? **the server is not running**. It only lives while you (or Claude Code) start it. Here's the startup, PowerShell-friendly:

```
...
cd C:\Users\avidu\Projects\badminton-highlight-indexer
python -m backend.main --server
...
```

Then open **http://localhost:8000** in your browser. Notes: run it from the repo root (static files resolve relative to where you launch), keep that terminal window open (Ctrl+C stops the server), and the `gemini` default segmenter needs your rotated `GEMINI\_API\_KEY` in `.env` ? which you've already verified working. Equally valid: just tell Claude Code "start the indexer server." Tomorrow morning: record ? start server ? paste the video's path ? process ? compile ? ? download.

On the WSL folder ? confession: I tried to look at the Cowork UI to point at the exact button, but the Claude app hides its own window while I'm controlling the computer, so I'm guiding from knowledge rather than sight. Three places to try, in order:

1. **Folder chips in this chat** ? the connected folders usually show as small pills near the message box or session header; hover the `projects` one (the wsl.localhost path) and an **??** should appear.
2. **The + / attachment button** in the composer ? it often opens a panel listing connected folders with remove options.
3. **The bulletproof fallback**: when you next start a **fresh** chat with me, the folder list is configurable at that point ? just drop the WSL entry there. Removing it never touches your files; it only revokes my access to that path.

If none of those surface an ?, send me a screenshot of the chat UI and I'll point at the pixel. Either way, nothing blocks tomorrow ? the CUJ runs on your machine, not my sandbox. ?

#### 551. user (2026-07-02T06:11:34.103Z)

ha the bulletproof is the only choice, I'll do it tomorrow or else I will again pull off a late nighter "code date" with you and miss my 7:00 AM badminton session! Appreciate the work!

#### 552. assistant (2026-07-02T06:11:34.103Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\session-handoff-2026-06-11.md
[/external_agent_tool_call]
```

**553. assistant (2026-07-02T06:11:34.103Z)**

```
[external_agent_tool_result]
The file C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\session-handoff-2026-06-11.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

**554. assistant (2026-07-02T06:11:34.103Z)**

Ha ? protecting the 7 AM session is the right call; the whole product exists so people \*play more badminton\*, not less. Everything's in place for tomorrow: PR-2 merged and live-verified, the runbook's in `scratch/PR2\_HANDOFF.md`, and the server is two lines away when you're back from court.

Go smash some rallies ? and record them, the pipeline's hungry. ? Appreciate you too, Avi. Good night!

**555. user (2026-07-02T06:11:34.103Z)**

```
<uploaded_files>
<file><file_path>C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\local_0303e96c-dada-4fe1-8bf4-03585eb32cde\uploads\Umiya
Velociti.m4a</file_path><file_uuid>30895475-f431-455a-8a57-2d650b6e4def</file_uuid></file>
</uploaded_files>
```

Tell me how much did you get from this audio recording. Additionally, I want a way for people to share their highlights as well on their profile and we provide Say 2GB free and after that a deletion policy as set by the user say FIFO, largest first, smallest first, least visited first etc. The strategy is to not use a per video or subscription modeling albeit use storage as the pricing point. What is the CUJ after uploading a video? Do we store it? Say we put a "upload" path max filesize 500 MB or point to some bucket? I want you to make a mental model of this and when I return I want to co design with you. Interesting to see if you pick up my "Hindi" in the audio.

**556. assistant (2026-07-02T06:11:34.103Z)**

```
[external_agent_tool_call: ToolSearch]
input: {"query":"select:DesignSync","max_results":3}
[/external_agent_tool_call]
```

**557. assistant (2026-07-02T06:11:34.103Z)**

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

**558. assistant (2026-07-02T06:11:34.103Z)**

```
[external_agent_tool_call: mcp__workspace__bash]
command: echo ping && ls -la /sessions/great-sharp-gauss/mnt/uploads/ 2>/dev/null | head -5
[/external_agent_tool_call]
```

**559. assistant (2026-07-02T06:11:34.104Z)**

```
[external_agent_tool_result: error]
UNC paths are not supported: \\wsl.localhost\ubuntu\home\avidullu\projects
[/external_agent_tool_result]
```

**560. assistant (2026-07-02T06:11:34.104Z)**

```
[external_agent_tool_call: TaskCreate]
description: docs/STORAGE_SHARING_MODEL.md ? 2GB free, user-chosen eviction, storage-anchored
```

pricing, post-upload CUJ, 500MB-vs-bucket, decision agenda for co-design.  
[/external\_agent\_tool\_call]

**561. assistant (2026-07-02T06:11:34.104Z)**

[external\_agent\_tool\_result]  
Task #23 created successfully: Draft storage-priced sharing mental model (co-design doc)  
[/external\_agent\_tool\_result]

**562. assistant (2026-07-02T06:11:34.104Z)**

[external\_agent\_tool\_call: Write]  
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE\_SHARING\_MODEL.md  
[/external\_agent\_tool\_call]

**563. assistant (2026-07-02T06:11:34.104Z)**

[external\_agent\_tool\_result]  
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE\_SHARING\_MODEL.md (file state is current in your context ? no need to Read it back)  
[/external\_agent\_tool\_result]

**564. assistant (2026-07-02T06:11:34.104Z)**

[external\_agent\_tool\_call: TaskUpdate]  
input: {"taskId":"23","status":"completed"}  
[/external\_agent\_tool\_call]

**565. assistant (2026-07-02T06:11:34.104Z)**

[external\_agent\_tool\_result]  
Updated task #23 status  
[/external\_agent\_tool\_result]

**566. assistant (2026-07-02T06:11:34.104Z)**

[external\_agent\_tool\_call: Edit]  
file: C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\session-handoff-2026-06-11.md  
[/external\_agent\_tool\_call]

**567. assistant (2026-07-02T06:11:34.104Z)**

[external\_agent\_tool\_result]  
The file C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\session-handoff-2026-06-11.md has been updated successfully. (file state is current in your context ? no need to Read it back)  
[/external\_agent\_tool\_result]

**568. assistant (2026-07-02T06:11:34.104Z)**

[external\_agent\_tool\_call: mcp\_cowork\_present\_files]  
input: {"files":[{"file\_path":"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\STORAGE\_SHARING\_MODEL.md"}]}\n[/external\_agent\_tool\_call]

**569. assistant (2026-07-02T06:11:34.104Z)**

[external\_agent\_tool\_result]  
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE\_SHARING\_MODEL.md

[/external\_agent\_tool\_result]

#### 570. assistant (2026-07-02T06:11:34.105Z)

The mental model is drafted and waiting for you. The four beats that shape everything:

**\*\*"Do we store it?" ? yes, but as two different things.\*\*** The raw match is a *\*transient processing input\** (kept through processing + correction, then auto-deleted on a retention deadline); the compiled reels and kept clips are the *\*durable library\** ? and the library is what "2 GB free" measures. That's the only way the math works (2 GB can't hold one raw match but holds ~25?40 reels), and it elegantly kills the free-CDN abuse problem: the library only ever contains pipeline outputs, so nobody can park random files in your storage product.

**\*\*Your eviction policies become a registry\*\*** ? ``EvictionPolicy` ABC` with ``fifo``, ``largest_first``, ``smallest_first``, ``least_visited``, exactly the pattern your segmenters already use. My one push-back for the co-design: default should be ``hard_stop`` (block new saves, warn loudly), not FIFO ? silently deleting someone's match memories is the one unforgivable UX sin in a memories product.

**\*\*The non-obvious economics: storage is nearly free, \*egress\* is the bill.\*\*** A 2 GB user costs under ?5/month to store anywhere ? but one mildly viral shared reel on GCS is ~\$10 of bandwidth, while on Cloudflare R2 it's \$0. So the hybrid: sources ? GCS (your credits + lifecycle auto-delete does retention for free), library/serving ? R2 (virality becomes marketing, not a bill). Storage-anchored pricing only has margins on zero-egress serving.

**\*\*500 MB vs bucket:\*\*** alpha keeps the chunked endpoint (but 500 MB kills full matches ? your real CUJ file sizes tomorrow should set the cap); at beta the bucket *\*is\** the upload path (signed resumable URLs, server never touches bytes ? straight out of your own PLATFORM P2).

The doc ends with a 10-item decision agenda ? that's our co-design session, ready when you are. And the audio is saved in memory as unfinished business: first fresh-chat session with a healed sandbox, I'll run Whisper on it and we'll find out what your Hindi added that the brief didn't. ?

#### 571. user (2026-07-02T06:11:34.105Z)

Let me also add another vector here that a free tier or a cheap tier would be something we never fall back to Gemini. That makes this also a model improvement based pricing structure because then it's two GB with the basic model is what you'll get. But if you pay a little more, then you get the best one. model. and ten GB of storage. There has to be a factor of being able to have a profile and others should be able to interact with you based on the videos. Say it will become a Facebook of videos. But the bar price to pay is this amount of videos you can store on that. Do you think you can explore this angle as well and connect the dots? We want to bake the cost of 1 time egress in the highlight storage. The end goal would be that the USB drive which the user puts in their system having all the videos, the system can process on device (Windows or use the GPU etc.) And customers who are serious will pay but there should be no abusers.

#### 572. assistant (2026-07-02T06:11:34.105Z)

Connecting these dots now ? they actually snap together into something more coherent than the sum of parts. Extending the doc first:

#### 573. assistant (2026-07-02T06:11:34.105Z)

[external\_agent\_tool\_call: Edit]

file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE\_SHARING\_MODEL.md

[/external\_agent\_tool\_call]

#### 574. assistant (2026-07-02T06:11:34.105Z)

[external\_agent\_tool\_result]

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE\_SHARING\_MODEL.md has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external\_agent\_tool\_result]

**575. assistant (2026-07-02T06:11:34.105Z)**

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE_SHARING_MODEL.md
[/external_agent_tool_call]
```

**576. assistant (2026-07-02T06:11:34.105Z)**

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE_SHARING_MODEL.md has
been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

**577. assistant (2026-07-02T06:11:34.105Z)**

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE_SHARING_MODEL.md
[/external_agent_tool_call]
```

**578. assistant (2026-07-02T06:11:34.105Z)**

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\STORAGE_SHARING_MODEL.md has
been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

**579. assistant (2026-07-02T06:11:34.105Z)**

```
[external_agent_tool_call: mcp__visualize__read_me]
input: {"modules":["diagram"],"platform":"desktop"}
[/external_agent_tool_call]
```

**580. assistant (2026-07-02T06:11:34.105Z)**

```
[external_agent_tool_result]
<persisted-output>
Output too large (61.5KB). Full output saved to: C:\Users\avidu\AppData\Roaming\Claude\local-
agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-
c9f3dd74258c\local_0303e96c-dada-4fe1-8bf4-03585eb32cde\.claude\projects\C--Users-avidu-AppData-
Roaming-Claude-local-agent-mode-sessions-87f748ce-e72b-44f3-8ef0-78e3212a99c3-b4180250-5146-
4b86-9001-c9f3dd74258c-local-0303e96c-dada-4fe1-8bf4-03585eb32cde-
outputs\3596c611-d592-407d-90be-16fe13094910\tool-results\toolu_01SZAmZtfN9tNsetr8sgpzja.json
```

Preview (first 2KB):

```
[
 {
 "type": "text",
 "text": "# Imagine ? Visual Creation Suite\n\n## Modules\n\nCall read_me again with the
modules parameter to load detailed guidance:\n- `diagram` ? SVG flowcharts, structural diagrams,
illustrative diagrams\n- `mockup` ? UI mockups, forms, cards, dashboards\n- `interactive` ?
interactive explainers with controls\n- `chart` ? charts, data analysis, geographic maps
(Char.js, D3 choropleth)\n- `art` ? illustration and generative art\n\nPick the closest fit. The
module includes all relevant design guidance.\n\n**Complexity budget ? hard limits:**\n- Box
subtitles: ?5 words. Detail goes in click-through (`sendPrompt`) or the prose below ? not the
box.\n- Colors: ?2 ramps per diagram. If colors encode meaning (states, tiers), add a 1-line
legend. Otherwise use one neutral ramp.\n- Horizontal tier: ?4 boxes at full width (~140px
each). 5+ boxes ? shrink to ?110px OR wrap to 2 rows OR split into overview + detail
diagrams.\n\nIf you catch yourself writing \"click to learn more\" in prose, the diagram itself
must ACTUALLY be sparse. Don't promise brevity then front-load everything.\n\n**Accessibility:**
For HTML widgets, begin with a visually-hidden `<h2 class=\"sr-only\">` containing a one-
sentence summary of the visualization for screen-reader users. (SVG widgets use `role=\"img\"`
with `<title>` and `<desc>` instead ? see SVG setup.)\n\nYou create rich visual content ? SVG
diagrams/illustrations and HTML interactive widgets ? that renders inline in conversation. The
best output feels like a natural extension of the chat.\n\n## Core Design System\n\nThese rules
```

apply to ALL use cases.\n\n### Philosophy\n- **\*\*Seamless\*\***: Users shouldn't notice where claude.ai ends and your widget begins.\n- **\*\*Flat\*\***: No gradients, mesh backgrounds, noise textures, or decorative effects. Clean flat surfaces.\n- **\*\*Compact\*\***: Show the essential inline. Explain the rest in text.\n- **\*\*Text goes in your response, visuals go in the tool\*\*** ? All explanatory text, descripti  
...  
</persisted-output>  
[/external\_agent\_tool\_result]

**581. assistant (2026-07-02T06:11:34.105Z)**

[external\_agent\_tool\_call: Read]  
file: C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\local\_0303e96c-dada-4fe1-8bf4-03585eb32cde\.claude\projects\C--Users-avidu-AppData-Roaming-Claude-local-agent-mode-sessions-87f748ce-e72b-44f3-8ef0-78e3212a99c3-b4180250-5146-4b86-9001-c9f3dd74258c-local-0303e96c-dada-4fe1-8bf4-03585eb32cde-outputs\3596c611-d592-407d-90be-16fe13094910\tool-results\toolu\_01SZAmZtfN9tNsetr8sgpzja.json  
[/external\_agent\_tool\_call]

**582. assistant (2026-07-02T06:11:34.106Z)**

[external\_agent\_tool\_result: error]  
File content (27116 tokens) exceeds maximum allowed tokens (25000). Use offset and limit parameters to read specific portions of the file, or search for specific content instead of reading the whole file.  
[/external\_agent\_tool\_result]

**583. assistant (2026-07-02T06:11:34.106Z)**

[external\_agent\_tool\_call: mcp\_\_workspace\_\_bash]  
command: grep -o '"text": "'.\*" '"/sessions/great-sharp-gauss/mnt/outputs/3596c611-d592-407d-90be-16fe13094910/tool-results/toolu\_01SZAmZtfN9tNsetr8sgpzja.json" 2>/dev/null | head -2  
[/external\_agent\_tool\_call]

**584. assistant (2026-07-02T06:11:34.106Z)**

[external\_agent\_tool\_result: error]  
UNC paths are not supported: \\wsl.localhost\ubuntu\home\avidullu\projects  
[/external\_agent\_tool\_result]

**585. assistant (2026-07-02T06:11:34.106Z)**

[external\_agent\_tool\_call: Grep]  
input: {"pattern":"var\\(--c","path":"C:\\Users\\avidu\\AppData\\Roaming\\Claude\\local-agent-mode-sessions\\87f748ce-e72b-44f3-8ef0-78e3212a99c3\\b4180250-5146-4b86-9001-c9f3dd74258c\\local\_0303e96c-dada-4fe1-8bf4-03585eb32cde\\.claude\\projects\\C--Users-avidu-AppData-Roaming-Claude-local-agent-mode-sessions-87f748ce-e72b-44f3-8ef0-78e3212a99c3-b4180250-5146-4b86-9001-c9f3dd74258c-local-0303e96c-dada-4fe1-8bf4-03585eb32cde-outputs\\3596c611-d592-407d-90be-16fe13094910\\tool-results\\toolu\_01SZAmZtfN9tNsetr8sgpzja.json","output\_mode":"content","-o":true,"head\_limit":40}  
[/external\_agent\_tool\_call]

**586. assistant (2026-07-02T06:11:34.106Z)**

[external\_agent\_tool\_result]  
4:var(--c  
4:var(--c  
4:var(--c  
4:var(--c  
[/external\_agent\_tool\_result]

**587. assistant (2026-07-02T06:11:34.106Z)**



```
[external_agent_tool_call: Grep]
input: {"pattern":"-c-[a-z0-9-]+", "path":"C:\\Users\\avidu\\AppData\\Roaming\\Claude\\local-agent-mode-sessions\\87f748ce-e72b-44f3-8ef0-78e3212a99c3\\b4180250-5146-4b86-9001-c9f3dd74258c\\local_0303e96c-dada-4fe1-8bf4-03585eb32cde\\.claude\\projects\\C--Users-avidu-AppData-Roaming-Claude-local-agent-mode-sessions-87f748ce-e72b-44f3-8ef0-78e3212a99c3-b4180250-5146-4b86-9001-c9f3dd74258c-local_0303e96c-dada-4fe1-8bf4-03585eb32cde-outputs\\3596c611-d592-407d-90be-16fe13094910\\tool-results\\toolu_01SZAmZtfN9tNsetr8sgpzja.json", "output_mode":"content", "-o":true, "head_limit":60}
[/external_agent_tool_call]
```

#### 588. assistant (2026-07-02T06:11:34.106Z)

```
[external_agent_tool_result]
No matches found
[/external_agent_tool_result]
```

#### 589. assistant (2026-07-02T06:11:34.106Z)

```
[external_agent_tool_call: mcp_visualize_show_widget]
input: {"title":"khelsutra_storage_model_tiers_flywheel", "loading_messages":["Stacking the tier ladder", "Spinning the corpus flywheel", "Wiring shelf to model"], "widget_code": "<h2 style='position:absolute;width:1px;height:1px;overflow:hidden;clip:rect(0 0 0 0)'>Three pricing tiers coupled to one storage meter, and the corpus flywheel where free users' corrections improve the owned model.</h2>\n<div style='font-family:var(--font-sans, system-ui, sans-serif); color:var(--text-primary, #1a1a1a); max-width:760px'\n <div style='font-weight:700; font-size:15px; margin-bottom:2px'\n >One meter (storage) . three coupled axes</div>\n <div style='font-size:12px; color:var(--text-secondary, #666); margin-bottom:12px'\n >The user only ever sees the shelf. Model quality and compute ride the tier.</div>\n\n <div style='display:flex; gap:10px; flex-wrap:wrap; margin-bottom:16px'\n >\n <div style='flex:1; min-width:170px; border:1.5px solid var(--border-strong, #c9c9c9); border-radius:10px; padding:12px'\n >\n <div style='font-weight:700; font-size:13px'\n >? Rally ? free</div>\n <div style='font-size:12px; margin-top:6px; line-height:1.7'\n >\n 2 GB library shelf
\n Owned model ? never Gemini
\n N matches/mo on our GPU
\n Full social\n </div>\n </div>\n <div style='flex:1; min-width:170px; border:1.5px solid var(--accent-primary, #2563eb); border-radius:10px; padding:12px; background:var(--surface-raised, rgba(37,99,235,0.05))'\n >\n <div style='font-weight:700; font-size:13px'\n >? Smash ? paid</div>\n <div style='font-size:12px; margin-top:6px; line-height:1.7'\n >\n 10 GB shelf
\n Best model path + priority
\n Higher caps, our GPU
\n Egress baked into price\n </div>\n </div>\n </div>\n </div>\n <div style='fle?
```

#### 590. assistant (2026-07-02T06:11:34.106Z)

```
[external_agent_tool_result]
Content rendered and shown to the user. Please do not duplicate the shown content in text because it's already visually represented.
[/external_agent_tool_result]
```

#### 591. assistant (2026-07-02T06:11:34.106Z)

```
[external_agent_tool_call: mcp_cowork_present_files]
input: {"files":[{"file_path":"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\STORAGE_SHARING_MODEL.md"}]}
[/external_agent_tool_call]
```

#### 592. assistant (2026-07-02T06:11:34.106Z)

```
[external_agent_tool_result]
C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\STORAGE_SHARING_MODEL.md
[/external_agent_tool_result]
```

#### 593. assistant (2026-07-02T06:11:34.106Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\session-handoff-2026-06-11.md
[/external_agent_tool_call]
```

#### 594. assistant (2026-07-02T06:11:34.106Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\87f748ce-e72b-44f3-8ef0-78e3212a99c3\b4180250-5146-4b86-9001-c9f3dd74258c\spaces\313cf772-aa7b-42d3-8233-50773dd063e6\memory\session-handoff-2026-06-11.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

#### 595. assistant (2026-07-02T06:11:34.106Z)

The dots connect better than you may have realized when you spoke them. Three observations from the synthesis:

\*\*\*"Free never touches Gemini" is the keystone, not a detail.\*\* It fuses three threads that until today existed for separate reasons ? the owned model (built for moat/cost), the correction loop (built for corpus), and storage pricing (built for revenue) ? into one self-reinforcing system: free users' corrections \*are\* the R&D budget for the free tier they use. The flywheel in the visual above is now the business model, not just the training strategy. And it quietly gives your open ship-gate decision (NEXT\_STEPS #2) its business meaning: the owned model ships the moment it's \*good enough to delight a free user\* ? a much more concrete bar than an F1 threshold alone.

\*\*Your 2026-06-09 licensing decision was secretly the end-game moat.\*\* The USB-drive vision ? serious users processing on their own Windows/GPU, cloud holding only the shareable shelf ? is only possible because the owned stack is Apache/MIT-clean. Gemini can never ship on-device; your weights can. The strictest guardrail in the repo turns out to be the product's final form.

\*\*One honest caveat baked into the doc:\*\* while the free tier is cloud-served, owned-model inference still costs \*our\* GPU minutes ? so the N-matches/month cap isn't stinginess, it's the bridge until on-device matures (academies first ? they have the GPUs and the seriousness; consumers after the WSL-free detector path exists).

The doc is now \*\*v2\*\* with all of it ? model-tier axis, the social layer scoped honestly (v1 = profiles + share links + view counts; the full "Facebook of videos" waits for proven retention), abuse economics showing no tier where an abuser extracts more than they pay, and the agenda grown to \*\*14 decisions\*\* grouped for the co-design session. Plus one item I added that you didn't ask for but will care about: minors hard-excluded from \*public\* profiles in v1 ? public sharing is a different risk class than the training-pool rule.

Ready to co-design whenever you walk back in. ?